SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received
MAR 18 2016
ND Oil & Gas Division

Well File No.
28557

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- ☐ Notice of Intent — Approximate Start Date:
- ☑ Report of Work Done — Date Work Completed: **September 9, 2015**
- ☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- ☐ Drilling Prognosis
- ☐ Redrilling or Repair
- ☐ Casing or Liner
- ☐ Plug Well
- ☐ Supplemental History
- ☐ Temporarily Abandon
- ☐ Other **Well is now on pump**
- ☐ Spill Report
- ☐ Shooting
- ☐ Acidizing
- ☐ Fracture Treatment
- ☐ Change Production Method
- ☐ Reclamation

Well Name and Number					
Wade Federal 5300 41-30 7T					
Footages		Qtr-Qtr	Section	Township	Range
877 F **S** L	**280** F **W** L	**LOT 4**	**30**	**153 N**	**100 W**
Field	Pool	County			
Baker	**Bakken**	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 09/09/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 8196.54'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 8130.84'

Company		Telephone Number
Oasis Petroleum North America LLC		**281 404-9436**
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	**TX**	**77002**
Signature	Printed Name	
	Jennifer Swenson	
Title	Date	
Regulatory Specialist	**March 17, 2016**	
Email Address		
jswenson@oasispetroleum.com		

FOR STATE USE ONLY

- ☑ Received
- ☐ Approved

Date: 4-7-2016

By:

Title: JARED THUNE
Engineering Technician

Wade Federal 5300 41-30 7T
End of Well Report

Wade Federal 5300 41-30 7T

877' FSL - 280' FWL
SEC 30, T153N, R100W
McKenzie County, ND

Prepared by:
Dylan E. Fowler, *well-site geologist*
Columbine Logging, Inc.
2385 S. Lipan St.
Denver CO 80223

Prepared for:

Brendan Hargrove, *operations geologist*
1001 Fannin, Suite 1500
Houston, Texas 77002

1.0 INTRODUCTION

Wade Federal 5200 41-30 7T is a East lateral Three Forks 1st Bench well located in SEC 30, T153N, R100W in McKenzie County, North Dakota. The primary pay zone was approximately 20' under the base of the Pronghorn member of the Bakken formation. This pay zone was picked for its production potential and quality of reservoir rock. The objective was to steer the well within the defined pay zone and within the legal requirements of the state. MWD services were performed by Schlumberger/Pathfinder and Directional services were provided by RPM. Heather A. Coutts, Jake Robertson and Dylan E. Fowler were the primary well site geologists; providing geo-steering and mud logging from Columbine Logging, Inc.

Well Information

API #:	33-053-05998	**Field:**	Baker
Spud Date:	09/01//2014	**TD Date:**	11/18/2014

Surface Location: 877' FSL & 280' FWL SEC 30, T153N, R100W, McKenzie County, North Dakota.

Intermediate Casing Point: 647' FSL & 878' FWL; SEC 30; 11,124' MD, 10,777.96' TVD

Bottom Hole Location: 547' FSL & 162' FEL; SEC 29; 20,530' MD, 10,881.61' TVD

Surface Elevation:	2045'	**KB:**	2070'
Casing Shoe:	11,124' MD, 10,778' TVD	**Total Depth Drilled:**	20,530'
Operator:	Oasis	**Rig:**	Patterson 488
Company man:	Tyler LaClaire, Bob Brown	**Well-site Geologist:**	Heather Coutts, Jake Robertson, Dylan E. Fowler
Mud logging:	Columbine Logging	**DD:**	RPM
Mud Service:	Reliable	**Drilling Mud:**	Invert, Brine
MWD:	Schlumberger/Pathfinder		

2.0 SERVICES

2.1 Well site Geology *(Columbine Logging, Inc.)*

Geological consulting and mud logging started on 10/03/2014. Services provided included; morning reports, evening reports, noon and midnight reports, sample examination, sample recording via pictures, production of vertical and horizontal mudlog, geo steering, sample collection and bagging, sample mailing and a final end of well report.

2.1.1 Geosteering

Our offset GR TVD logs were from the Wade Federal 5300 41-30 8T2 well, located on the same well pad as Wade Federal 5300 41-30 7T. Within the Three Forks 1st Bench, the primary objective was to stay near the middle gamma markers which marks the inter layering between limy sandstones and siltstone. Gamma patterns were compared with the offset log and a TVD log was created while landing the curve to in order to land in the targeted zone. Steering in the lateral was accomplished by calculating dip from relevant gamma markers, as well as by using lithology, total gas and ROP to determine our position within the formation.

2.1.2 Gamma and Surveys

Gamma and survey MWD services were provided by Schlumberger/Pathfinder. The majority of the well was drilled within the target area of the Three Forks 1st Bench.

2.2 Mud Logging *(Columbine Logging, Inc.)*

2.2.1 Sample Examination

Samples were collected every 30 ft in the straight hole and build section, and every 30 ft while drilling the lateral. Descriptions included; mineralogy, color, firmness, argillaceous content, structure, texture, allochems, porosity, oil stain, and hydrocarbon fluorescence. Carbonate identification was determined with 10% dilute HCl^-, alizarin red and calcimeter. Hydrocarbon fluorescence was determined using a fluoroscope with a UV lamp.

2.2.2 Gas Detection

Gas was logged using a Bloodhound total gas/chromatograph system. The gas detection system uses an infra-red detector to measure total gas and the chromatograph separates and measures gases C1, C2, C3, iC4 and nC4. Gas was recorded in units where 1 unit equals 100 ppm. The gas detection system measured gases: C1, C2, C3, iC4, nC4, H2S, O2 and CO_2.

The Bloodhound Gas Detection and Chromatograph system use digital signal processing techniques and non-dispersive infrared and chemical sensors for gas detection. The system uses a proprietary chromatograph, which has the capability to detect from 0 to 10,000 gas units. This translates as 0 to 100% typical naturally-occurring hydrocarbon gas mixtures. Calibration is performed using National Institute of Standards and Technology (NIST) traceable calibration gases. Lab calibration points include 0%, 2.5%, and 100% pure methane. Complete immunity to saturation or damage in the presence of high concentrations of both light and heavy hydrocarbon gases precludes the necessity of constant re-calibration or zero referencing. This allows the Bloodhound to react to hydrocarbon based gases from zero to 100% in concentration without dilution.

Lag time was approximated from a calculation of annular velocity based on: pump output, open-hole diameter, cased hole diameter, collar diameter, drill pipe diameter and bottom hole assembly. Connection gases were monitored to confirm lag time calculations and adjustments were made when necessary.

3.0 GEOLOGY

3.1 Formation Tops

Formation tops were picked using ROP, lithology, and gamma ray to identify markers in the curve and lateral (Table 3.1).

Formation/Marker Beds	ACTUAL				Prognosis	
Vertical Section	Top MD (ft)	Top TVD (ft)	THICKNESS (ft)	Difference (ft)	TVD KB/DF(ft)	TVDSS (ft)
Kibbey Lime	8347	8347	145	2	8349	-6279
Charles Salt	8492	8492	684	0	8492	-6422
Base Last Salt	9176	9176	214	0	9176	-7106
Mission Canyon	9391	9390	544	1	9391	-7321
Lodgepole	9935	9934	721	1	9935	-7865
False Bakken	10744	10655	21	9	10664	-8594
Upper Bakken Shale	10762	10676	18	-1	10675	-8605
Middle Bakken	10795	10694	41	0	10694	-8624
Lower Bakken Shale	10885	10735	10	0	10735	-8665
Pronghorn	10912	10745	17	0	10745	-8675
Threeforks	10976	10762	~28	-1	10761	-8691

Table 3.1 Wade Federal 5300 41-30 7T Formation Tops

3.2 Lithology

Sample analysis began at 8,000' MD in the Otter Formation.

3.3 Formation Dip

The formation had an average dip of 89.35°.

3.4 Shows

The vertical-build section was drilled with invert mud and the lateral was drilled with brine/production water. The oil-based mud contributed a background gas of 200-300 units, and saturated cuttings with oil, making all cuttings in the vertical show the same cut and fluorescence. Gas shows were around 2000+ units during the drilling of the lateral.

3.5 Oil Shows

Invert mud was used in the vertical, masking any oil shows. In the lateral part of the well the oil shows were consistently a moderately yellow to yellow green fluorescence with a fast milky blue cut and a medium brown residue ring.

4.0 WELLBORE

The surface location is 877' FSL & 280' FWL SEC 30, T153N, R100W, McKenzie County, North Dakota. Ground elevation is 2,045’ and KB elevation was 2,070’, referenced to the Kelly bushing of Patterson 488. The curve was landed in the Middle Bakken at 11,124’ MD; 10,777.96’ TVD. The lateral was drilled to TD at 20,530’ MD, 10,881.61’ TVD, 547’ FSL & 162’ FEL; SEC 30. Figure 4.1 shows a cross-section of the lateral.

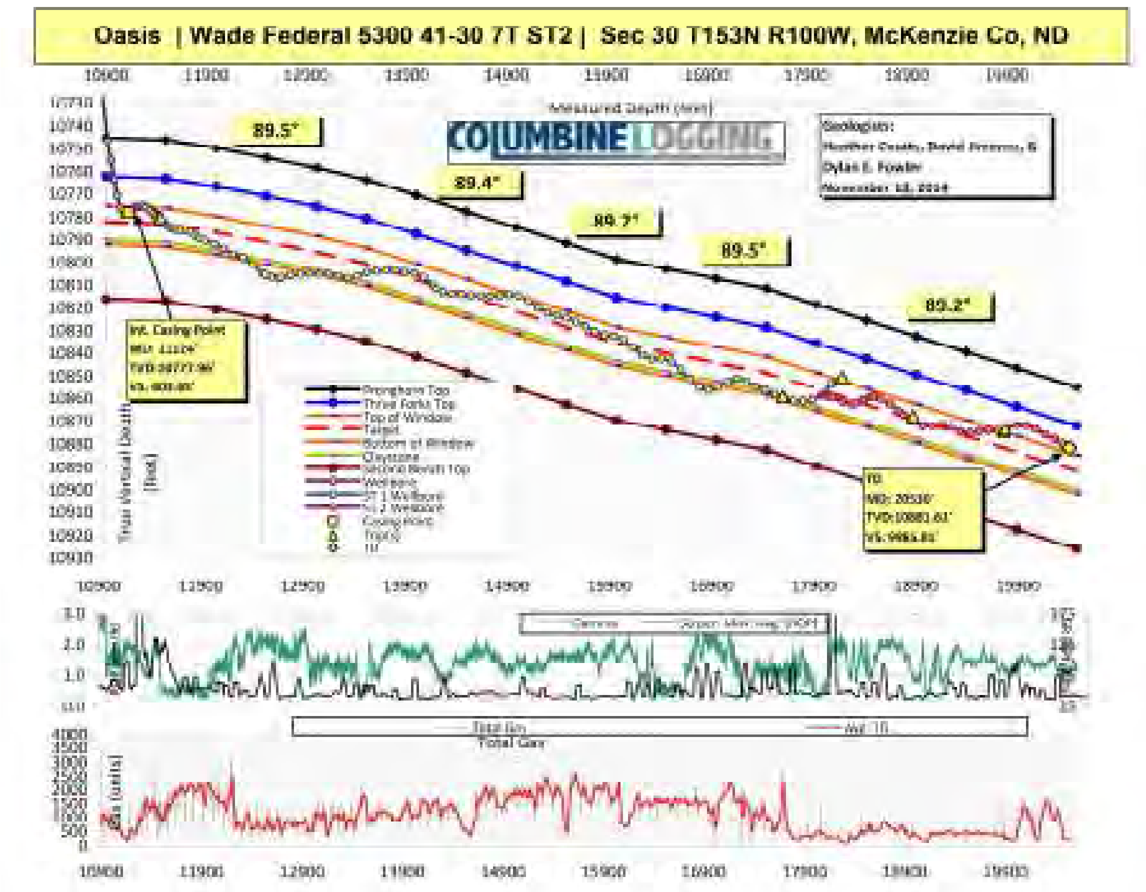


Figure 4.1 Wellbore Cross Section

5.0 SUMMARY AND CONCLUSION

Wade Federal 5300 41-30 7T is a East lateral Three Forks 1st Bench well located in SEC 30, T153N, R100W in McKenzie County, North Dakota. The primary pay zone was 20' under the bottom of the Pronghorn member of the Bakken formation. This pay zone was picked for its production potential and quality of reservoir rock. The objective was to steer the well within the defined pay zone and within the legal requirements of the state.

The primary objective was to stay near the middle gamma markers which marks the inter layering between limy sandstones and siltstones. Gamma patterns were compared with the offset log and a TVD log was created while landing the curve to in order to land in the targeted zone. Steering in the lateral was accomplished by calculating dip from relevant gamma markers, as well as by using lithology, total gas and ROP to determine our position within the formation.

The formation had an average dip of 89.35°. Refer to Figure 4.1 on page 6 for a detailed formation dip profile.

Currently the well is awaiting completion.

Dylan E. Fowler, Well-Site Geologist
Columbine Logging, Inc.
2385 S. Lipan St.
Denver CO 80223
(303) 289-7764

COLUMBINE LOGGING

TH

WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)

JUL 2015
RECEIVED
ND OIL & GAS
DIVISION

Well File No. ~~28978~~ 28557

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion

- ☑ Oil Well
- ☐ Gas Well
- ☐ EOR Well
- ☐ SWD Well
- ☐ Recompletion
- ☐ Water Supply Well
- ☐ Deepened Well
- ☐ Other:
- ☐ Added Horizontal Leg
- ☐ Extended Horizontal Leg

Well Name and Number: **Wade Federal 5300 41-30 7T**	Spacing Unit Description: **Sec. 30/29 T153N R100W**
Operator: **Oasis Petroleum North America** — Telephone Number: **(281) 404-9591**	Field: **Baker**
Address: **1001 Fannin, Suite 1500**	Pool: **Bakken**
City: **Houston** — State: **TX** — Zip Code: **77002**	Permit Type: ☐ Wildcat ☑ Development ☐ Extension

LOCATION OF WELL

At Surface	Qtr-Qtr	Section	Township	Range	County
877 F S L 280 F W L	Lot 4	30	153 N	100 W	McKenzie

Spud Date	Date TD Reached	Drilling Contractor and Rig Number	KB Elevation (Ft)	Graded Elevation (Ft)
September 1, 2014	November 19, 2014	Patterson 488	2077	2045

Type of Electric and Other Logs Run (See Instructions)
MWD/GR from KOP to TD; CBL from int. TD to surface

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	String Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	13 3/8	0	2070	17 1/2	54.5			969	0
Vertical Hole	Intermediate	9 5/8	0	6080	13 1/2	36			1105	
Vertical Hole	Intermediate	7	0	11167	8 3/4	32			801	6400
Lateral1	Liner	4 1/2	10269	20517	6	13.5			507	

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top	Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	~~20530~~	Perforations	11167	~~20517~~	10283		04/24/2015			
	11509			11501						
ST 1	18174		11270	18174						
ST 2	20530		18053	20530						

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft): **Lateral 1- 11167' to 20517'**	Name of Zone (If Different from Pool Name)

Date Well Completed (SEE INSTRUCTIONS)	Producing Method	Pumping-Size & Type of Pump	Well Status (Producing or Shut-In)
June 9, 2015	Flowing		Producing

Date of Test	Hours Tested	Choke Size	Production for Test	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Oil Gravity-API (Corr.)	Disposition of Gas
07/05/2015	24	26 /64		832	23	1108	42.0 °	Sold

Flowing Tubing Pressure (PSI)	Flowing Casing Pressure (PSI)	Calculated 24-Hour Rate	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Gas-Oil Ratio
850	1925		832	23	1108	28

GEOLOGICAL MARKERS

Formation	MD (Ft)	TVD (Ft)
Dakota	5391	5391
Kibbey Lime	8347	8347
First Charles Salt	8492	8492
UB		
Base Last Salt	9176	9176
Ratcliffe		
Mission Canyon	9391	9390
Lodgepole	9935	9934
False Bakken	10744	10655
Scallion		
Upper Bakken	10763	10676
Middle Bakken	10795	10694
Lower Bakken	10885	10735
Pronghorn	10912	10745
Three Forks	10976	10762
Claystone	10976	10789

PLUG BACK INFORMATION

Well Bore	Type of Plug	Top (Ft)	Bottom (Ft)	Sacks Cement

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
04/24/2015	Three Forks	11167	20517	36	134353	Barrels

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)
Sand Frac		3973130	9218	39.0

Details
40/70 White: 1437920
20/40 Ceramic: 1859940
20/40 Resin Coated: 605340
100 Mesh White: 159020

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address **jswenson@oasispetroleum.com**	Date **07/16/2015**
Signature	Printed Name **Jennifer Swenson**	Title **Regulatory Specialist**

AH

AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE – Form 8

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)

Well File No.	NDIC CTB No.
28557	To be assigned 228394

RECEIVED JUN 2015 ND OIL & GAS DIVISION

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number	Qtr-Qtr	Section	Township	Range	County
WADE FEDERAL 5300 41-30 7T	LOT4	30	153	100	~~Williams~~ McKenzie

Operator	Telephone Number	Field
Oasis Petroleum North America LLC	(281) 404-9573	BAKER

Address	City	State	Zip Code
1001 Fannin, Suite 1500	Houston	TX	77002

Name of First Purchaser	Telephone Number	% Purchased	Date Effective
Oasis Petroleum Marketing LLC	(281) 404-9627	100%	June 1, 2015
Principal Place of Business	**City**	**State**	**Zip Code**
1001 Fannin, Suite 1500	Houston	TX	77002
Field Address	**City**	**State**	**Zip Code**

Transporter	Telephone Number	% Transported	Date Effective
Hiland Crude, LLC	(580) 616-2058	75%	June 1, 2015
Address	**City**	**State**	**Zip Code**
P.O. Box 3886	Enid	OK	73702

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	**% Purchased**	**Date Effective**
Other Transporters Transporting From This Lease	**% Transported**	**Date Effective**
Hofmann Trucking	25%	June 1, 2015
Other Transporters Transporting From This Lease	**% Transported**	**Date Effective**
		June 1, 2015

Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date: June 15, 2015
Signature: Dina Barron	Printed Name: Dina Barron	Title: Mktg. Contracts Administrator

Above Signature Witnessed By:

Signature	Printed Name	Title
Alexa Cardona	Alexa Cardona	Marketing Analyst

FOR STATE USE ONLY

Date Approved	JUN 19 2015
By	Evie Roberson
Title	Oil & Gas Production Analyst

Industrial Commission of North Dakota
Oil and Gas Division
Verbal Approval To Purchase and Transport Oil

Well or Facility No **28557**

Tight Hole **Yes**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	**Todd Hanson**	**(701) 577-1632**

WELL INFORMATION

Well Name
WADE FEDERAL 5300 41-30 7T

Inspector
Richard Dunn

Well Location	QQ	Sec	Twp		Rng	
	LOT4	**30**	**153**	**N**	**100**	**W**

County
MCKENZIE

Footages
877 Feet From the **S** Line
280 Feet From the **W** Line

Field
BAKER

Pool
BAKKEN

Date of First Production Through Permanent Wellhead **6/9/2015** **This Is The First Sales**

PURCHASER / TRANSPORTER

Purchaser	Transporter
OASIS PETROLEUM MARKETING LLC	**HOFMANN TRUCKING, LLC**

TANK BATTERY

Central Tank Battery Number : 228394-01

SALES INFORMATION **This Is The First Sales**

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD		DATE
15000	BBLS	**241**	BBLS	**6/9/2015**
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	

DETAILS

Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.

Start Date	**6/9/2015**
Date Approved	**6/17/2015**
Approved By	**Richard Dunn**

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. **28557**

MAY 2015 RECEIVED ND OIL & GAS DIVISION

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **May 18, 2015**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [x] Other **Change well status to CONFIDENTIAL**
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number				
Wade Federal 5300 41-30 ~~7B~~ 7T				
Footages		Qtr-Qtr	Section	Township	Range
877 F **S** L	**280** F **W** L	**LOT4**	**30**	**153 N**	**100 W**
Field: **Baker**	Pool: **Bakken**	County: **McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective immediately, we request CONFIDENTIAL STATUS for the above referenced well.

This well has not been completed

OFF CONFIDENTIAL 11/19/15.

Company	Telephone Number
Oasis Petroleum North America LLC	**281-404-9436**

Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	**TX**	**77002**

Signature	Printed Name
[signature]	**Jennifer Swenson**
Title	Date
Regulatory Specialist	**May 18, 2015**
Email Address	
jswenson@oasispetroleum.com	

FOR STATE USE ONLY

- [ ] Received
- [x] Approved

Date: 5/21/15

By: Alice D. Webber

Title: **Engineering Technician**

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

JAN 2015 RECEIVED ND OIL & GAS DIVISION

CTB

Well File No. 228394-01

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☑ Notice of Intent — Approximate Start Date: **February 15, 2015**

☐ Report of Work Done — Date Work Completed:

☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

☐ Drilling Prognosis ☐ Spill Report
☐ Redrilling or Repair ☐ Shooting
☐ Casing or Liner ☐ Acidizing
☐ Plug Well ☐ Fracture Treatment
☐ Supplemental History ☐ Change Production Method
☐ Temporarily Abandon ☐ Reclamation
☑ Other **Central production facility-commingle pf**

Well Name and Number					
(see details)					
Footages: F L F L	Qtr-Qtr	Section **30**	Township **153 N**	Range **100 W**	
Field	Pool **Bakken**	County **McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America, LLC respectfully requests approval to commingle oil and gas in a central production facility known as 5300 30-29 CTB with common ownership for the following wells:
Well file #28554 Wade Federal 5300 31-30 2B Lot3 Sec. 30 T153N R100W API 33-053-05995
Well file #28394 Wade Federal 5300 41-30 4T Lot4 Sec. 30 T153N R100W API 33-053-05943
Well file #28556 Wade Federal 5300 41-30 5T2 Lot4 Sec. 30 T153N R100W API 33-053-05997
Well file #28425 Wade Federal 5300 41-30 6B Lot4 Sec. 30 T153N R100W API 33-053-05954
Well file #28557 Wade Federal 5300 41-30 7T Lot4 Sec. 30 T153N R100W API 33-053-05998
Well file #28555 Wade Federal 5300 41-30 3T2 is being reevaluated and will not be commingled.
Please find the following attachments:
1. A schematic drawing of the facility which diagrams the testing, treating, routing, and transferring of production. 2. A plat showing the location of the central facility. 3. Affidavit of title indicating common ownership. Oasis will allocate production measured at the central production facility to the various wells on the basis of isolated production tests utilizing oil, gas, and water meters on a test separator at the central production facility. Oasis will measure the production from each well separately each month for a minimum of three days. Oasis believes that such allocation will result in an accurate determination of production from each well. Tank vapor gas is being recovered and burned by a 98% DRE enclosed combuster.

Company		Telephone Number
Oasis Petroleum North America, LLC		**(713) 770-6430**
Address **1001 Fannin Suite 1500**		
City **Houston**	State **TX**	Zip Code **77002**
Signature	Printed Name **David Copeland**	
Title **Regulatory Specialist**	Date **January 24, 2015**	
Email Address **dcopeland@oasispetroleum.com**		

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 2-2-2015	
By David Tabor	
Title **PETROLEUM ENGINEER**	

COMMINGLING AFFIDAVIT

STATE OF NORTH DAKOTA)
) ss.
COUNTY OF MCKENZIE)

The under signed, Alex Haley, of lawful age, being first duly sworn on her oath states that she is a duly authorized agent of Oasis Petroleum North America LLC, and that she has personal knowledge of the facts hereinafter set forth to make this Affidavit.

1. Sections 29 & 30, Township 153 North, Range 100 West, McKenzie County North Dakota constitute a spacing unit in accordance with the applicable orders for the Bakken pool.

2. Six wells have been drilled in the spacing unit, which are known as the Wade Federal 5300 31-30 2B, Wade Federal 5300 41-30 4T, Wade Federal 5300 41-30 5T2, Wade Federal 5300 41-30 6B, and the Wade Federal 5300 41-30 7T

3. By NDIC Order 23339 dated March 18, 2014, all oil and gas interest within the aforementioned spacing unit were pooled.

4. All Working Interests, Royalty Interests and Overriding Royalty Interests in the Wade Federal 5300 31-30 2B, -Wade Federal 5300 41-30 4T, Wade Federal 5300 41-30 5T2, Wade Federal 5300 41-30 6B, and the Wade Federal 5300 41-30 7T will be in common.

Dated this 22nd day of January, 2015

alex Haley

Alexandra Haley, Landman

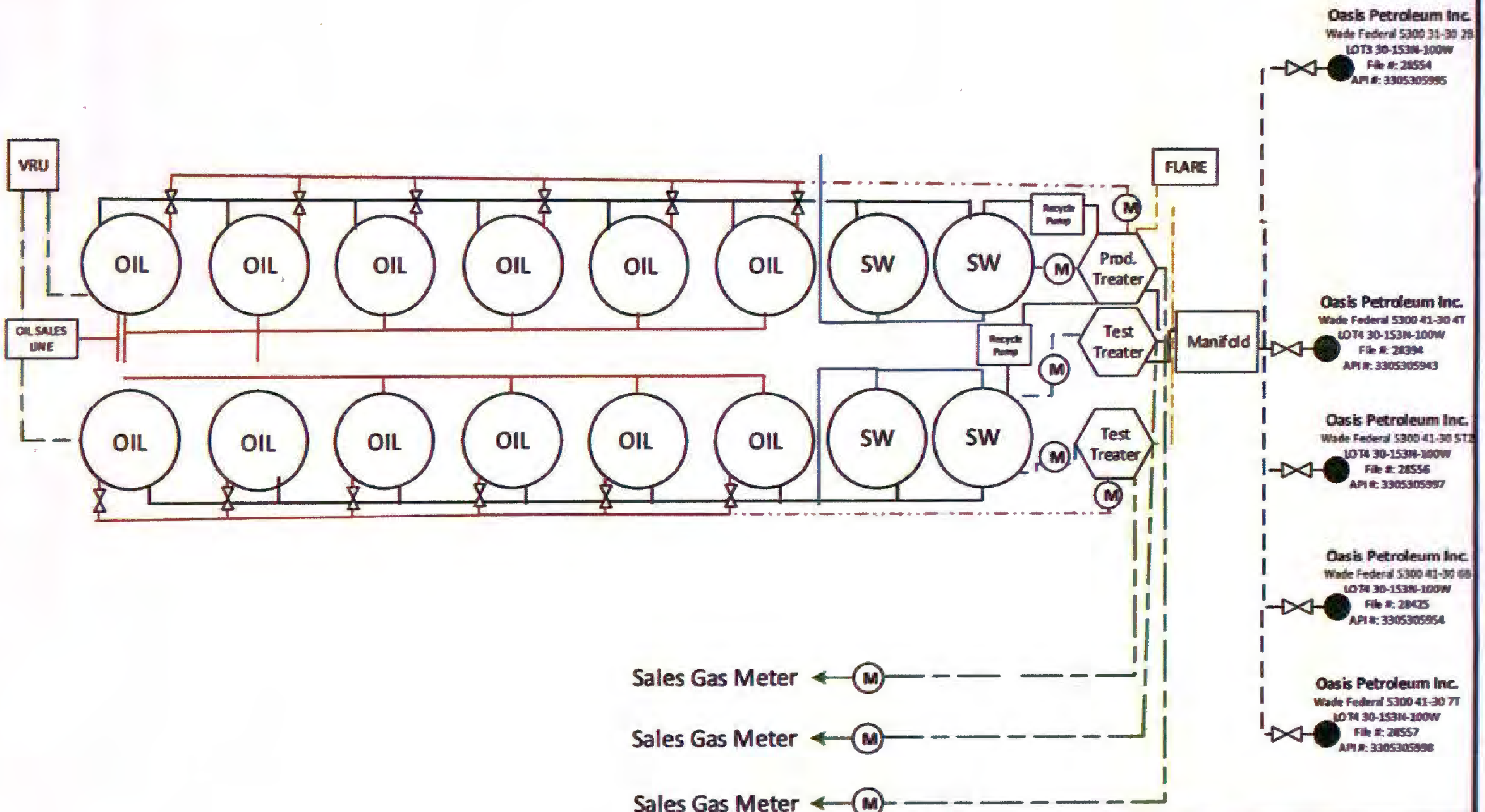
STATE OF TEXAS)
) ss.
COUNTY OF HARRIS)

Subscribed to and sworn before me this 22nd day of January, 2015

Phinetta Roberta Chinn

Notary Public
State of Texas
My Commission Expires: October 31, 2018

PHINETTA ROBERTA CHINN
Notary Public, State of Texas
My Commission Expires
October 31, 2018



OASIS PETROLEUM

WADE FEDERAL 5300 30-29
CENTRAL TANK BATTERY 2

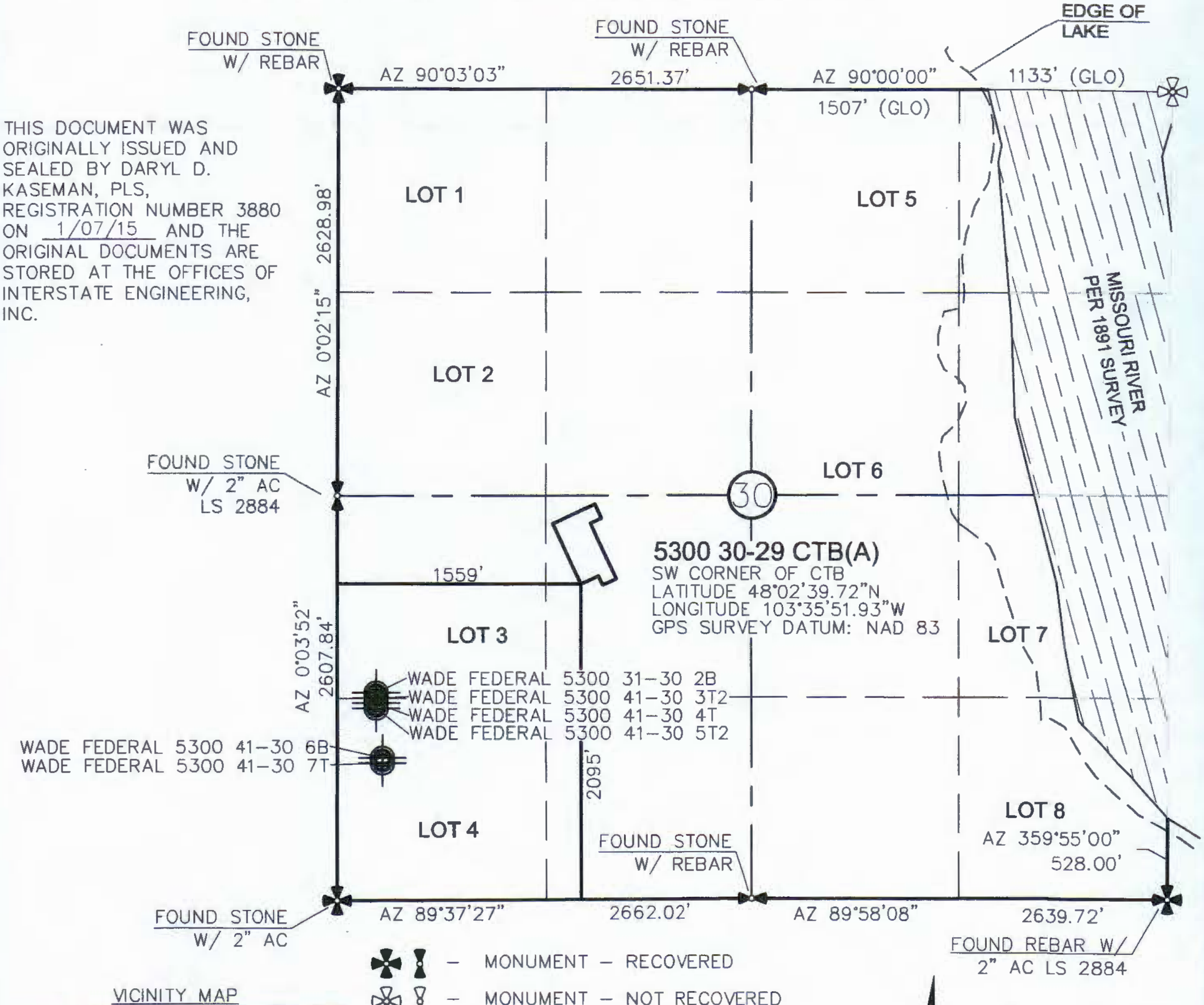
DATE	REV.	BY	APPR.	SCALE
JANUARY 11, 2014	0	LEE		NA

LOCATION	FIELD
NORTH DAKOTA	BAKER

BATTERY LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"5300 30-29 CTB(A)"
SECTION 30, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 1/07/15 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.



VICINITY MAP

STAKED ON 3/25/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

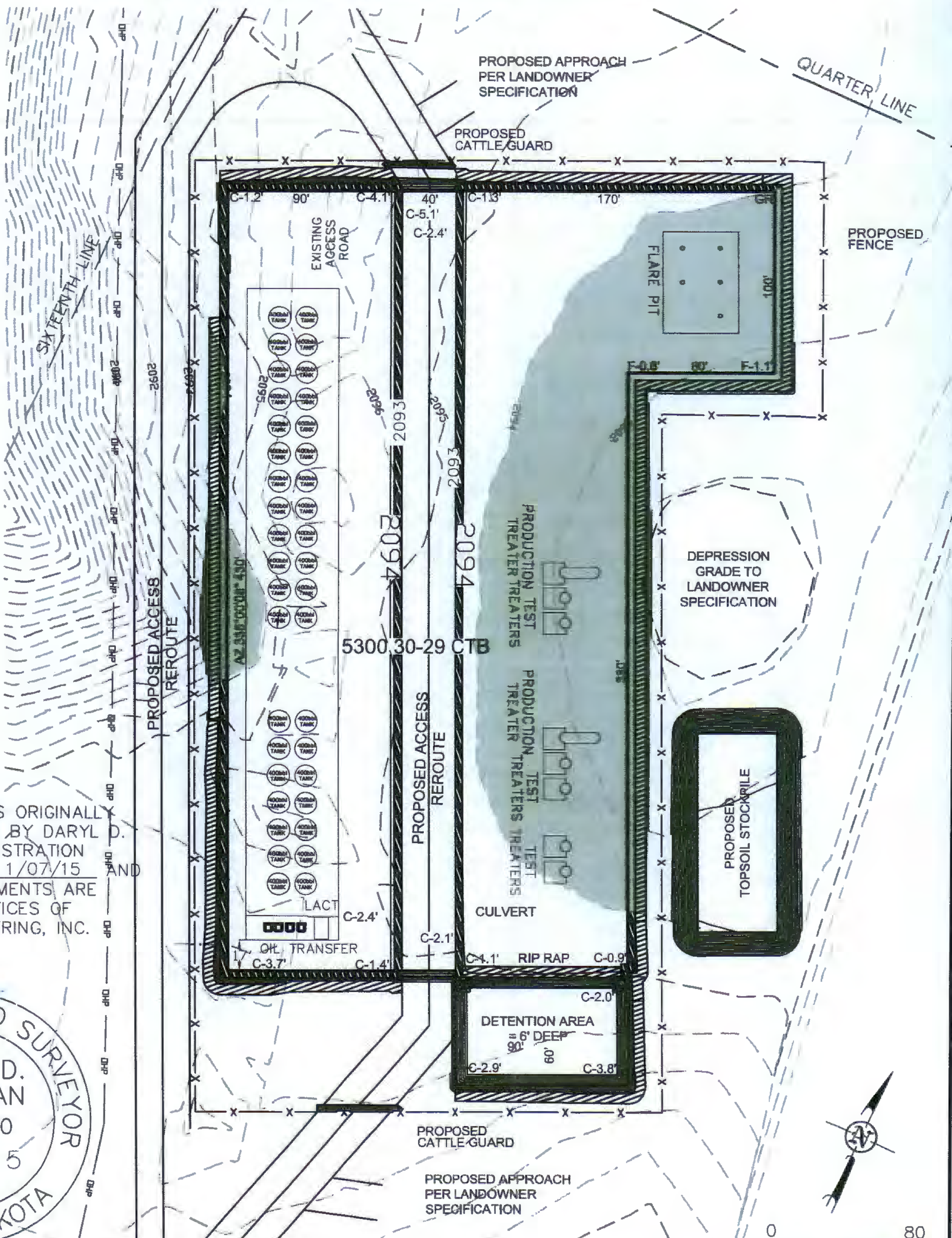
DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 1/7/15
NORTH DAKOTA

SHEET NO.					
1/5	INTERSTATE ENGINEERING Professionals you need, people you trust	Interstate Engineering, Inc. P.O. Box 648 425 East Main Street Sidney, Montana 59270 Ph (406) 433-5617 Fax (406) 433-5618 www.interstateeng.com Other offices in Minnesota, North Dakota and South Dakota	OASIS PETROLEUM NORTH AMERICA, LLC BATTERY LOCATION PLAT SECTION 30, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA Drawn By: J.J.S. Project No.: S15-09-003 Checked By: D.D.K. Date: JAN 2015	Revision No. / Date / By	Description

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"5300 30-29 CTB(A)"
SECTION 30, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 1/07/15 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 1/7/15
NORTH DAKOTA

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

2/5
SHEET NO.

INTERSTATE ENGINEERING
Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

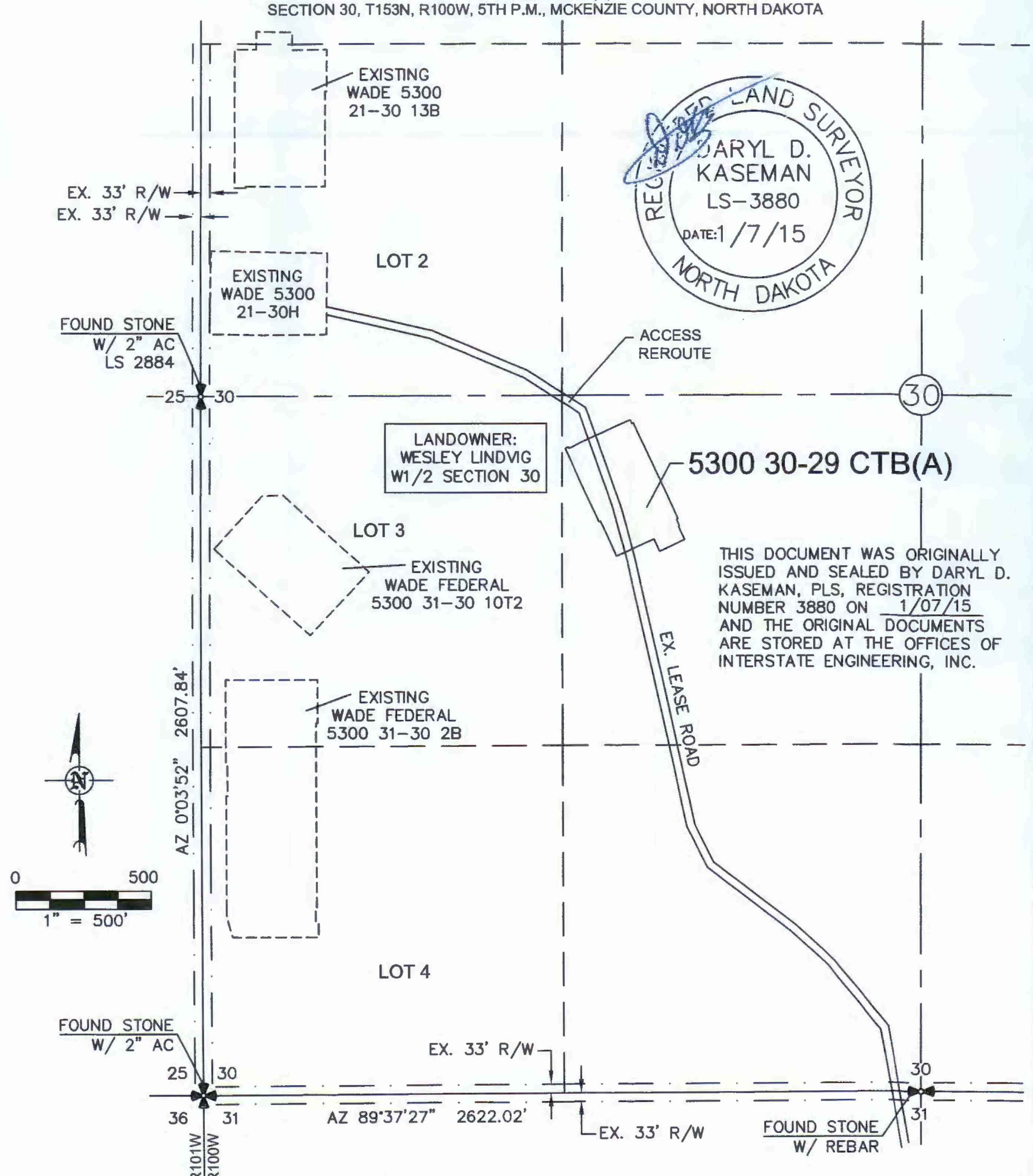
OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	J.J.S.	Project No.:	S15-09-003
Checked By:	D.D.K.	Date:	JAN 2015

Revision No.	Date	By	Description

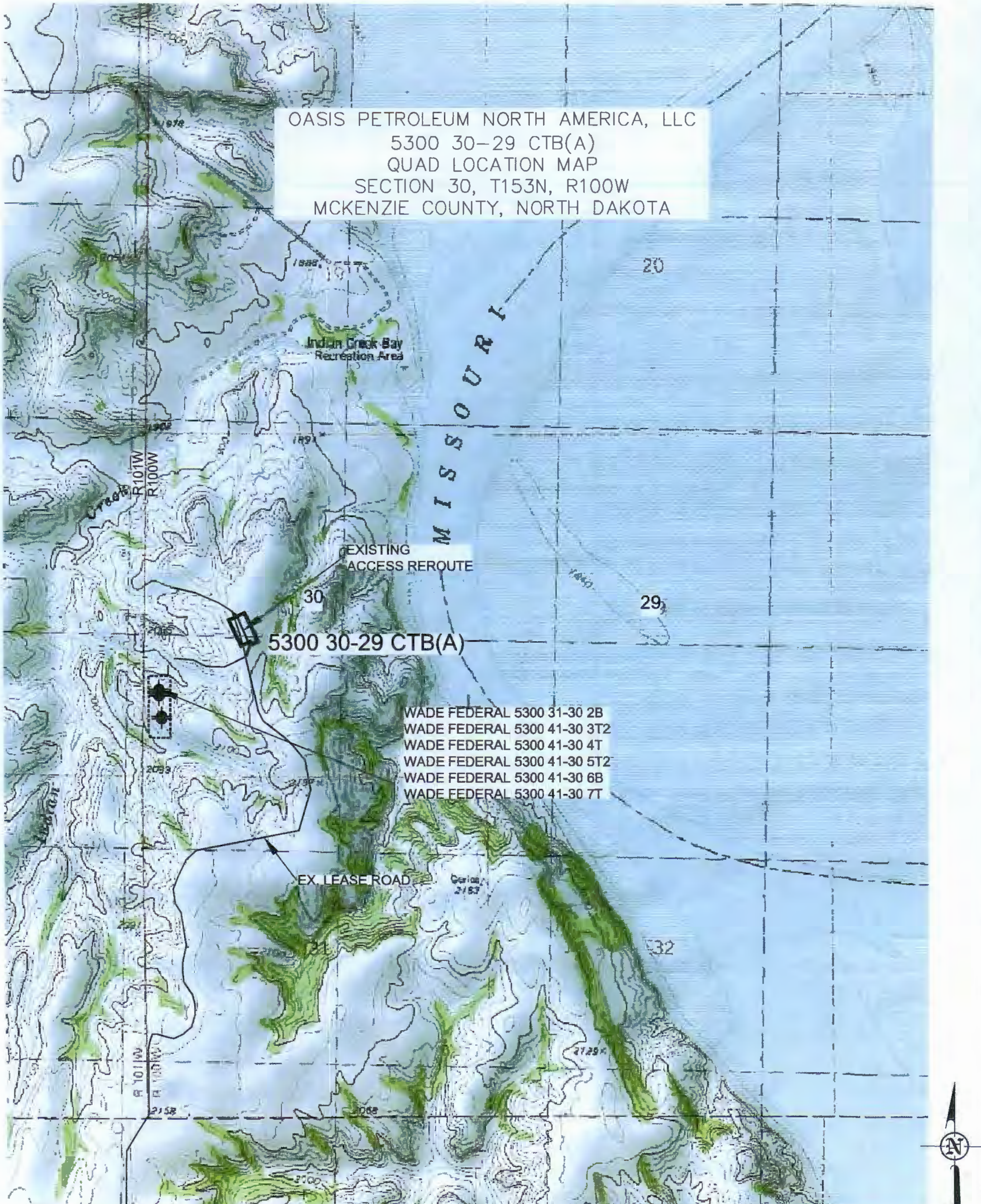
ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"5300 30-29 CTB(A)"
SECTION 30, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

3/5 SHEET NO.	INTERSTATE ENGINEERING Professionals you need, people you trust	Interstate Engineering, Inc. P.O. Box 648 425 East Main Street Sidney, Montana 59270 Ph (406) 433-5617 Fax (406) 433-5618 www.interstateeng.com Other offices in Minnesota, North Dakota and South Dakota	OASIS PETROLEUM NORTH AMERICA, LLC ACCESS APPROACH SECTION 30, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA Drawn By: J.J.S. Project No.: S15-09-003 Checked By: D.D.K. Date: JAN 2015	Revision No. / Date / By / Description



4/5

SHEET NO.

INTERSTATE ENGINEERING

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Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

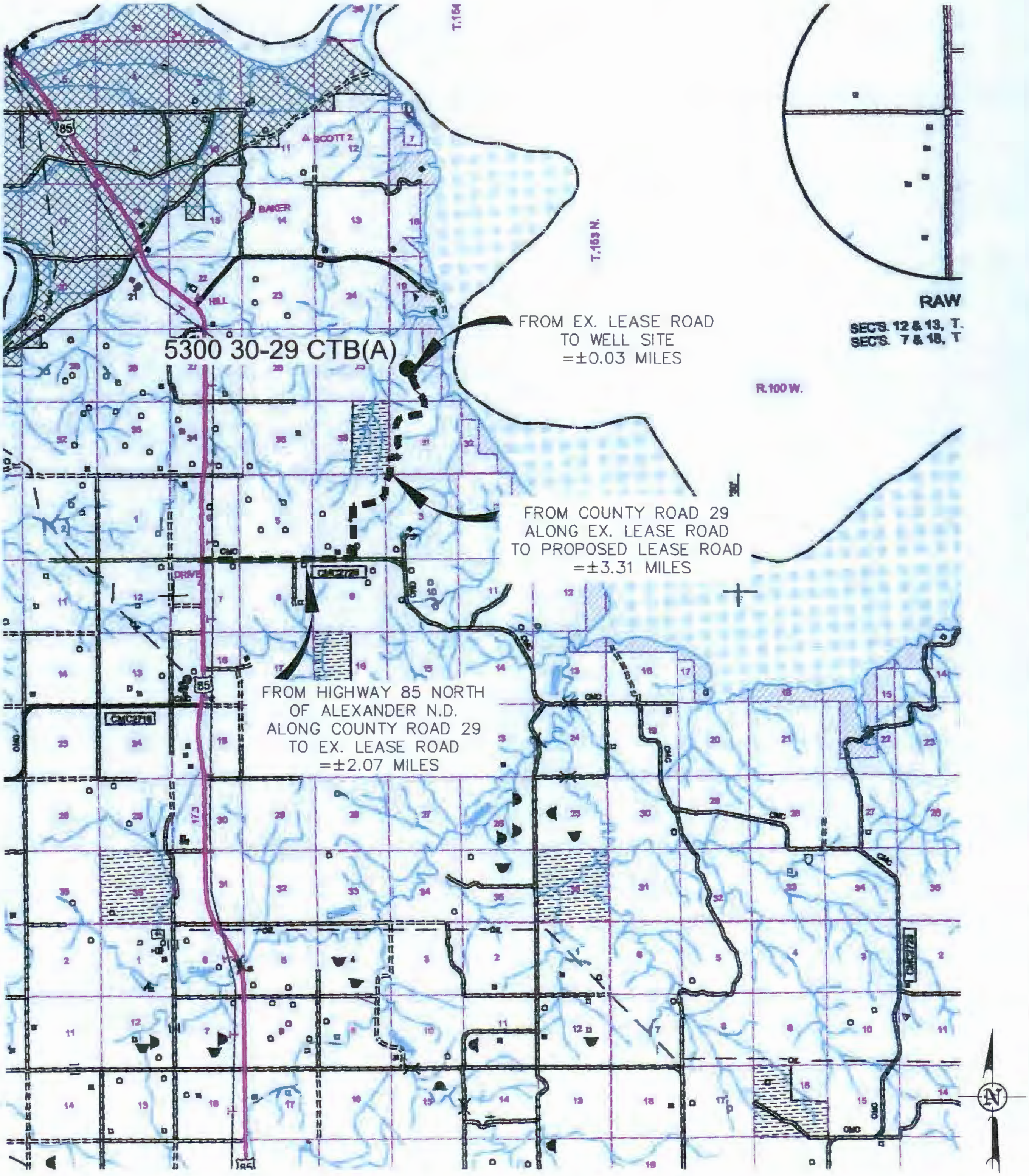
OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	J.J.S.	Project No.:	S15-09-003
Checked By:	D.D.K.	Date:	JAN 2015

Revision No.	Date	By	Description

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"5300 30-29 CTB(A)"
SECTION 30, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

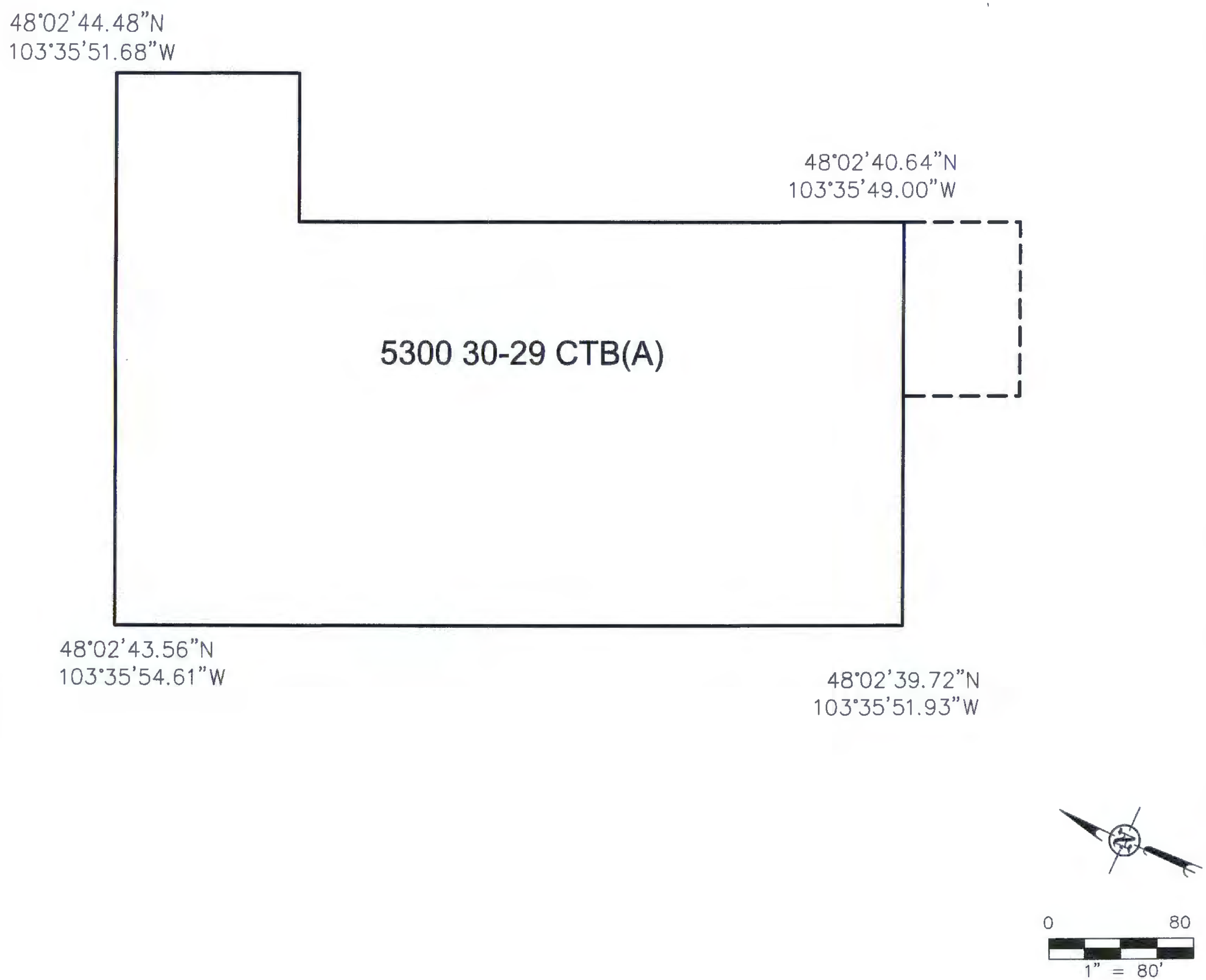


SCALE: 1" = 2 MILE

5/5 SHEET NO.	INTERSTATE ENGINEERING — Professionals you need, people you trust	Interstate Engineering, Inc. P.O. Box 648 425 East Main Street Sidney, Montana 59270 Ph (406) 433-5617 Fax (406) 433-5618 www.interstateeng.com Other offices in Minnesota, North Dakota and South Dakota	OASIS PETROLEUM NORTH AMERICA, LLC COUNTY ROAD MAP SECTION 30, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA Drawn By: J.J.S. Project No.: S15-09-003 Checked By: D.D.K. Date: JAN 2015

Revision No.	Date	By	Description

LAT/LONG PAD CORNERS



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. 28976

28557

RECEIVED AUG 2014 ND OIL & GAS DIVISION

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **September 14, 2014**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [ ] Other **offsite pit**
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [x] Reclamation

Well Name and Number					
Wade Federal 5300 21-30 12T					
Footages		Qtr-Qtr	Section	Township	Range
1640 F N L	270 F W L	SWNW	30	153 N	100 W
Field	Pool	County			
Baker	Bakken	McKenzie			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC respectfully requests to use an offsite pit for this well. The following wells will also use this pit:

Wade Federal 5300 31-30 2B – 28554
Wade Federal 5300 41-30 3T2 – 28555
Wade Federal 5300 41-30 4T – 28394
Wade Federal 5300 41-30 5T2 – 28556
Wade Federal 5300 41-30 6B – 28425 – TH
Wade Federal 5300 41-30 7T – 28557
Wade Federal 5300 41-30 8T2 – 28558 – TH
Wade Federal 5300 41-30 9B – 28744
Wade Federal 5300 21-30 13B – 28978
Wade Federal 5300 21-30 14T2 – 28977

Attached are the plats for the offsite pit location.

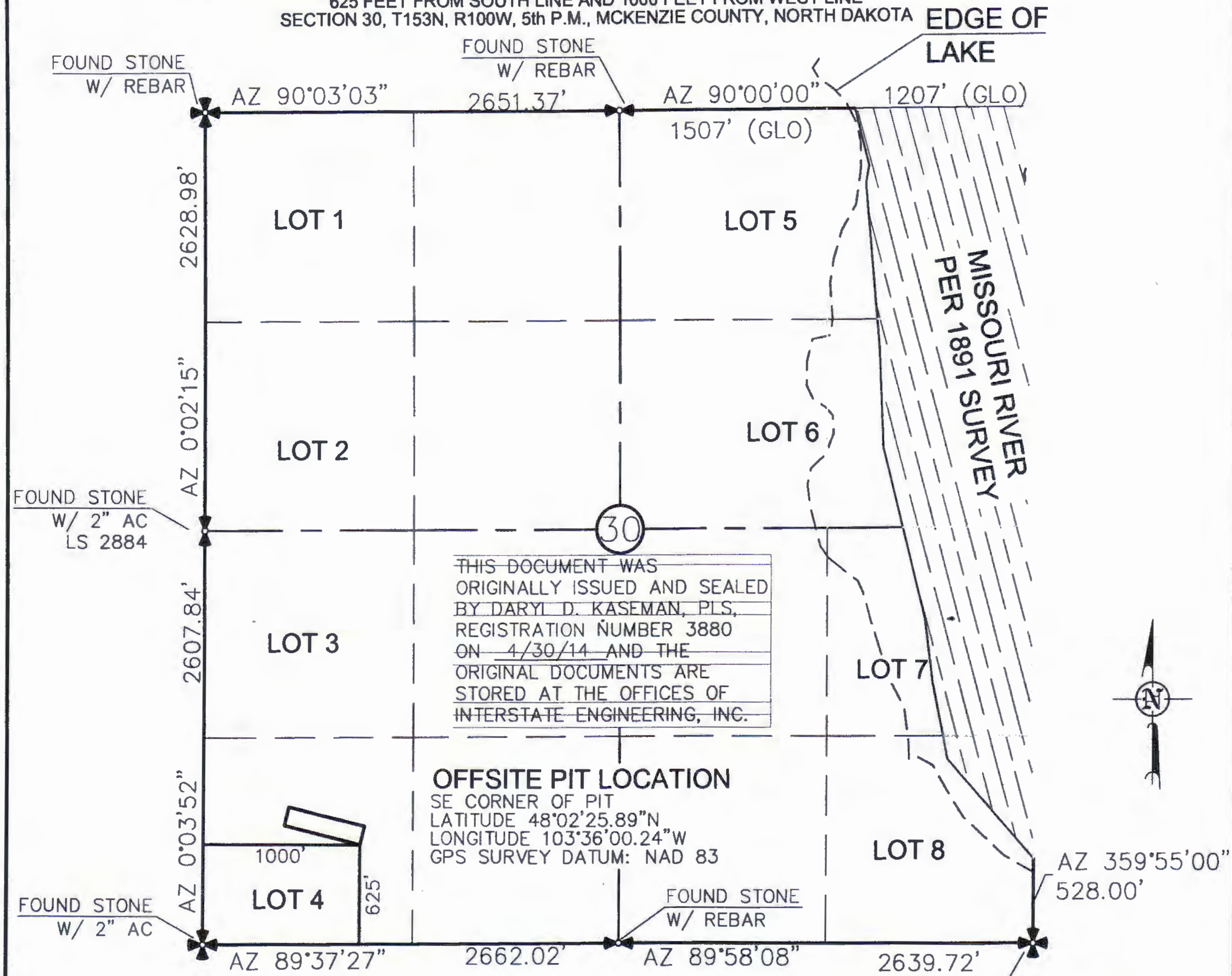
Company		Telephone Number
Oasis Petroleum North America LLC		281-404-9589
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	TX	77002
Signature	Printed Name	
Sonja Rolfes	Sonja Rolfes	
Title	Date	
Regulatory Analyst	August 20, 2014	
Email Address		
srolfes@oasispetroleum.com		

FOR STATE USE ONLY

[ ] Received	[x] Approved
Date	2-9-15
By	[signature]
Title	[illegible]

OFFSITE PIT LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



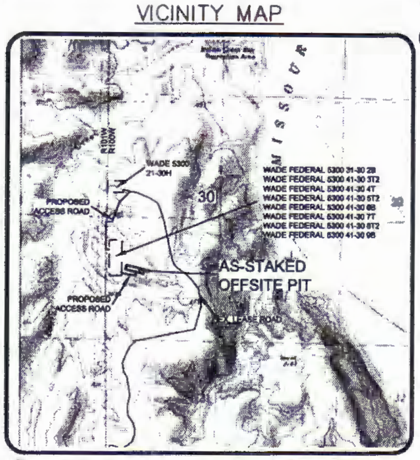
– MONUMENT – RECOVERED

– MONUMENT – NOT RECOVERED

0 1000

1" = 1000'

VICINITY MAP



STAKED ON 4/30/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

1/3
SHEET NO.

INTERSTATE ENGINEERING
Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

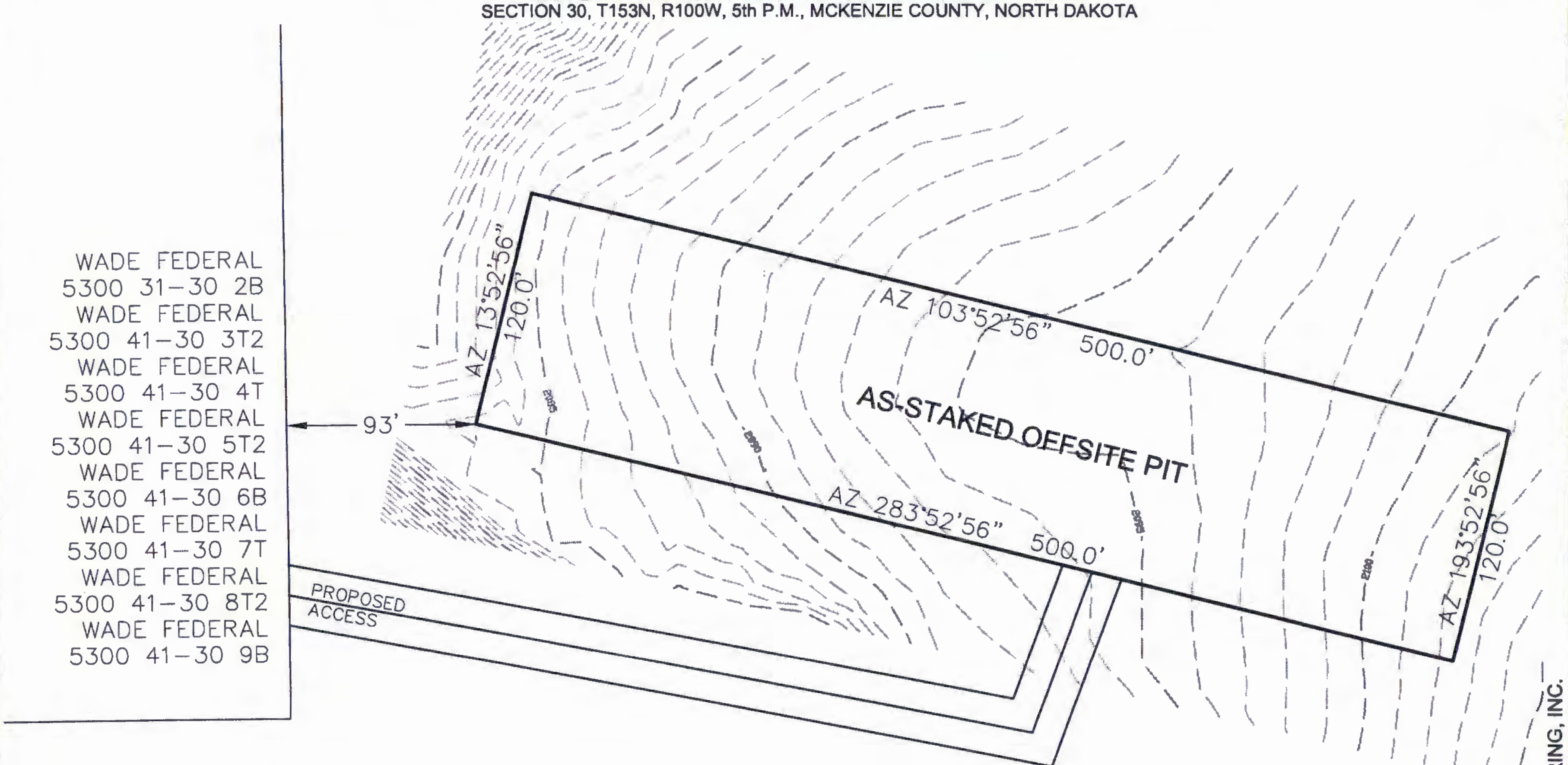
OASIS PETROLEUM NORTH AMERICA, LLC
OFFSITE PIT LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S13-09-381.09
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description

PAD LAYOUT

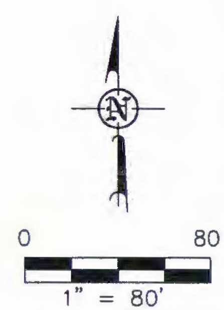
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2,
WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B,
WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 4/30/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.09
Checked By: D.D.K. Date: APRIL 2014

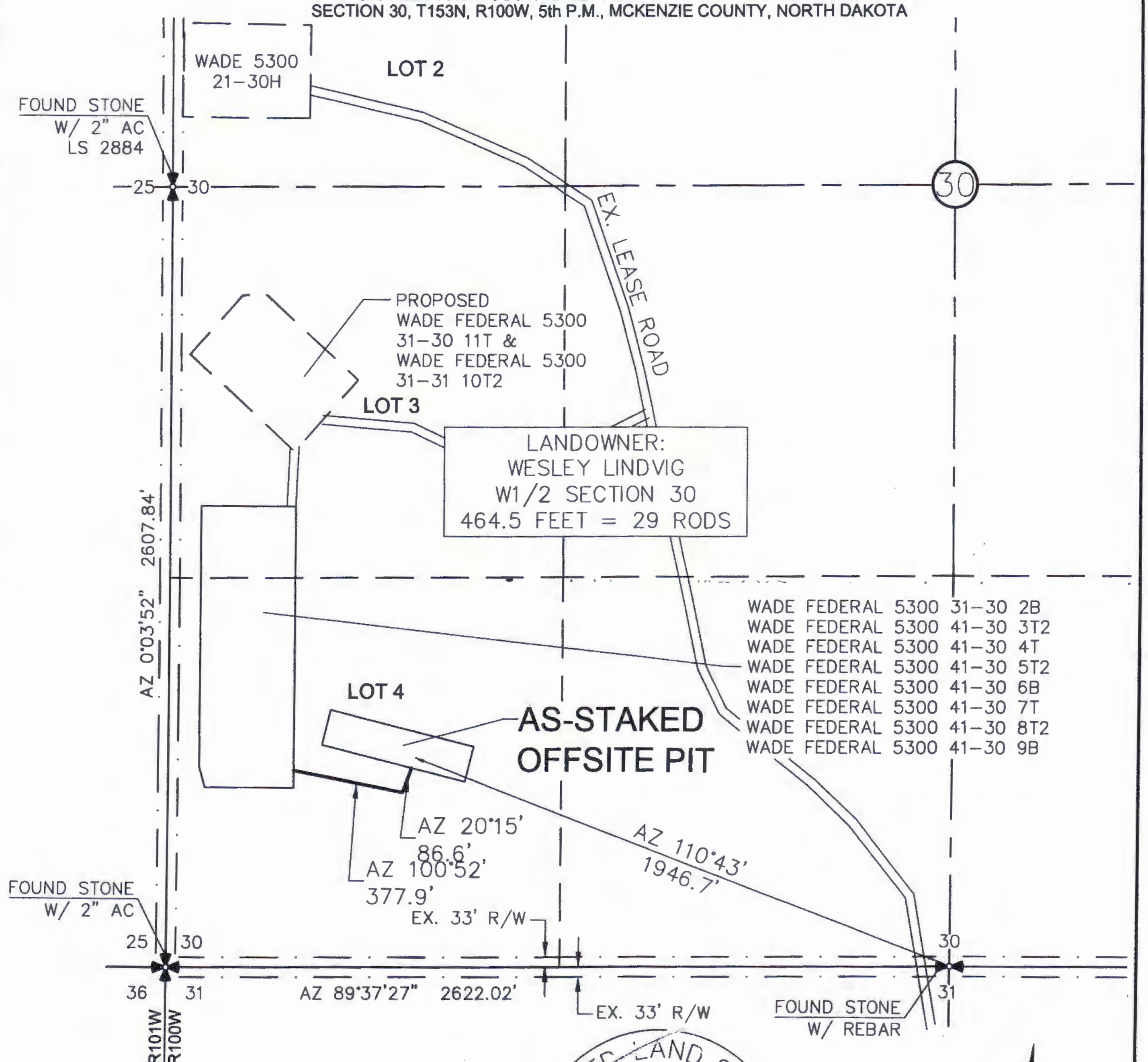
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

INTERSTATE ENGINEERING
Professionals you need, people you trust

SHEET NO. 2/3

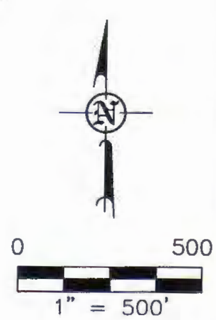
ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 4/30/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

SHEET NO.				
3/3	INTERSTATE ENGINEERING Professionals you need, people you trust	Interstate Engineering, Inc. P.O. Box 648 425 East Main Street Sidney, Montana 59270 Ph (406) 433-5617 Fax (406) 433-5618 www.interstateeng.com Other offices in Minnesota, North Dakota and South Dakota	OASIS PETROLEUM NORTH AMERICA, LLC ACCESS APPROACH SECTION 30, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA Drawn By: B.H.H. Project No.: S13-09-381.09 Checked By: D.D.K. Date: APRIL 2014	Revision No. / Date / By / Description

OFF-SITE PIT AGREEMENT

In consideration of the sum of [redacted] paid by Oasis Petroleum North America LLC ("Oasis") the undersigned surface owners, Wesley Lindvig and Barbara Lindvig, for themselves and their heirs, successors, administrators and assigns, hereby acknowledge the receipt and sufficiency of said payment in full and complete settlement for and as a release of all claim for loss, damage or injury to the hereafter described surface property arising out of the off-site cuttings pit, in which the cuttings from the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells will be buried, located on the approximately two (2.0) acre tract of land identified on the plat attached hereto as Exhibit "A" and which is situated on the following described real property located in McKenzie County, State of North Dakota, towit:

Township 153 North, Range 100 West, 5th P.M.
Section 30: Lots 3 & 4 a/k/a W½SW¼

The undersigned knows that Oasis Petroleum North America LLC is the operator and will be drilling the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells. The undersigned further states that they are fully aware that the cuttings generated from the drilling of the **Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2** wells will be buried in the pit on the above described location.

Dated this 19 day of May, 2014.

SURFACE OWNER(S)

Wesley Lindvig
Wesley Lindvig

Barbara J. Lindvig
Barbara Lindvig

BJL W.G.L.

* Location will be fenced after Construction.
* Pit will be reclaimed to Owners Satisfaction

BJL W.G.L.

ACKNOWLEDGMENT INDIVIDUAL

State of North Dakota)
)
County of McKenzie)

BE IT REMEMBERED, That on this 19 day of May, 2014 before me, a Notary Public, in and for said County and State, personally appeared Wesley Lindvig and Barbara Lindvig, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me to that they executed the same as their free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year last above written.

My Commission expires:

JAMES DEMORRETT
Notary Public
State of North Dakota
My Commission Expires Aug. 16, 2018

James DeMorrett

Notary Public

28394
28425
28554

SUNDRY NOTICE AND REPORTS ON WELLS - FORM

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No 28555

28556
28557
28558
28744

MAY 2014 RECEIVED ND OIL & GAS DIVISION

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [ ] Notice of Intent — Approximate Start Date:
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [ ] Other
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number					
Wade Federal 5300 41-30 4T + See Details					
Footages		Qtr-Qtr	Section	Township	Range
1263 F S L	240 F W L	SWSW	30	153 N	100 W
Field	Pool		County		
	Bakken		McKenzie		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests to use an offsite pit for the wells listed below. We are requesting to use an offiste pit because a pit wont fit on location with a rig anchor and the land adjacent is too rough. Attached are the plats.

Wade Federal 5300 31-30 2B - 28554
Wade Federal 5300 41-30 3T2 - 28555
Wade Federal 5300 41-30 4T - 28394
Wade Federal 5300 41-30 5T2 - 28556
Wade Federal 5300 41-30 6B - 28425
Wade Federal 5300 41-30 7T - 28557
Wade Federal 5300 41-30 8T2 - 28558
Wade Federal 5300 41-30 9B - 28744

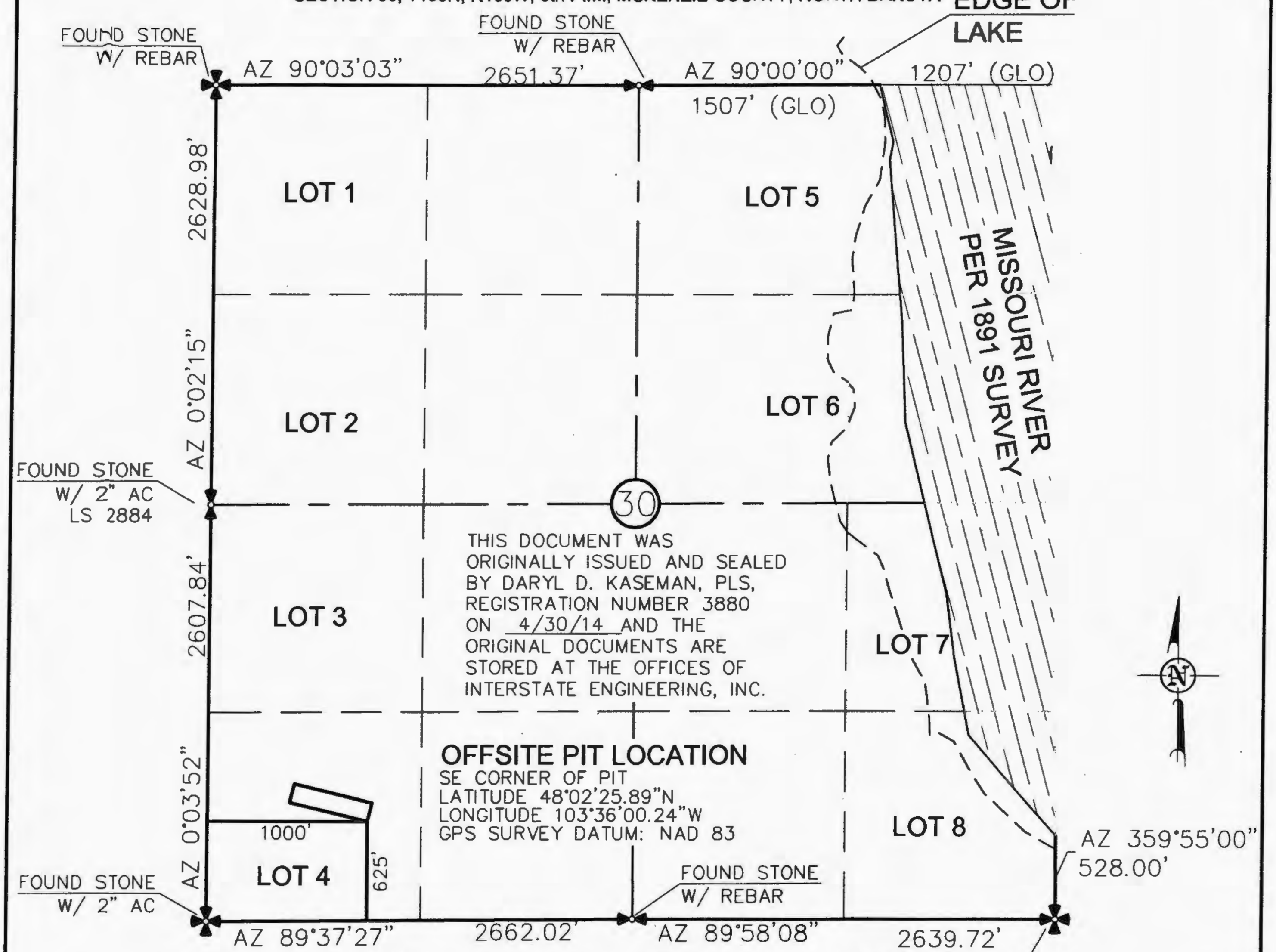
Company		Telephone Number
Oasis Petroleum North America LLC		281-404-9491
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	TX	77002
Signature	Printed Name	
Brandi Terry (signature)	Brandi Terry	
Title	Date	
Regulatory Specialist	May 12, 2014	
Email Address		
bterry@oasispetroleum.com		

FOR STATE USE ONLY

[X] Received	[ ] Approved
Date	12-23-14
By	[signature]
Title	[signature]

OFFSITE PIT LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



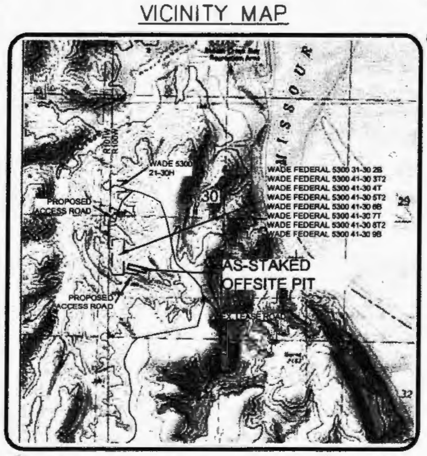
– MONUMENT – RECOVERED

– MONUMENT – NOT RECOVERED

0 1000

1" = 1000'

VICINITY MAP



STAKED ON 4/30/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

1/3
SHEET NO.

INTERSTATE ENGINEERING
Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
OFFSITE PIT LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S13-09-381.09
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2,
WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B,
WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
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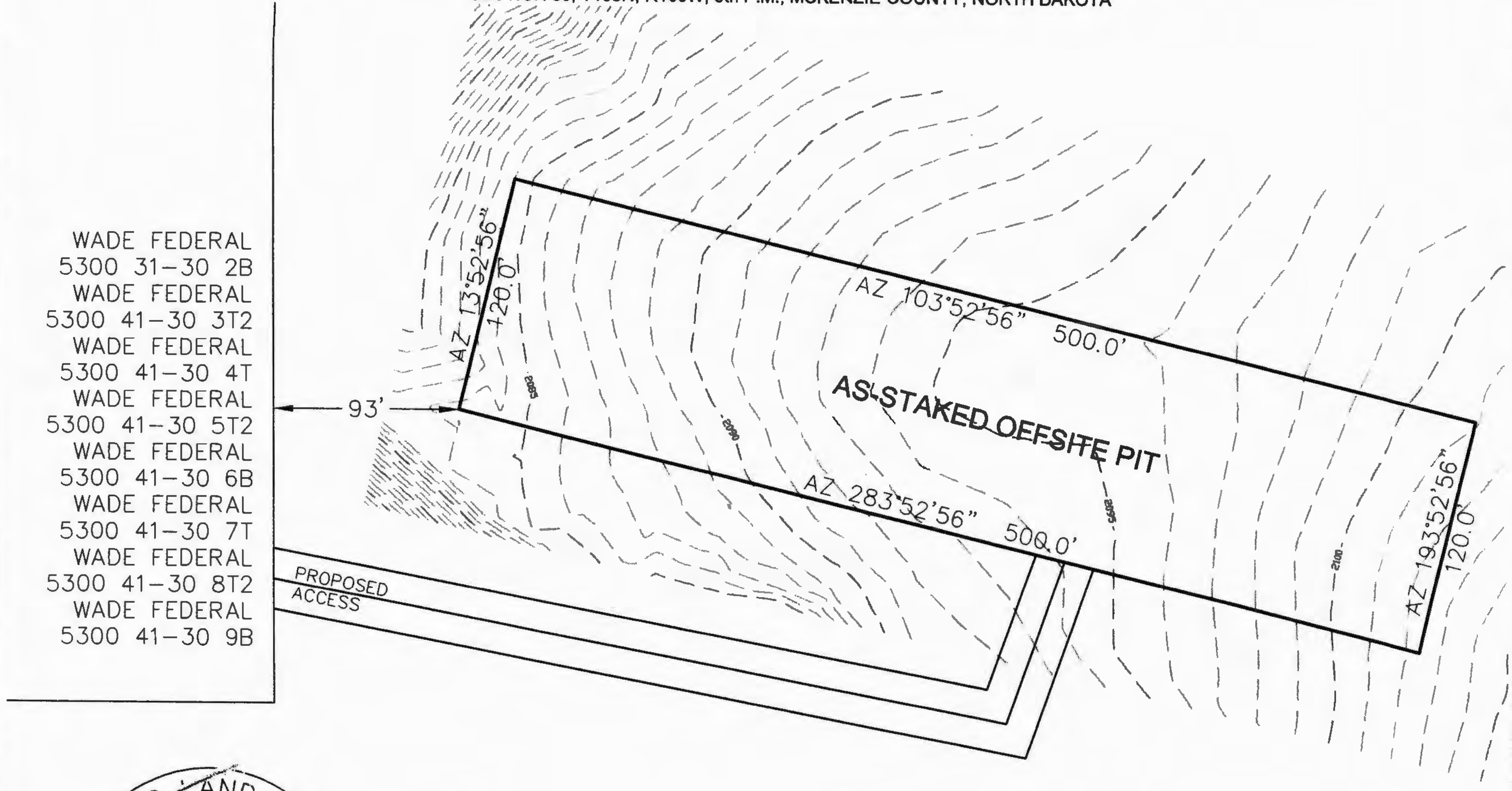

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 4/30/14
NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 4/30/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

0 80
1" = 80'

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.09
Checked By: D.D.K. Date: APRIL 2014

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
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Fax (406) 433-5618
www.interstateeng.com
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SHEET NO. 2/3

ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
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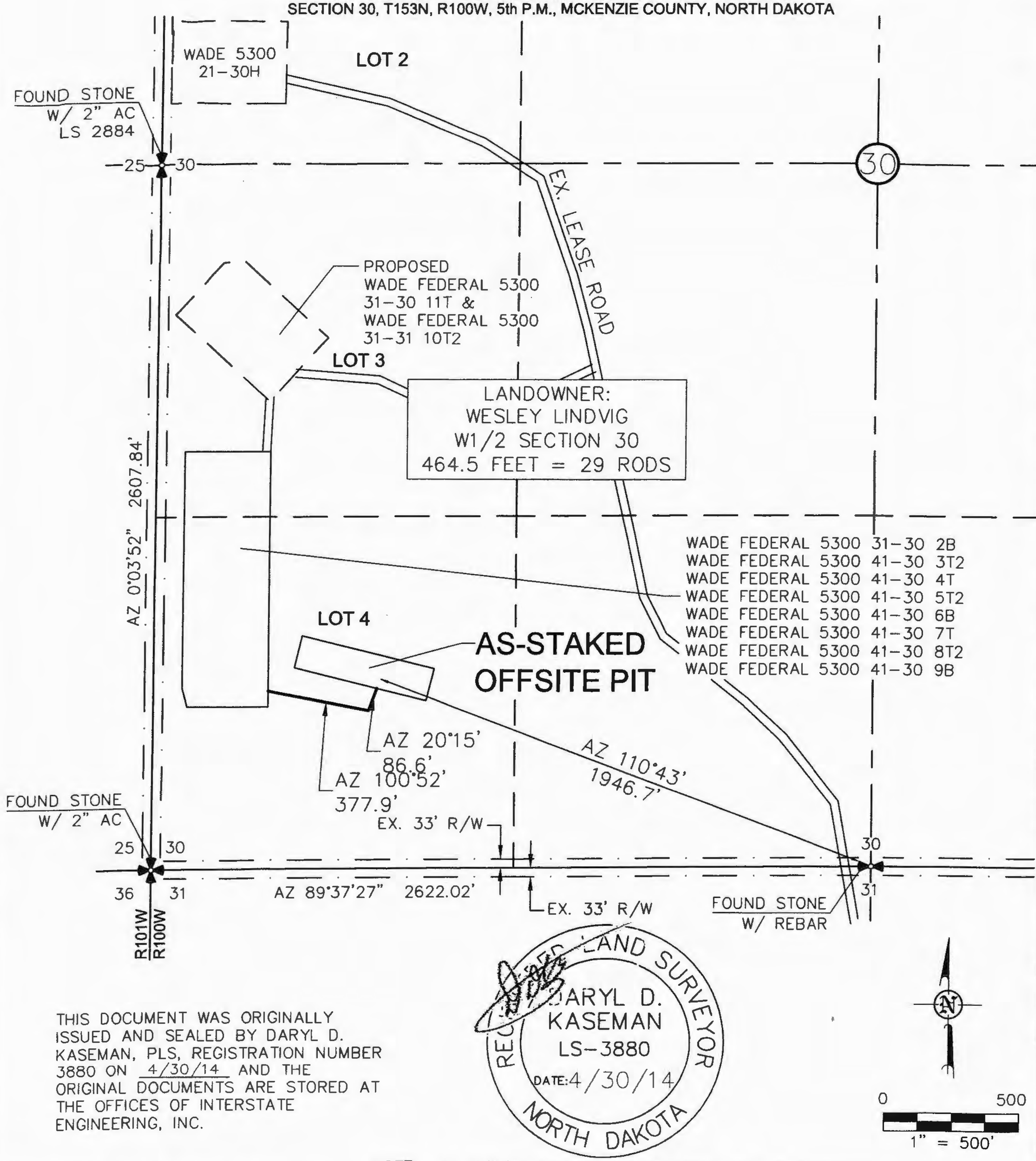

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3/3
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SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S13-09-381.09
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

DEC 2014
RECEIVED
ND OIL & GAS
DIVISION

Well File No. **28557**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **December 9, 2014**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [ ] Other **Waiver from tubing/packer requirement**
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number					
Wade Federal 5300 41-30 7T					
Footages		Qtr-Qtr	Section	Township	Range
877 F **S** L	**280** F **W** L	**LOT 4**	**30**	**153 N**	**100 W**
Field: **Baker**	Pool: **Bakken**		County: **McKenzie**		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will ise a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by montioring equipement;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company		Telephone Number
Oasis Petroleum North America LLC		**281-404-9436**
Address: **1001 Fannin, Suite 1500**		
City: **Houston**	State: **TX**	Zip Code: **77002**
Signature:	Printed Name: **Jennifer Swenson**	
Title: **Regulatory Assistant**	Date: **December 9, 2014**	
Email Address: **jswenson@oasispetroleum.com**		

FOR STATE USE ONLY

[ ] Received	[x] Approved
Date	December 11, 2014
By	
Title	PETROLEUM ENGINEER

A Schlumberger Company
9251 E 104th Ave.
Commerce City, CO 80640
(303) 439-5500

Directional Survey Certification Form

Wade Federal 5300 41-30 #7T

Oasis Petroleum	**Original Hole**	**5-Dec-2014**
Company	Well Name	Final report Date
	ND, McKenzie County (NAD 83 NZ) Oasis	**33-053-05998**
PathFinder Job Number	County / State	API Number
N 48° 2' 28.32000" **N 48.0412**	**W 103° 36' 10.82000"** **W 103.60300556**	**877 ft FSL & 500 ft FEL** **Sec 30 Twn 153 N Rng 100 W**
Surface Latitude	Surface Logitude	Surface Section - Township - Range
NAD83 ND State Plane, N Zone, Ft	**Patterson 488**	**KB 32ft @ 2077.00 ft / GL: 2045.00 ft MSL**
Datum & Coordinate System	Rig Contractor	Height Reference

Survey Depth **112.00** (Depth From) to **11423.00** (Depth To)

Measurement While Drilling
Type of Survey

Survey Depth **11423.00** (Depth From) to **11501.00** (Depth To)

Straight line projection to Bit/TD
Type of Survey

Site Supervisors

	Brittany Feddersen - MWD
Directional Driller 1	MWD Surveyor 1
	Dennis Lipcovski - MWD
Directional Driller 2	MWD Surveyor 2

The data submitted in this report conforms to the standards and procedures as set forth by Schlumberger. This report represents a true and correct directional wellbore survey based on original survey data obtained at the well site.

Matt VanderSchaaf — 12/5/2014 15:46

Matt VanderSchaaf — Date
PathFinder Well Planner II

Wade Federal 5300 41-30 #7T MWD 0' to 11501' Definitive Survey Geodetic Report

(Def Survey)

Report Date:	December 05, 2014 - 03:37 PM	**Survey / DLS Computation:**	Minimum Curvature / Lubinski
Client:	Oasis Petroleum	**Vertical Section Azimuth:**	91.850 ° (True North)
Field:	ND, McKenzie County (NAD 83 NZ) Oasis 2014	**Vertical Section Origin:**	0.000 ft, 0.000 ft
Structure / Slot:	Oasis 30-153N-100W (Wade Federal 5300 41-30 Pad) - Patterson 488 / Wade Federal 5300 41-30 #7T	**TVD Reference Datum:**	KB 32ft
Well:	Wade Federal 5300 41-30 #7T	**TVD Reference Elevation:**	2077.000 ft above MSL
Borehole:	Original Hole	**Seabed / Ground Elevation:**	2045.000 ft above MSL
UWI / API#:	Unknown / Unknown	**Magnetic Declination:**	8.395 °
Survey Name:	Wade Federal 5300 41-30 #7T OH MWD 0' to 11501' Definitive	**Total Gravity Field Strength:**	1000.0448mgn (9.80665 Based)
Survey Date:	December 05, 2014	**Gravity Model:**	GARM
Tort / AHD / DDI / ERD Ratio:	144.137 ° / 1078.695 ft / 5.220 / 0.100	**Total Magnetic Field Strength:**	56237.474 nT
Coordinate Reference System:	NAD83 North Dakota State Plane, Northern Zone, Feet	**Magnetic Dip Angle:**	72.904 °
Location Lat / Long:	N 48° 2' 28.32000", W 103° 36' 10.82000"	**Declination Date:**	December 05, 2014
Location Grid N/E Y/X:	N 395087.826 ft, E 1209642.216 ft	**Magnetic Declination Model:**	BGGM 2014
CRS Grid Convergence Angle:	-2.3091 °	**North Reference:**	True North
Grid Scale Factor:	0.99993613	**Grid Convergence Used:**	0.0000 °
Version / Patch:	2.8.572.0	**Total Corr Mag North->True North:**	8.3953 °
		Local Coord Referenced To:	Well Head

Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
Surface	0.00	0.00	0.00	0.00	0.00	0.00	0.00	N/A	395087.83	1209642.22	N 48 2 28.32	W 103 36 10.82
Begin MWD Survey	112.00	0.79	207.45	112.00	-0.33	-0.69	-0.36	0.71	395087.16	1209641.83	N 48 2 28.31	W 103 36 10.83
	203.00	0.70	234.77	202.99	-1.05	-1.56	-1.10	0.40	395086.31	1209641.05	N 48 2 28.30	W 103 36 10.84
	294.00	0.88	239.07	293.98	-2.08	-2.24	-2.15	0.21	395085.67	1209639.97	N 48 2 28.30	W 103 36 10.85
	386.00	0.53	258.70	385.97	-3.09	-2.69	-3.18	0.46	395085.27	1209638.93	N 48 2 28.29	W 103 36 10.87
	476.00	0.62	246.59	475.97	-3.93	-2.96	-4.03	0.17	395085.03	1209638.07	N 48 2 28.29	W 103 36 10.88
	566.00	0.62	284.48	565.96	-4.85	-3.04	-4.95	0.45	395084.99	1209637.15	N 48 2 28.29	W 103 36 10.89
	657.00	0.26	42.87	656.96	-5.19	-2.76	-5.29	0.85	395085.28	1209636.82	N 48 2 28.29	W 103 36 10.90
	747.00	0.62	338.81	746.96	-5.25	-2.16	-5.32	0.62	395085.89	1209636.81	N 48 2 28.30	W 103 36 10.90
	838.00	0.70	15.23	837.95	-5.31	-1.16	-5.35	0.46	395086.88	1209636.82	N 48 2 28.31	W 103 36 10.90
	928.00	0.79	359.66	927.95	-5.21	-0.01	-5.21	0.25	395088.03	1209637.01	N 48 2 28.32	W 103 36 10.90
	1019.00	0.97	22.26	1018.94	-4.97	1.33	-4.93	0.43	395089.35	1209637.35	N 48 2 28.33	W 103 36 10.89
	1109.00	0.53	129.04	1108.93	-4.37	1.77	-4.31	1.37	395089.77	1209637.98	N 48 2 28.34	W 103 36 10.88
	1200.00	1.14	140.54	1199.92	-3.44	0.81	-3.41	0.69	395088.77	1209638.84	N 48 2 28.33	W 103 36 10.87
	1290.00	0.44	255.36	1289.92	-3.18	0.03	-3.18	1.54	395087.98	1209639.04	N 48 2 28.32	W 103 36 10.87
	1385.00	0.70	257.20	1384.91	-4.09	-0.19	-4.10	0.27	395087.80	1209638.12	N 48 2 28.32	W 103 36 10.88
	1480.00	0.97	275.41	1479.90	-5.45	-0.24	-5.46	0.40	395087.80	1209636.75	N 48 2 28.32	W 103 36 10.90
	1575.00	0.79	105.82	1574.90	-5.62	-0.35	-5.63	1.85	395087.71	1209636.57	N 48 2 28.32	W 103 36 10.90
	1670.00	1.32	116.09	1669.88	-3.99	-1.01	-4.02	0.59	395086.98	1209638.16	N 48 2 28.31	W 103 36 10.88
	1765.00	1.06	115.50	1764.86	-2.18	-1.87	-2.24	0.27	395086.05	1209639.90	N 48 2 28.30	W 103 36 10.85
	1860.00	0.62	62.53	1859.85	-0.93	-2.01	-0.99	0.89	395085.86	1209641.14	N 48 2 28.30	W 103 36 10.83
	1956.00	0.53	10.99	1955.85	-0.41	-1.33	-0.45	0.53	395086.51	1209641.71	N 48 2 28.31	W 103 36 10.83
	2051.00	0.70	1.86	2050.84	-0.34	-0.32	-0.35	0.21	395087.52	1209641.86	N 48 2 28.32	W 103 36 10.83
9 5/8" Casing Point	*2100.00*	*0.82*	*54.14*	*2099.84*	*-0.06*	*0.18*	*-0.05*	*1.39*	*395088.01*	*1209642.17*	*N 48 2 28.32*	*W 103 36 10.82*
	2116.00	0.97	64.88	2115.84	0.15	0.31	0.16	1.39	395088.13	1209642.39	N 48 2 28.32	W 103 36 10.82
	2179.00	1.06	333.00	2178.83	0.35	1.06	0.38	2.32	395088.87	1209642.64	N 48 2 28.33	W 103 36 10.81
	2274.00	1.23	326.04	2273.81	-0.67	2.68	-0.59	0.23	395090.53	1209641.74	N 48 2 28.35	W 103 36 10.83
	2368.00	0.44	17.14	2367.80	-1.17	3.87	-1.04	1.08	395091.73	1209641.33	N 48 2 28.36	W 103 36 10.84
	2463.00	0.62	33.42	2462.80	-0.80	4.64	-0.65	0.25	395092.49	1209641.75	N 48 2 28.37	W 103 36 10.83
	2557.00	0.62	30.16	2556.79	-0.29	5.51	-0.12	0.04	395093.33	1209642.32	N 48 2 28.37	W 103 36 10.82
	2652.00	0.62	151.47	2651.79	0.21	5.50	0.39	1.14	395093.31	1209642.82	N 48 2 28.37	W 103 36 10.81
	2747.00	0.88	159.74	2746.78	0.74	4.36	0.88	0.30	395092.15	1209643.28	N 48 2 28.36	W 103 36 10.81
	2843.00	1.06	173.61	2842.77	1.15	2.79	1.24	0.31	395090.56	1209643.57	N 48 2 28.35	W 103 36 10.80
	2938.00	0.97	179.58	2937.75	1.31	1.11	1.34	0.15	395088.88	1209643.60	N 48 2 28.33	W 103 36 10.80
	3033.00	0.97	208.96	3032.74	0.97	-0.39	0.96	0.52	395087.39	1209643.16	N 48 2 28.32	W 103 36 10.81
	3128.00	0.79	208.77	3127.73	0.31	-1.67	0.25	0.19	395086.15	1209642.40	N 48 2 28.30	W 103 36 10.82
	3223.00	1.06	200.08	3222.71	-0.26	-3.07	-0.36	0.32	395084.77	1209641.73	N 48 2 28.29	W 103 36 10.83
	3318.00	1.41	138.77	3317.70	0.26	-4.78	0.11	1.36	395083.05	1209642.13	N 48 2 28.27	W 103 36 10.82
	3413.00	1.23	135.93	3412.67	1.79	-6.39	1.59	0.20	395081.38	1209643.54	N 48 2 28.26	W 103 36 10.80
	3508.00	1.32	135.28	3507.65	3.32	-7.90	3.07	0.10	395079.81	1209644.96	N 48 2 28.24	W 103 36 10.77
	3603.00	1.06	114.63	3602.63	4.92	-9.04	4.63	0.52	395078.61	1209646.48	N 48 2 28.23	W 103 36 10.75
	3698.00	0.88	104.92	3697.61	6.44	-9.60	6.14	0.26	395077.99	1209647.96	N 48 2 28.23	W 103 36 10.73
	3793.00	0.97	108.74	3792.60	7.92	-10.04	7.60	0.11	395077.49	1209649.41	N 48 2 28.22	W 103 36 10.71
	3888.00	0.70	112.43	3887.59	9.24	-10.52	8.90	0.29	395076.96	1209650.69	N 48 2 28.22	W 103 36 10.69
	3983.00	0.26	103.07	3982.59	9.99	-10.79	9.65	0.47	395076.66	1209651.42	N 48 2 28.21	W 103 36 10.68
	4078.00	0.35	124.52	4077.59	10.45	-11.00	10.10	0.15	395076.42	1209651.86	N 48 2 28.21	W 103 36 10.67
	4173.00	0.35	6.14	4172.58	10.71	-10.88	10.37	0.63	395076.54	1209652.14	N 48 2 28.21	W 103 36 10.67
	4268.00	0.35	17.10	4267.58	10.81	-10.32	10.48	0.07	395077.10	1209652.28	N 48 2 28.22	W 103 36 10.67
	4363.00	0.35	31.54	4362.58	11.03	-9.79	10.72	0.09	395077.61	1209652.53	N 48 2 28.22	W 103 36 10.66
	4458.00	0.44	31.65	4457.58	11.36	-9.23	11.06	0.09	395078.16	1209652.90	N 48 2 28.23	W 103 36 10.66
	4553.00	0.35	46.30	4552.58	11.74	-8.72	11.47	0.14	395078.65	1209653.32	N 48 2 28.23	W 103 36 10.65
	4648.00	0.26	77.31	4647.58	12.15	-8.47	11.89	0.19	395078.88	1209653.75	N 48 2 28.24	W 103 36 10.65
	4743.00	0.44	119.27	4742.57	12.69	-8.60	12.41	0.32	395078.73	1209654.27	N 48 2 28.24	W 103 36 10.64
	4838.00	0.44	66.55	4837.57	13.34	-8.64	13.07	0.41	395078.67	1209654.92	N 48 2 28.23	W 103 36 10.63
	4933.00	0.70	52.95	4932.57	14.12	-8.14	13.86	0.31	395079.13	1209655.74	N 48 2 28.24	W 103 36 10.62
	5028.00	0.62	63.88	5027.56	15.03	-7.57	14.79	0.16	395079.67	1209656.69	N 48 2 28.25	W 103 36 10.60
	5123.00	0.70	87.05	5122.55	16.06	-7.31	15.83	0.29	395079.88	1209657.74	N 48 2 28.25	W 103 36 10.59
	5219.00	0.53	110.55	5218.55	17.06	-7.44	16.83	0.31	395079.72	1209658.73	N 48 2 28.25	W 103 36 10.57
	5313.00	0.62	75.27	5312.54	17.96	-7.46	17.73	0.38	395079.66	1209659.63	N 48 2 28.25	W 103 36 10.56
	5409.00	0.62	65.24	5408.54	18.92	-7.11	18.70	0.11	395079.97	1209660.62	N 48 2 28.25	W 103 36 10.54
	5503.00	0.53	69.03	5502.53	19.78	-6.74	19.57	0.10	395080.30	1209661.50	N 48 2 28.25	W 103 36 10.53
	5598.00	0.70	83.29	5597.53	20.76	-6.52	20.56	0.24	395080.49	1209662.49	N 48 2 28.26	W 103 36 10.52
	5693.00	0.70	82.46	5692.52	21.90	-6.37	21.71	0.01	395080.58	1209663.65	N 48 2 28.26	W 103 36 10.50
	5788.00	0.62	74.88	5787.52	22.97	-6.16	22.78	0.12	395080.75	1209664.73	N 48 2 28.26	W 103 36 10.48
	5883.00	0.44	84.56	5882.51	23.82	-5.99	23.64	0.21	395080.89	1209665.60	N 48 2 28.26	W 103 36 10.47
	5978.00	0.35	77.81	5977.51	24.47	-5.90	24.29	0.11	395080.95	1209666.25	N 48 2 28.26	W 103 36 10.46
	6041.00	0.35	72.62	6040.51	24.83	-5.80	24.66	0.05	395081.04	1209666.62	N 48 2 28.26	W 103 36 10.46
	6105.00	0.26	101.75	6104.51	25.16	-5.77	24.99	0.28	395081.05	1209666.95	N 48 2 28.26	W 103 36 10.45
	6168.00	0.09	88.18	6167.51	25.35	-5.80	25.18	0.28	395081.02	1209667.14	N 48 2 28.26	W 103 36 10.45
	6263.00	0.35	70.91	6262.51	25.70	-5.70	25.53	0.28	395081.10	1209667.49	N 48 2 28.26	W 103 36 10.44
	6359.00	0.35	102.62	6358.50	26.26	-5.67	26.09	0.20	395081.11	1209668.05	N 48 2 28.26	W 103 36 10.44
	6454.00	0.35	111.41	6453.50	26.82	-5.84	26.64	0.06	395080.92	1209668.60	N 48 2 28.26	W 103 36 10.43
	6549.00	0.53	50.41	6548.50	27.42	-5.66	27.25	0.50	395081.07	1209669.22	N 48 2 28.26	W 103 36 10.42
	6644.00	0.53	68.09	6643.50	28.15	-5.22	28.00	0.17	395081.48	1209669.98	N 48 2 28.27	W 103 36 10.41
	6739.00	0.35	107.63	6738.49	28.83	-5.14	28.68	0.36	395081.53	1209670.67	N 48 2 28.27	W 103 36 10.40
	6834.00	0.44	121.00	6833.49	29.43	-5.42	29.27	0.13	395081.23	1209671.24	N 48 2 28.27	W 103 36 10.39

Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
	6929.00	1.14	58.32	6928.48	30.54	-5.11	30.39	1.07	395081.49	1209672.37	N 48 2 28.27	W 103 36 10.37
	7024.00	1.23	64.30	7023.46	32.23	-4.17	32.11	0.16	395082.36	1209674.13	N 48 2 28.28	W 103 36 10.35
	7119.00	0.97	77.69	7118.44	33.91	-3.56	33.82	0.38	395082.91	1209675.86	N 48 2 28.28	W 103 36 10.32
	7214.00	1.14	90.32	7213.43	35.64	-3.39	35.55	0.30	395083.00	1209677.59	N 48 2 28.29	W 103 36 10.30
	7309.00	1.06	88.69	7308.41	37.46	-3.38	37.37	0.09	395082.94	1209679.42	N 48 2 28.29	W 103 36 10.27
	7404.00	0.70	91.57	7403.40	38.92	-3.37	38.83	0.38	395082.89	1209680.87	N 48 2 28.29	W 103 36 10.25
	7500.00	0.79	89.38	7499.39	40.16	-3.38	40.08	0.10	395082.83	1209682.12	N 48 2 28.29	W 103 36 10.23
	7595.00	0.70	97.27	7594.38	41.40	-3.45	41.31	0.14	395082.72	1209683.35	N 48 2 28.29	W 103 36 10.21
	7690.00	0.62	139.61	7689.38	42.32	-3.92	42.22	0.51	395082.21	1209684.24	N 48 2 28.28	W 103 36 10.20
	7785.00	0.62	148.12	7784.37	42.95	-4.74	42.82	0.10	395081.36	1209684.81	N 48 2 28.27	W 103 36 10.19
	7880.00	0.53	140.35	7879.37	43.53	-5.52	43.37	0.13	395080.57	1209685.33	N 48 2 28.27	W 103 36 10.18
	7975.00	0.18	158.50	7974.37	43.88	-5.99	43.71	0.38	395080.08	1209685.64	N 48 2 28.26	W 103 36 10.18
	8070.00	0.26	182.81	8069.37	43.93	-6.35	43.75	0.13	395079.72	1209685.67	N 48 2 28.26	W 103 36 10.18
	8164.00	0.09	4.08	8163.36	43.93	-6.49	43.75	0.37	395079.58	1209685.66	N 48 2 28.26	W 103 36 10.18
	8259.00	0.18	22.11	8258.36	43.99	-6.28	43.81	0.10	395079.79	1209685.73	N 48 2 28.26	W 103 36 10.18
	8354.00	0.79	53.43	8353.36	44.55	-5.75	44.39	0.68	395080.30	1209686.34	N 48 2 28.26	W 103 36 10.17
	8449.00	0.97	43.90	8448.35	45.60	-4.78	45.47	0.24	395081.22	1209687.46	N 48 2 28.27	W 103 36 10.15
	8544.00	1.14	59.21	8543.33	46.94	-3.71	46.84	0.34	395082.23	1209688.87	N 48 2 28.28	W 103 36 10.13
	8640.00	1.06	65.29	8639.32	48.54	-2.85	48.47	0.15	395083.02	1209690.53	N 48 2 28.29	W 103 36 10.11
	8735.00	1.06	59.68	8734.30	50.07	-2.04	50.03	0.11	395083.77	1209692.12	N 48 2 28.30	W 103 36 10.08
	8830.00	0.97	65.04	8829.29	51.53	-1.26	51.51	0.14	395084.49	1209693.63	N 48 2 28.31	W 103 36 10.06
	8924.00	0.88	64.06	8923.27	52.88	-0.61	52.88	0.10	395085.09	1209695.03	N 48 2 28.31	W 103 36 10.04
	9019.00	0.79	55.32	9018.26	54.05	0.08	54.08	0.16	395085.73	1209696.25	N 48 2 28.32	W 103 36 10.02
	9114.00	0.70	53.95	9113.25	55.03	0.80	55.09	0.10	395086.40	1209697.29	N 48 2 28.33	W 103 36 10.01
	9209.00	0.62	52.76	9208.25	55.89	1.45	55.97	0.09	395087.02	1209698.19	N 48 2 28.33	W 103 36 10.00
	9304.00	0.62	140.15	9303.24	56.63	1.37	56.70	0.90	395086.91	1209698.93	N 48 2 28.33	W 103 36 9.99
	9399.00	0.70	153.57	9398.24	57.25	0.45	57.29	0.18	395085.97	1209699.48	N 48 2 28.32	W 103 36 9.98
	9494.00	0.79	149.01	9493.23	57.88	-0.63	57.89	0.11	395084.87	1209700.03	N 48 2 28.31	W 103 36 9.97
	9589.00	0.79	149.18	9588.22	58.59	-1.75	58.56	0.00	395083.72	1209700.65	N 48 2 28.30	W 103 36 9.96
	9684.00	0.44	166.64	9683.22	59.04	-2.67	58.98	0.41	395082.78	1209701.04	N 48 2 28.29	W 103 36 9.95
	9779.00	0.62	196.33	9778.21	59.00	-3.52	58.92	0.34	395081.94	1209700.94	N 48 2 28.29	W 103 36 9.95
	9874.00	0.53	225.02	9873.21	58.57	-4.32	58.46	0.31	395081.15	1209700.45	N 48 2 28.28	W 103 36 9.96
	9969.00	0.26	61.95	9968.21	58.46	-4.53	58.34	0.82	395080.95	1209700.33	N 48 2 28.28	W 103 36 9.96
	10063.00	0.35	42.55	10062.20	58.83	-4.22	58.73	0.14	395081.24	1209700.72	N 48 2 28.28	W 103 36 9.96
	10157.00	0.35	31.35	10156.20	59.16	-3.76	59.07	0.07	395081.69	1209701.08	N 48 2 28.28	W 103 36 9.95
	10202.00	0.44	25.98	10201.20	59.30	-3.49	59.22	0.22	395081.95	1209701.24	N 48 2 28.29	W 103 36 9.95
	10219.00	0.35	8.06	10218.20	59.33	-3.38	59.25	0.89	395082.06	1209701.28	N 48 2 28.29	W 103 36 9.95
	10251.00	0.62	66.09	10250.20	59.50	-3.21	59.42	1.65	395082.22	1209701.46	N 48 2 28.29	W 103 36 9.95
	10283.00	4.48	106.04	10282.16	60.87	-3.49	60.78	12.58	395081.89	1209702.81	N 48 2 28.29	W 103 36 9.93
	10314.00	9.41	106.74	10312.93	64.49	-4.55	64.38	15.91	395080.68	1209706.35	N 48 2 28.28	W 103 36 9.87
	10346.00	13.37	107.80	10344.29	70.58	-6.44	70.41	12.39	395078.56	1209712.30	N 48 2 28.26	W 103 36 9.78
	10378.00	14.95	107.39	10375.32	78.11	-8.80	77.87	4.95	395075.89	1209719.66	N 48 2 28.23	W 103 36 9.67
	10409.00	17.24	112.10	10405.10	86.28	-11.73	85.94	8.50	395072.65	1209727.61	N 48 2 28.20	W 103 36 9.56
	10441.00	21.90	113.99	10435.24	96.26	-15.94	95.79	14.69	395068.04	1209737.28	N 48 2 28.16	W 103 36 9.41
	10473.00	27.17	115.37	10464.35	108.49	-21.50	107.86	16.56	395062.00	1209749.11	N 48 2 28.11	W 103 36 9.23
	10504.00	30.34	115.28	10491.52	122.17	-27.88	121.34	10.23	395055.08	1209762.32	N 48 2 28.04	W 103 36 9.03
	10536.00	34.38	114.63	10518.54	137.92	-35.10	136.86	12.67	395047.24	1209777.54	N 48 2 27.97	W 103 36 8.81
	10568.00	38.51	114.74	10544.28	155.44	-43.04	154.13	12.91	395038.62	1209794.48	N 48 2 27.90	W 103 36 8.55
	10600.00	40.80	114.20	10568.92	174.29	-51.49	172.72	7.24	395029.42	1209812.71	N 48 2 27.81	W 103 36 8.28
	10631.00	43.44	113.06	10591.91	193.60	-59.82	191.77	8.87	395020.33	1209831.40	N 48 2 27.73	W 103 36 8.00
	10663.00	46.78	112.20	10614.49	214.79	-68.54	212.69	10.61	395010.78	1209851.96	N 48 2 27.64	W 103 36 7.69
	10694.00	50.12	110.69	10635.05	236.65	-77.01	234.28	11.37	395001.44	1209873.19	N 48 2 27.56	W 103 36 7.37
	10726.00	52.76	111.42	10654.99	260.27	-86.00	257.63	8.44	394991.52	1209896.16	N 48 2 27.47	W 103 36 7.03
	10758.00	53.73	112.37	10674.14	284.36	-95.56	281.42	3.85	394981.01	1209919.54	N 48 2 27.38	W 103 36 6.68
	10790.00	56.98	111.33	10692.33	309.09	-105.36	305.85	10.50	394970.24	1209943.55	N 48 2 27.28	W 103 36 6.32
	10821.00	61.64	112.50	10708.15	334.12	-115.31	330.57	15.38	394959.30	1209967.85	N 48 2 27.18	W 103 36 5.96
	10853.00	65.60	113.45	10722.37	360.86	-126.50	356.96	12.66	394947.06	1209993.77	N 48 2 27.07	W 103 36 5.57
	10885.00	67.36	114.04	10735.14	388.08	-138.32	383.81	5.75	394934.17	1210020.12	N 48 2 26.96	W 103 36 5.17
	10917.00	71.58	113.82	10746.36	415.84	-150.47	411.20	13.20	394920.92	1210046.99	N 48 2 26.84	W 103 36 4.77
	10948.00	74.74	114.38	10755.33	443.30	-162.58	438.28	10.34	394907.73	1210073.56	N 48 2 26.72	W 103 36 4.37
	10980.00	75.62	113.63	10763.52	471.95	-175.17	466.54	3.56	394894.02	1210101.29	N 48 2 26.59	W 103 36 3.95
	11011.00	78.26	112.75	10770.52	500.08	-187.06	494.30	8.95	394881.02	1210128.54	N 48 2 26.47	W 103 36 3.55
7" Casing Point	*11036.00*	*81.21*	*112.72*	*10774.98*	*523.06*	*-196.56*	*516.98*	*11.81*	*394870.61*	*1210150.83*	*N 48 2 26.38*	*W 103 36 3.21*
	11043.00	82.04	112.71	10776.00	529.53	-199.24	523.37	11.81	394867.68	1210157.10	N 48 2 26.35	W 103 36 3.12
	11075.00	87.93	112.41	10778.79	559.33	-211.46	552.80	18.43	394854.28	1210186.01	N 48 2 26.23	W 103 36 2.69
	11107.00	92.42	111.83	10778.70	589.34	-223.51	582.43	14.15	394841.05	1210215.13	N 48 2 26.11	W 103 36 2.25
	11147.00	92.51	112.56	10776.98	626.81	-238.60	619.43	1.84	394824.48	1210251.50	N 48 2 25.97	W 103 36 1.70
	11178.00	91.01	112.02	10776.02	655.85	-250.36	648.10	5.14	394811.58	1210279.67	N 48 2 25.85	W 103 36 1.28
	11208.00	91.28	111.69	10775.42	684.03	-261.52	675.94	1.42	394799.30	1210307.03	N 48 2 25.74	W 103 36 0.87
	11239.00	89.78	111.36	10775.14	713.22	-272.89	704.78	4.95	394786.78	1210335.38	N 48 2 25.63	W 103 36 0.45
	11270.00	89.87	110.86	10775.23	742.49	-284.06	733.70	1.64	394774.46	1210363.83	N 48 2 25.52	W 103 36 0.02
	11300.00	88.81	110.92	10775.58	770.84	-294.76	761.72	3.54	394762.64	1210391.40	N 48 2 25.41	W 103 35 59.61
	11331.00	87.76	110.50	10776.51	800.16	-305.71	790.71	3.65	394750.53	1210419.91	N 48 2 25.30	W 103 35 59.18
	11361.00	87.85	109.22	10777.65	828.67	-315.90	818.90	4.27	394739.21	1210447.67	N 48 2 25.20	W 103 35 58.77
	11392.00	88.29	109.82	10778.70	858.19	-326.25	848.10	2.40	394727.70	1210476.43	N 48 2 25.10	W 103 35 58.34
Last MWD Survey	11423.00	88.29	107.43	10779.62	887.86	-336.14	877.46	7.71	394716.63	1210505.37	N 48 2 25.00	W 103 35 57.91
Projection to Bit	11501.00	88.29	107.43	10781.95	962.96	-359.50	951.85	0.00	394690.30	1210578.75	N 48 2 24.77	W 103 35 56.81

Survey Type: Def Survey

Survey Error Model: ISCWSA Rev 0 *** 3-D 95.000% Confidence 2.7955 sigma

Survey Program:

Description	Part	MD From (ft)	MD To (ft)	EOU Freq (ft)	Hole Size (in)	Casing Diameter (in)	Survey Tool Type	Borehole / Survey
	1	0.000	32.000	1/98.425	30.000	30.000	SLB_MWD-STD-Depth Only	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	32.000	32.000	Act Stns	30.000	30.000	SLB_MWD-STD-Depth Only	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	32.000	11423.000	Act Stns	30.000	30.000	SLB_MWD-STD	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	11423.000	11501.000	Act Stns	30.000	30.000	SLB_BLIND+TREND	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to

A Schlumberger Company
9251 E 104th Ave.
Commerce City, CO 80640
(303) 439-5500

Directional Survey Certification Form

Oasis Petroleum	**Wade Federal 5300 41-30 #7T ST1**	**8-Dec-2014**
Company	Well Name	Final report Date
	ND, McKenzie County (NAD 83 NZ) Oasis	**33-053-05998**
PathFinder Job Number	County / State	API Number
N 48° 2' 28.32000" **N 48.0412**	**W 103° 36' 10.82000"** **W 103.60300556**	**877 ft FSL & 500 ft FEL** **Sec 30 Twn 153 N Rng 100 W**
Surface Latitude	Surface Logitude	Surface Section - Township - Range
NAD83 ND State Plane, N Zone, Ft	**Patterson 488**	**KB 32ft @ 2077.00 ft / GL: 2045.00 ft MSL**
Datum & Coordinate System	Rig Contractor	Height Reference

Survey Depth **11297.00** (Depth From) to **18148.00** (Depth To)

Measurement While Drilling
Type of Survey

Survey Depth **18148.00** (Depth From) to **18174.00** (Depth To)

Straight line projection to Bit/TD
Type of Survey

Site Supervisors

	Brittany Peddersen - MWD
Directional Driller 1	MWD Surveyor 1
	Brittany Peddersen - MWD
Directional Driller 2	MWD Surveyor 2

The data submitted in this report conforms to the standards and procedures as set forth by Schlumberger. This report represents a true and correct directional wellbore survey based on original survey data obtained at the well site.

Matt VanderSchaaf
Matt VanderSchaaf
PathFinder Well Planner II

12/8/2014 8:32
Date

Wade Federal 5300 41-30 #7T ST1 11270' to 18174' Definitive Survey
Geodetic Report
(Def Survey)

Report Date:	December 08, 2014 - 08:23 AM	**Survey / DLS Computation:**	Minimum Curvature / Lubinski
Client:	Oasis Petroleum	**Vertical Section Azimuth:**	91.850 ° (True North)
Field:	ND, McKenzie County (NAD 83 NZ) Oasis 2014	**Vertical Section Origin:**	0.000 ft, 0.000 ft
Structure / Slot:	Oasis 30-153N-100W (Wade Federal 5300 41-30 Pad) - Patterson 488 / Wade Federal 5300 41-30 #7T	**TVD Reference Datum:**	KB 32ft
Well:	Wade Federal 5300 41-30 #7T	**TVD Reference Elevation:**	2077.000 ft above MSL
Borehole:	ST1	**Seabed / Ground Elevation:**	2045.000 ft above MSL
UWI / API#:	Unknown / Unknown	**Magnetic Declination:**	8.395 °
Survey Name:	Wade Federal 5300 41-30 #7T ST1 11270' to 18174' Definitive	**Total Gravity Field Strength:**	1000.0448mgn (9.80665 Based)
Survey Date:	December 05, 2014	**Gravity Model:**	GARM
Tort / AHD / DDI / ERD Ratio:	254.337 ° / 7750.128 ft / 6.518 / 0.714	**Total Magnetic Field Strength:**	56237.474 nT
Coordinate Reference System:	NAD83 North Dakota State Plane, Northern Zone, Feet	**Magnetic Dip Angle:**	72.904 °
Location Lat / Long:	N 48° 2' 28.32000", W 103° 36' 10.82000"	**Declination Date:**	December 05, 2014
Location Grid N/E Y/X:	N 395087.826 ft, E 1209642.216 ft	**Magnetic Declination Model:**	BGGM 2014
CRS Grid Convergence Angle:	-2.3091 °	**North Reference:**	True North
Grid Scale Factor:	0.99993613	**Grid Convergence Used:**	0.0000 °
Version / Patch:	2.8.572.0	**Total Corr Mag North->True North:**	8.3953 °
		Local Coord Referenced To:	Well Head

Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
Tie-Into OH Survey	11270.00	89.87	110.86	10775.23	742.49	-284.06	733.70	N/A	394774.46	1210363.83	N 48 2 25.52	W 103 36 0.02
Begin ST1 Survey	11297.00	87.76	111.19	10775.79	767.98	-293.74	758.89	7.91	394763.77	1210388.61	N 48 2 25.42	W 103 35 59.65
	11331.00	87.93	108.15	10777.07	800.32	-305.18	790.88	8.95	394751.05	1210420.11	N 48 2 25.31	W 103 35 59.18
	11361.00	87.49	106.03	10778.27	829.24	-313.99	819.53	7.21	394741.10	1210448.38	N 48 2 25.22	W 103 35 58.76
	11392.00	88.46	105.32	10779.36	859.32	-322.36	849.36	3.88	394731.53	1210477.85	N 48 2 25.14	W 103 35 58.32
	11423.00	87.58	104.44	10780.43	889.51	-330.31	879.30	4.01	394722.38	1210507.44	N 48 2 25.06	W 103 35 57.88
	11453.00	87.23	102.20	10781.79	918.88	-337.22	908.46	7.55	394714.31	1210536.30	N 48 2 24.99	W 103 35 57.45
	11484.00	88.20	98.42	10783.03	949.51	-342.76	938.93	12.58	394707.54	1210566.52	N 48 2 24.94	W 103 35 57.00
	11514.00	88.29	96.72	10783.95	979.34	-346.71	968.66	5.67	394702.40	1210596.06	N 48 2 24.90	W 103 35 56.57
	11545.00	88.81	92.21	10784.73	1010.29	-349.12	999.54	14.64	394698.74	1210626.82	N 48 2 24.87	W 103 35 56.11
	11576.00	89.25	90.40	10785.26	1041.29	-349.83	1030.53	6.01	394696.79	1210657.75	N 48 2 24.87	W 103 35 55.66
	11606.00	89.34	87.77	10785.63	1071.25	-349.35	1060.52	8.77	394696.06	1210687.74	N 48 2 24.87	W 103 35 55.21
	11637.00	89.69	86.52	10785.89	1102.14	-347.80	1091.48	4.19	394696.35	1210718.73	N 48 2 24.89	W 103 35 54.76
	11698.00	90.57	87.48	10785.75	1162.92	-344.61	1152.40	2.13	394697.09	1210779.72	N 48 2 24.92	W 103 35 53.86
	11740.00	87.67	90.39	10786.40	1204.85	-343.83	1194.38	9.78	394696.18	1210821.69	N 48 2 24.93	W 103 35 53.24
	11790.00	88.02	90.76	10788.28	1254.81	-344.33	1244.34	1.02	394693.66	1210871.59	N 48 2 24.92	W 103 35 52.51
	11832.00	88.99	87.89	10789.37	1296.75	-343.84	1286.32	7.21	394692.47	1210913.55	N 48 2 24.93	W 103 35 51.89
	11885.00	89.08	88.19	10790.26	1349.62	-342.03	1339.28	0.59	394692.15	1210966.54	N 48 2 24.94	W 103 35 51.11
	11981.00	89.08	87.97	10791.81	1445.40	-338.81	1435.21	0.23	394691.49	1211062.52	N 48 2 24.98	W 103 35 49.70
	12076.00	88.55	87.76	10793.77	1540.15	-335.27	1530.12	0.60	394691.20	1211157.49	N 48 2 25.01	W 103 35 48.30
	12171.00	88.55	88.18	10796.17	1634.90	-331.91	1625.03	0.44	394690.74	1211252.46	N 48 2 25.04	W 103 35 46.91
	12267.00	89.60	88.20	10797.72	1730.69	-328.88	1720.97	1.09	394689.91	1211348.43	N 48 2 25.07	W 103 35 45.50
	12362.00	86.88	89.55	10800.64	1825.51	-327.01	1815.90	3.20	394687.95	1211443.35	N 48 2 25.09	W 103 35 44.10
	12458.00	88.11	87.95	10804.84	1921.27	-324.92	1911.78	2.10	394686.17	1211539.23	N 48 2 25.11	W 103 35 42.69
	12489.00	89.96	87.82	10805.36	1952.19	-323.77	1942.75	5.98	394686.07	1211570.22	N 48 2 25.12	W 103 35 42.23
	12553.00	88.46	86.32	10806.24	2015.96	-320.50	2006.66	3.31	394686.76	1211634.20	N 48 2 25.16	W 103 35 41.29
	12617.00	90.40	88.75	10806.88	2079.77	-317.75	2070.59	4.86	394686.94	1211698.19	N 48 2 25.18	W 103 35 40.35
	12648.00	90.75	88.85	10806.57	2110.72	-317.10	2101.58	1.17	394686.34	1211729.18	N 48 2 25.19	W 103 35 39.89
	12744.00	90.31	88.68	10805.68	2206.58	-315.03	2197.55	0.49	394684.54	1211825.15	N 48 2 25.21	W 103 35 38.48
	12839.00	90.22	90.62	10805.24	2301.50	-314.45	2292.55	2.04	394681.29	1211920.09	N 48 2 25.22	W 103 35 37.08
	12934.00	89.96	91.36	10805.09	2396.49	-316.09	2387.53	0.83	394675.82	1212014.92	N 48 2 25.20	W 103 35 35.69
	13029.00	90.04	91.38	10805.09	2491.49	-318.37	2482.50	0.09	394669.73	1212109.72	N 48 2 25.18	W 103 35 34.29
	13124.00	89.87	90.80	10805.17	2586.48	-320.17	2577.49	0.64	394664.10	1212204.55	N 48 2 25.16	W 103 35 32.89
	13219.00	89.16	91.01	10805.97	2681.46	-321.67	2672.47	0.78	394658.77	1212299.39	N 48 2 25.14	W 103 35 31.49
	13315.00	89.60	89.60	10807.01	2777.42	-322.18	2768.46	1.54	394654.39	1212395.27	N 48 2 25.14	W 103 35 30.08
	13410.00	91.45	87.92	10806.14	2872.27	-320.13	2863.43	2.63	394652.62	1212490.24	N 48 2 25.16	W 103 35 28.68
	13505.00	90.57	89.39	10804.46	2967.10	-317.90	2958.38	1.80	394651.02	1212585.20	N 48 2 25.18	W 103 35 27.29
	13601.00	90.04	90.89	10803.95	3063.06	-318.13	3054.38	1.66	394646.92	1212681.10	N 48 2 25.18	W 103 35 25.87
	13696.00	90.57	91.34	10803.45	3158.05	-319.98	3149.36	0.73	394641.25	1212775.92	N 48 2 25.16	W 103 35 24.48
	13791.00	89.34	91.54	10803.52	3253.04	-322.37	3244.33	1.31	394635.03	1212870.71	N 48 2 25.14	W 103 35 23.08
	13885.00	89.78	92.42	10804.24	3347.04	-325.62	3338.27	1.05	394628.01	1212964.44	N 48 2 25.10	W 103 35 21.70
	13979.00	89.34	90.59	10804.97	3441.03	-328.09	3432.23	2.00	394621.75	1213058.22	N 48 2 25.08	W 103 35 20.31
	14074.00	88.72	90.31	10806.57	3535.99	-328.83	3527.21	0.72	394617.18	1213153.09	N 48 2 25.07	W 103 35 18.92
	14169.00	86.97	90.10	10810.15	3630.88	-329.17	3622.14	1.86	394613.02	1213247.92	N 48 2 25.07	W 103 35 17.52
	14200.00	87.85	90.36	10811.55	3661.83	-329.30	3653.10	2.96	394611.65	1213278.86	N 48 2 25.07	W 103 35 17.06
	14264.00	89.25	90.60	10813.17	3725.79	-329.83	3717.08	2.22	394608.53	1213342.75	N 48 2 25.06	W 103 35 16.12
	14359.00	89.96	90.09	10813.82	3820.76	-330.40	3812.08	0.92	394604.13	1213437.64	N 48 2 25.06	W 103 35 14.72
	14454.00	89.87	89.35	10813.96	3915.69	-329.94	3907.07	0.78	394600.77	1213532.58	N 48 2 25.06	W 103 35 13.33
	14549.00	89.60	90.29	10814.40	4010.63	-329.64	4002.07	1.03	394597.24	1213627.50	N 48 2 25.06	W 103 35 11.93
	14645.00	89.96	89.74	10814.77	4106.58	-329.67	4098.07	0.68	394593.35	1213723.42	N 48 2 25.06	W 103 35 10.52
	14740.00	90.13	89.87	10814.70	4201.52	-329.34	4193.07	0.23	394589.85	1213818.34	N 48 2 25.07	W 103 35 9.12
	14835.00	90.13	89.63	10814.48	4296.45	-328.93	4288.07	0.25	394586.43	1213913.28	N 48 2 25.07	W 103 35 7.72
	14930.00	90.22	89.31	10814.19	4391.37	-328.05	4383.06	0.35	394583.48	1214008.22	N 48 2 25.08	W 103 35 6.32
	15025.00	89.34	88.85	10814.55	4486.26	-326.52	4478.05	1.05	394581.18	1214103.19	N 48 2 25.09	W 103 35 4.92
	15121.00	87.93	89.73	10816.84	4582.13	-325.33	4574.01	1.73	394578.50	1214199.11	N 48 2 25.10	W 103 35 3.51
	15216.00	88.02	89.66	10820.20	4677.00	-324.83	4668.95	0.12	394575.18	1214293.99	N 48 2 25.11	W 103 35 2.11
	15311.00	89.43	88.97	10822.31	4771.88	-323.69	4763.92	1.65	394572.49	1214388.92	N 48 2 25.12	W 103 35 0.72
	15406.00	89.16	89.25	10823.48	4866.77	-322.22	4858.90	0.41	394570.14	1214483.88	N 48 2 25.13	W 103 34 59.32
	15502.00	89.34	89.01	10824.74	4962.65	-320.76	4954.88	0.31	394567.73	1214579.83	N 48 2 25.15	W 103 34 57.91
	15597.00	88.90	88.97	10826.20	5057.52	-319.09	5049.85	0.47	394565.58	1214674.79	N 48 2 25.16	W 103 34 56.51
	15691.00	88.81	88.97	10828.08	5151.38	-317.40	5143.82	0.10	394563.48	1214768.74	N 48 2 25.18	W 103 34 55.13
	15786.00	88.46	88.65	10830.34	5246.22	-315.42	5238.77	0.50	394561.62	1214863.69	N 48 2 25.20	W 103 34 53.73
	15882.00	89.78	88.60	10831.81	5342.06	-313.12	5334.73	1.38	394560.06	1214959.66	N 48 2 25.22	W 103 34 52.32
	15977.00	89.60	88.25	10832.33	5436.89	-310.51	5429.69	0.41	394558.84	1215054.64	N 48 2 25.25	W 103 34 50.92
	16073.00	88.55	89.11	10833.88	5532.73	-308.30	5525.65	1.41	394557.18	1215150.61	N 48 2 25.27	W 103 34 49.51
	16168.00	87.23	90.36	10837.37	5627.59	-307.86	5620.58	1.91	394553.80	1215245.47	N 48 2 25.27	W 103 34 48.11
	16263.00	88.46	89.42	10840.95	5722.47	-307.68	5715.51	1.63	394550.16	1215340.33	N 48 2 25.28	W 103 34 46.71
	16358.00	89.60	91.67	10842.56	5817.42	-308.58	5810.49	2.65	394545.43	1215435.18	N 48 2 25.27	W 103 34 45.32
	16453.00	90.48	91.63	10842.49	5912.42	-311.32	5905.45	0.93	394538.87	1215529.95	N 48 2 25.24	W 103 34 43.92
	16549.00	88.11	91.33	10843.67	6008.40	-313.80	6001.40	2.49	394532.53	1215625.72	N 48 2 25.21	W 103 34 42.51
	16643.00	87.76	91.59	10847.06	6102.34	-316.19	6095.31	0.46	394526.35	1215719.45	N 48 2 25.19	W 103 34 41.13
	16737.00	87.05	92.45	10851.31	6196.24	-319.50	6189.15	1.19	394519.26	1215813.08	N 48 2 25.16	W 103 34 39.74
	16831.00	88.37	91.44	10855.07	6290.16	-322.69	6283.02	1.77	394512.30	1215906.73	N 48 2 25.13	W 103 34 38.36
	16924.00	90.13	90.83	10856.29	6383.14	-324.53	6375.99	2.00	394506.71	1215999.55	N 48 2 25.11	W 103 34 37.00
	17018.00	91.28	90.30	10855.13	6477.11	-325.45	6469.98	1.35	394502.00	1216093.41	N 48 2 25.10	W 103 34 35.61
	17113.00	91.98	91.08	10852.43	6572.05	-326.60	6564.93	1.10	394497.03	1216188.24	N 48 2 25.09	W 103 34 34.21

Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
	17207.00	89.87	90.12	10850.91	6666.01	-327.58	6658.91	2.47	394492.26	1216282.09	N 48 2 25.08	W 103 34 32.83
	17300.00	88.29	91.28	10852.40	6758.97	-328.72	6751.88	2.11	394487.38	1216374.94	N 48 2 25.06	W 103 34 31.46
	17394.00	89.16	89.90	10854.49	6852.93	-329.69	6845.85	1.74	394482.63	1216468.79	N 48 2 25.05	W 103 34 30.08
	17488.00	88.90	90.70	10856.09	6946.88	-330.18	6939.84	0.89	394478.35	1216562.67	N 48 2 25.05	W 103 34 28.70
	17583.00	88.90	90.37	10857.91	7041.83	-331.06	7034.81	0.35	394473.64	1216657.53	N 48 2 25.04	W 103 34 27.30
	17677.00	88.90	90.24	10859.71	7135.78	-331.56	7128.79	0.14	394469.35	1216751.41	N 48 2 25.03	W 103 34 25.92
	17771.00	89.60	90.45	10860.94	7229.74	-332.13	7222.78	0.78	394465.00	1216845.29	N 48 2 25.03	W 103 34 24.53
	17865.00	90.31	89.55	10861.02	7323.69	-332.13	7316.78	1.22	394461.22	1216939.21	N 48 2 25.03	W 103 34 23.15
	17959.00	90.40	89.00	10860.44	7417.59	-330.94	7410.77	0.59	394458.62	1217033.17	N 48 2 25.04	W 103 34 21.77
	18053.00	92.15	87.89	10858.34	7511.40	-328.39	7504.71	2.20	394457.38	1217127.12	N 48 2 25.06	W 103 34 20.39
Last ST1 Survey	18148.00	93.39	89.33	10853.75	7606.13	-326.09	7599.57	2.00	394455.86	1217221.99	N 48 2 25.09	W 103 34 18.99
Projection to Bit	18174.00	93.39	89.33	10852.22	7632.06	-325.79	7625.52	0.00	394455.12	1217247.93	N 48 2 25.09	W 103 34 18.61

Survey Type: Def Survey

Survey Error Model: ISCWSA Rev 0 *** 3-D 95.000% Confidence 2.7955 sigma
Survey Program:

Description	Part	MD From (ft)	MD To (ft)	EOU Freq (ft)	Hole Size (in)	Casing Diameter (in)	Survey Tool Type	Borehole / Survey
	1	0.000	32.000	1/98.425	30.000	30.000	SLB_MWD-STD-Depth Only	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	32.000	32.000	Act Stns	30.000	30.000	SLB_MWD-STD-Depth Only	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	32.000	11270.000	Act Stns	30.000	30.000	SLB_MWD-STD	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	11270.000	18148.000	Act Stns	30.000	30.000	SLB_MWD-STD	ST1 / Wade Federal 5300 41-30 #7T ST1 11270' to 18174'
	1	18148.000	18174.000	Act Stns	30.000	30.000	SLB_BLIND+TREND	ST1 / Wade Federal 5300 41-30 #7T ST1 11270' to 18174'

PATHFINDER
A Schlumberger Company

9251 E 104th Ave.
Commerce City, CO 80640
(303) 439-5500

Directional Survey Certification Form

Oasis Petroleum	**Wade Federal 5300 41-30 #7T ST2**	**8-Dec-2014**
Company	Well Name	Final report Date
	ND, McKenzie County (NAD 83 NZ) Oasis	**33-053-05998**
PathFinder Job Number	County / State	API Number
N 48° 2' 28.32000" **N 48.0412**	**W 103° 36' 10.82000"** **W 103.60300556**	**877 ft FSL & 500 ft FEL** **Sec 30 Twn 153 N Rng 100 W**
Surface Latitude	Surface Logitude	Surface Section - Township - Range
NAD83 ND State Plane, N Zone, Ft	**Patterson 488**	**KB 32ft @ 2077.00 ft / GL: 2045.00 ft MSL**
Datum & Coordinate System	Rig Contractor	Height Reference

Survey Depth **18068.00** (Depth From) to **20457.00** (Depth To)

Measurement While Drilling
Type of Survey

Survey Depth **20457.00** (Depth From) to **20530.00** (Depth To)

Straight line projection to Bit/TD
Type of Survey

Site Supervisors

	Brittany Feddersen - MWD
Directional Driller 1	MWD Surveyor 1
	Brittany Feddersen - MWD
Directional Driller 2	MWD Surveyor 2

The data submitted in this report conforms to the standards and procedures as set forth by Schlumberger. This report represents a true and correct directional wellbore survey based on original survey data obtained at the well site.

Matt VanderSchaaf — 12/8/2014 8:41

Matt VanderSchaaf — Date
PathFinder Well Planner II

Wade Federal 5300 41-30 #7T ST2 18053' to 20530' Definitive Survey
Geodetic Report

(Non-Def Survey)

Report Date:	December 08, 2014 - 08:38 AM	**Survey / DLS Computation:**	Minimum Curvature / Lubinski
Client:	Oasis Petroleum	**Vertical Section Azimuth:**	91.850 ° (True North)
Field:	ND, McKenzie County (NAD 83 NZ) Oasis 2014	**Vertical Section Origin:**	0.000 ft, 0.000 ft
Structure / Slot:	Oasis 30-153N-100W (Wade Federal 5300 41-30 Pad) - Patterson 488 / Wade Federal 5300 41-30 #7T	**TVD Reference Datum:**	KB 32ft
Well:	Wade Federal 5300 41-30 #7T	**TVD Reference Elevation:**	2077.000 ft above MSL
Borehole:	ST2	**Seabed / Ground Elevation:**	2045.000 ft above MSL
UWI / API#:	Unknown / Unknown	**Magnetic Declination:**	8.395 °
Survey Name:	Wade Federal 5300 41-30 #7T ST2 18053' to 20530' Definitive	**Total Gravity Field Strength:**	1000.0448mgn (9.80665 Based)
Survey Date:	December 05, 2014	**Gravity Model:**	GARM
Tort / AHD / DDI / ERD Ratio:	291.164 ° / 10105.834 ft / 6.744 / 0.929	**Total Magnetic Field Strength:**	56237.474 nT
Coordinate Reference System:	NAD83 North Dakota State Plane, Northern Zone, Feet	**Magnetic Dip Angle:**	72.904 °
Location Lat / Long:	N 48° 2' 28.32000", W 103° 36' 10.82000"	**Declination Date:**	December 05, 2014
Location Grid N/E Y/X:	N 395087.826 ft, E 1209642.216 ft	**Magnetic Declination Model:**	BGGM 2014
CRS Grid Convergence Angle:	-2.3091 °	**North Reference:**	True North
Grid Scale Factor:	0.99993613	**Grid Convergence Used:**	0.0000 °
Version / Patch:	2.8.572.0	**Total Corr Mag North->True North:**	8.3953 °
		Local Coord Referenced To:	Well Head

Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
Tie-Into ST1	18053.00	92.15	87.89	10858.34	7511.40	-328.39	7504.71	N/A	394457.38	1217127.12	N 48 2 25.06	W 103 34 20.39
Begin ST2 Survey	18068.00	89.78	90.11	10858.09	7526.38	-328.13	7519.70	21.65	394457.04	1217142.11	N 48 2 25.07	W 103 34 20.17
	18100.00	88.64	90.71	10858.53	7558.36	-328.36	7551.70	4.03	394455.52	1217174.07	N 48 2 25.06	W 103 34 19.69
	18131.00	89.25	90.45	10859.10	7589.35	-328.67	7582.69	2.14	394453.96	1217205.03	N 48 2 25.06	W 103 34 19.24
	18163.00	89.34	91.00	10859.50	7621.34	-329.08	7614.69	1.74	394452.26	1217236.98	N 48 2 25.06	W 103 34 18.77
	18259.00	88.55	91.04	10861.27	7717.31	-330.79	7710.65	0.82	394446.69	1217332.79	N 48 2 25.04	W 103 34 17.36
	18354.00	90.22	90.43	10862.29	7812.29	-332.00	7805.64	1.87	394441.65	1217427.64	N 48 2 25.03	W 103 34 15.96
	18386.00	90.40	90.12	10862.11	7844.27	-332.16	7837.64	1.12	394440.20	1217459.61	N 48 2 25.03	W 103 34 15.49
	18449.00	91.54	89.94	10861.05	7907.23	-332.19	7900.63	1.83	394437.63	1217522.54	N 48 2 25.03	W 103 34 14.56
	18474.00	90.75	89.49	10860.55	7932.21	-332.07	7925.62	3.64	394436.75	1217547.52	N 48 2 25.03	W 103 34 14.19
	18544.00	91.54	89.71	10859.15	8002.14	-331.58	7995.60	1.17	394434.42	1217617.46	N 48 2 25.03	W 103 34 13.16
	18576.00	89.08	89.69	10858.97	8034.12	-331.41	8027.60	7.69	394433.30	1217649.44	N 48 2 25.03	W 103 34 12.69
	18640.00	89.16	89.87	10859.96	8098.07	-331.16	8091.59	0.31	394430.97	1217713.38	N 48 2 25.03	W 103 34 11.75
	18735.00	88.11	89.74	10862.22	8192.98	-330.84	8186.56	1.11	394427.46	1217808.28	N 48 2 25.04	W 103 34 10.35
	18830.00	87.85	89.63	10865.57	8287.85	-330.32	8281.50	0.30	394424.16	1217903.16	N 48 2 25.04	W 103 34 8.96
	18871.00	88.02	89.05	10867.05	8328.78	-329.85	8322.47	1.47	394422.98	1217944.11	N 48 2 25.05	W 103 34 8.35
	18967.00	88.20	87.81	10870.21	8424.56	-327.22	8418.38	1.30	394421.74	1218040.04	N 48 2 25.07	W 103 34 6.94
	19062.00	89.52	88.36	10872.10	8519.33	-324.05	8513.31	1.51	394421.09	1218135.02	N 48 2 25.10	W 103 34 5.54
	19158.00	90.57	88.20	10872.03	8615.15	-321.16	8609.27	1.11	394420.10	1218231.00	N 48 2 25.13	W 103 34 4.13
	19252.00	90.22	90.25	10871.38	8709.04	-319.89	8703.25	2.21	394417.59	1218324.96	N 48 2 25.14	W 103 34 2.75
	19347.00	88.37	89.64	10872.55	8803.98	-319.80	8798.24	2.05	394413.85	1218419.86	N 48 2 25.14	W 103 34 1.35
	19443.00	89.16	89.20	10874.62	8899.87	-318.83	8894.21	0.94	394410.95	1218515.79	N 48 2 25.15	W 103 33 59.94
	19536.00	90.40	89.28	10874.97	8992.77	-317.60	8987.20	1.34	394408.44	1218608.75	N 48 2 25.16	W 103 33 58.57
	19630.00	90.48	90.57	10874.25	9086.71	-317.47	9081.19	1.37	394404.78	1218702.67	N 48 2 25.17	W 103 33 57.19
	19725.00	89.87	90.38	10873.96	9181.68	-318.26	9176.19	0.67	394400.16	1218797.55	N 48 2 25.16	W 103 33 55.79
	19812.00	89.87	92.25	10874.16	9268.67	-320.26	9263.16	2.15	394394.66	1218884.36	N 48 2 25.14	W 103 33 54.51
	19906.00	91.54	91.38	10873.00	9362.66	-323.23	9357.10	2.00	394387.91	1218978.10	N 48 2 25.11	W 103 33 53.13
	20000.00	90.13	91.48	10871.63	9456.65	-325.58	9451.06	1.50	394381.78	1219071.88	N 48 2 25.08	W 103 33 51.74
	20094.00	89.69	90.62	10871.78	9550.64	-327.30	9545.05	1.03	394376.27	1219165.72	N 48 2 25.07	W 103 33 50.36
	20189.00	88.29	90.04	10873.45	9645.59	-327.85	9640.03	1.60	394371.90	1219260.59	N 48 2 25.06	W 103 33 48.96
	20284.00	88.99	89.30	10875.71	9740.49	-327.30	9735.00	1.07	394368.62	1219355.50	N 48 2 25.07	W 103 33 47.57
	20378.00	88.64	87.84	10877.65	9834.31	-324.96	9828.94	1.60	394367.17	1219449.46	N 48 2 25.09	W 103 33 46.18
Last ST2 Survey	20457.00	88.46	87.33	10879.65	9913.07	-321.63	9907.85	0.68	394367.32	1219528.43	N 48 2 25.12	W 103 33 45.02
Projection to Bit	20530.00	88.46	87.33	10881.61	9985.81	-318.23	9980.74	0.00	394367.78	1219601.40	N 48 2 25.15	W 103 33 43.95

Survey Type: Non-Def Survey

Survey Error Model: ISCWSA Rev 0 *** 3-D 95.000% Confidence 2.7955 sigma
Survey Program:

Description	Part	MD From (ft)	MD To (ft)	EOU Freq (ft)	Hole Size (in)	Casing Diameter (in)	Survey Tool Type	Borehole / Survey
	1	0.000	32.000	1/98.425	30.000	30.000	SLB_MWD-STD-Depth Only	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	32.000	32.000	Act Stns	30.000	30.000	SLB_MWD-STD-Depth Only	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	32.000	11270.000	Act Stns	30.000	30.000	SLB_MWD-STD	Original Hole / Wade Federal 5300 41-30 #7T OH MWD 0' to
	1	11270.000	18053.000	Act Stns	30.000	30.000	SLB_MWD-STD	ST1 / Wade Federal 5300 41-30 #7T ST1 11270' to 18174'
	1	18053.000	20457.000	Act Stns	30.000	30.000	SLB_MWD-STD	ST2 / Wade Federal 5300 41-30 #7T ST2 18053' to 20530'
	1	20457.000	20530.000	Act Stns	30.000	30.000	SLB_BLIND+TREND	ST2 / Wade Federal 5300 41-30 #7T ST2 18053' to 20530'

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. **28557**

RECEIVED JUL 2014 ND OIL & GAS DIVISION

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **September 1, 2014**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [x] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [ ] Other
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number				
Wade Federal 5300 41-30 7T				
Footages: **877** F **S** L **280** F **W** L	Qtr-Qtr: **lot4**	Section: **30**	Township: **153 N**	Range: **100 W**
Field:	Pool: **Bakken**	County: **McKenzie**		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests to revise the casing plan for the subject well as follows:

13 3/8" 54.5# surface casing will be ran to 2,050'
Contingency 9 5/8" 40# will be ran to 6,400' in order to isolate the Dakota
7" 32# intermediate casing will be ran to 11,041'
4.5" 13.5# liner will be ran to 20,590'

Attached is a revised drill plan, directional plan/plot and well summary.

Company		Telephone Number
Oasis Petroleum North America LLC		**281-404-9491**
Address: **1001 Fannin, Suite 1500**		
City: **Houston**	State: **TX**	Zip Code: **77002**
Signature:	Printed Name: **Brandi Terry**	
Title: **Regulatory Specialist**	Date: **July 30, 2014**	
Email Address: **bterry@oasispetroleum.com**		

FOR STATE USE ONLY

- [ ] Received
- [x] Approved

Date: 8-18-14

By: Nathaniel Erbele

Title: Petroleum Resource Specialist

DRILLING PLAN

OPERATOR	Oasis Petroleum	**COUNTY/STATE**	McKenzie Co., ND
WELL NAME	Wade Federal 5300 41-30 7T	**RIG**	Nabors 149
WELL TYPE	Horizontal Three Forks		
LOCATION	SW SW 30-153N-100W	Surface Location (survey plat):	877' FSL 280' FWL
EST. T.D.	20,590'	**GROUND ELEV:**	2,045'
TOTAL LATERAL:	9,549'	**KB ELEV:**	2,070'
		Sub Height:	25'

MARKER		**TVD**	**Subsea TVD**
Pierre	NDIC MAP	1,920	150
Greenhorn		4,566	-2,496
Mowry		4,969	-2,899
Dakota		5,391	-3,321
Rierdon		6,407	-4,337
Dunham Salt		6,896	-4,826
Dunham Salt Base		6,942	-4,872
Pine Salt		7,205	-5,135
Pine Salt Base		7,229	-5,159
Opeche Salt		7,291	-5,221
Opeche Salt Base		7,371	-5,301
Amsden		7,615	-5,545
Tyler		7,771	-5,701
Otter/Base Minnelusa		7,994	-5,924
Kibbey Lime		8,336	-6,266
Charles Salt		8,484	-6,414
Base Last Salt		9,163	-7,093
Mission Canyon		9,377	-7,307
Lodgepole		9,926	-7,856
False Bakken		10,656	-8,586
Upper Bakken Shale		10,668	-8,598
Middle Bakken		10,683	-8,613
Lower Bakken Shale		10,719	-8,649
Pronghorn		10,728	-8,658
Threeforks		10,744	-8,674
Threeforks(Top of Target)		10,761	-8,691
Threeforks(Base of Target)		10,770	-8,700
Claystone		10,770	-8,700

Est. Dip Rate: [illegible]

Max. Anticipated BHP: 4667

LOGS:

	Type	**Interval**
OH Logs:	Request Log waiver based on the Wade Federal 5300 21-30H 2,150' N of surface location	
CBL/GR:	Above top of cement/GR to base of casing	
MWD GR:	KOP to lateral TD	

DEVIATION:

Surf:	3 deg. max., 1 deg / 100'; srvy every 500'
Prod:	5 deg. max., 1 deg / 100'; srvy every 100'

DST'S: None planned

CORES: None planned

MUDLOGGING: Two-Man: Begin 200' above Kibbey
30' samples in curve and lateral

BOP: 11" 5000 psi blind, pipe & annular

Surface Formation: *Glacial till*

MUD:	**Interval**		**Type**	**WT**	**Vis**	**WL**	**Remarks**
Surface:	0' -	2,050'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks
Intermediate:	2,050' -	11,041'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks
Lateral:	11,041' -	20,590'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks

CASING:	**Size**	**Wt ppf**	**Hole**	**Depth**		**Cement**	**WOC**	**Remarks**
Surface:	13-3/8"	54.5#	17-1/2"	2,050'		To Surface	12	100' into Pierre
Inermediate (Dakota):	9-5/8"	40#	12-1/4"	6,400'		To Surface	24	Set Casing across Dakota
Intermediate:	7"	32#	8-3/4"	11,041'		3891	24	1500' above Dakota
Production Liner:	4.5"	13.5#	6"	20,590'	TOL @ 10,243'			50' above KOP

PROBABLE PLUGS, IF REQ'D:

OTHER:	**MD**	**TVD**	**FNL/FSL**	**FEL/FWL**	**S-T-R**	**AZI**
Surface:	2,050	2,050	877 FSL	280 FWL	30-T153-R100	
KOP:	10,293'	10,293'	877 FSL	340 FWL	30-T153-R100	
EOC:	11,039'	10,770'	896 FSL	779 FWL	30-T153-R100	112.4
Casing Point:	11,041'	10,770'	896 FSL	780 FWL	30-T153-R100	112.4
Middle Bakken Lateral TD:	20,590'	10,828'	553 FSL	210 FEL	29-T153-R100	90.0

Survey Company:
Build Rate: 12 deg /100'

Comments:

Request Log waiver based on the Wade Federal 5300 21-30H 2,150' N of surface location

No frac string planned

35 packers and 25 sleeves planned 3.6MM lbs 30% ceramic

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)

OASIS PETROLEUM

Geology: N. Gabelman 1/20/2014 **Engineering:** SMG3.26

Oasis

Indian Hills
153N-100W-29/30
Wade Federal 5300 41-30 7T

Wade Federal 5300 41-30 7T

Plan: Design #1

Standard Planning Report

19 May, 2014

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Project	Indian Hills		
Map System:	US State Plane 1983	**System Datum:**	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-29/30					
Site Position:			**Northing:**	395,521.43 ft	**Latitude:**	48° 2' 32.580 N
From:	Lat/Long		**Easting:**	1,209,621.64 ft	**Longitude:**	103° 36' 11.410 W
Position Uncertainty:		0.0 ft	**Slot Radius:**	13.200 in	**Grid Convergence:**	-2.31 °

Well	Wade Federal 5300 41-30 7T					
Well Position	**+N/-S**	-431.7 ft	**Northing:**	395,088.50 ft	**Latitude:**	48° 2' 28.320 N
	+E/-W	40.1 ft	**Easting:**	1,209,644.31 ft	**Longitude:**	103° 36' 10.820 W
Position Uncertainty		0.0 ft	**Wellhead Elevation:**		**Ground Level:**	2,045.0 ft

Wellbore Wade Federal 5300 41-30 7T

Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	1/28/2014	8.18	72.95	56,489

Design Design #1

Audit Notes:

Version:		**Phase:**	PROTOTYPE	**Tie On Depth:**	0.0

Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)
	0.0	0.0	0.0	91.85

Plan Sections

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,050.0	0.00	0.00	2,050.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,060.0	0.50	90.00	2,060.0	0.0	0.0	5.00	5.00	0.00	90.00	
8,925.6	0.50	90.00	8,925.3	0.0	60.0	0.00	0.00	0.00	0.00	
8,935.6	0.00	0.00	8,935.3	0.0	60.0	5.00	-5.00	0.00	180.00	
10,000.3	0.00	0.00	10,000.0	0.0	60.0	0.00	0.00	0.00	0.00	
10,292.8	0.00	0.00	10,292.5	0.0	60.0	0.00	0.00	0.00	0.00	
11,039.9	89.65	112.38	10,770.0	-180.7	498.8	12.00	12.00	0.00	112.38	
11,041.2	89.65	112.38	10,770.0	-181.2	500.0	0.00	0.00	0.00	0.00	
11,782.3	89.65	90.00	10,774.6	-324.1	1,222.4	3.02	0.00	-3.02	269.93	
20,590.1	89.65	90.00	10,828.4	-324.1	10,030.0	0.00	0.00	0.00	0.00	

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,920.0	0.00	0.00	1,920.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,020.0	0.00	0.00	2,020.0	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 5.00									
2,030.0	0.00	0.00	2,030.0	0.0	0.0	0.0	0.00	0.00	0.00
Start 6865.6 hold at 2030.0 MD									
2,050.0	0.00	0.00	2,050.0	0.0	0.0	0.0	0.00	0.00	0.00
9 5/8"									
2,060.0	0.50	90.00	2,060.0	0.0	0.0	0.0	5.00	5.00	0.00
2,100.0	0.50	90.00	2,100.0	0.0	0.4	0.4	0.00	0.00	0.00
2,200.0	0.50	90.00	2,200.0	0.0	1.3	1.3	0.00	0.00	0.00
2,300.0	0.50	90.00	2,300.0	0.0	2.1	2.1	0.00	0.00	0.00
2,400.0	0.50	90.00	2,400.0	0.0	3.0	3.0	0.00	0.00	0.00
2,500.0	0.50	90.00	2,500.0	0.0	3.9	3.9	0.00	0.00	0.00
2,600.0	0.50	90.00	2,600.0	0.0	4.8	4.8	0.00	0.00	0.00
2,700.0	0.50	90.00	2,700.0	0.0	5.6	5.6	0.00	0.00	0.00
2,800.0	0.50	90.00	2,800.0	0.0	6.5	6.5	0.00	0.00	0.00
2,900.0	0.50	90.00	2,900.0	0.0	7.4	7.4	0.00	0.00	0.00
3,000.0	0.50	90.00	3,000.0	0.0	8.2	8.2	0.00	0.00	0.00
3,100.0	0.50	90.00	3,100.0	0.0	9.1	9.1	0.00	0.00	0.00
3,200.0	0.50	90.00	3,200.0	0.0	10.0	10.0	0.00	0.00	0.00
3,300.0	0.50	90.00	3,300.0	0.0	10.9	10.9	0.00	0.00	0.00
3,400.0	0.50	90.00	3,399.9	0.0	11.7	11.7	0.00	0.00	0.00
3,500.0	0.50	90.00	3,499.9	0.0	12.6	12.6	0.00	0.00	0.00
3,600.0	0.50	90.00	3,599.9	0.0	13.5	13.5	0.00	0.00	0.00
3,700.0	0.50	90.00	3,699.9	0.0	14.4	14.3	0.00	0.00	0.00
3,800.0	0.50	90.00	3,799.9	0.0	15.2	15.2	0.00	0.00	0.00
3,900.0	0.50	90.00	3,899.9	0.0	16.1	16.1	0.00	0.00	0.00
4,000.0	0.50	90.00	3,999.9	0.0	17.0	17.0	0.00	0.00	0.00
4,100.0	0.50	90.00	4,099.9	0.0	17.8	17.8	0.00	0.00	0.00
4,200.0	0.50	90.00	4,199.9	0.0	18.7	18.7	0.00	0.00	0.00
4,300.0	0.50	90.00	4,299.9	0.0	19.6	19.6	0.00	0.00	0.00
4,400.0	0.50	90.00	4,399.9	0.0	20.5	20.5	0.00	0.00	0.00
4,500.0	0.50	90.00	4,499.9	0.0	21.3	21.3	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,566.1	0.50	90.00	4,566.0	0.0	21.9	21.9	0.00	0.00	0.00
Greenhorn									
4,600.0	0.50	90.00	4,599.9	0.0	22.2	22.2	0.00	0.00	0.00
4,700.0	0.50	90.00	4,699.9	0.0	23.1	23.1	0.00	0.00	0.00
4,800.0	0.50	90.00	4,799.9	0.0	24.0	23.9	0.00	0.00	0.00
4,900.0	0.50	90.00	4,899.9	0.0	24.8	24.8	0.00	0.00	0.00
4,969.1	0.50	90.00	4,969.0	0.0	25.4	25.4	0.00	0.00	0.00
Mowry									
5,000.0	0.50	90.00	4,999.9	0.0	25.7	25.7	0.00	0.00	0.00
5,100.0	0.50	90.00	5,099.9	0.0	26.6	26.6	0.00	0.00	0.00
5,200.0	0.50	90.00	5,199.9	0.0	27.4	27.4	0.00	0.00	0.00
5,300.0	0.50	90.00	5,299.9	0.0	28.3	28.3	0.00	0.00	0.00
5,391.1	0.50	90.00	5,391.0	0.0	29.1	29.1	0.00	0.00	0.00
Dakota									
5,400.0	0.50	90.00	5,399.9	0.0	29.2	29.2	0.00	0.00	0.00
5,500.0	0.50	90.00	5,499.9	0.0	30.1	30.0	0.00	0.00	0.00
5,600.0	0.50	90.00	5,599.9	0.0	30.9	30.9	0.00	0.00	0.00
5,700.0	0.50	90.00	5,699.9	0.0	31.8	31.8	0.00	0.00	0.00
5,800.0	0.50	90.00	5,799.9	0.0	32.7	32.7	0.00	0.00	0.00
5,900.0	0.50	90.00	5,899.9	0.0	33.6	33.5	0.00	0.00	0.00
6,000.0	0.50	90.00	5,999.8	0.0	34.4	34.4	0.00	0.00	0.00
6,100.0	0.50	90.00	6,099.8	0.0	35.3	35.3	0.00	0.00	0.00
6,200.0	0.50	90.00	6,199.8	0.0	36.2	36.2	0.00	0.00	0.00
6,300.0	0.50	90.00	6,299.8	0.0	37.0	37.0	0.00	0.00	0.00
6,400.0	0.50	90.00	6,399.8	0.0	37.9	37.9	0.00	0.00	0.00
6,407.2	0.50	90.00	6,407.0	0.0	38.0	38.0	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	90.00	6,499.8	0.0	38.8	38.8	0.00	0.00	0.00
6,600.0	0.50	90.00	6,599.8	0.0	39.7	39.6	0.00	0.00	0.00
6,700.0	0.50	90.00	6,699.8	0.0	40.5	40.5	0.00	0.00	0.00
6,800.0	0.50	90.00	6,799.8	0.0	41.4	41.4	0.00	0.00	0.00
6,896.2	0.50	90.00	6,896.0	0.0	42.2	42.2	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.50	90.00	6,899.8	0.0	42.3	42.3	0.00	0.00	0.00
6,942.2	0.50	90.00	6,942.0	0.0	42.6	42.6	0.00	0.00	0.00
Dunham Salt Base									
7,000.0	0.50	90.00	6,999.8	0.0	43.2	43.1	0.00	0.00	0.00
7,100.0	0.50	90.00	7,099.8	0.0	44.0	44.0	0.00	0.00	0.00
7,200.0	0.50	90.00	7,199.8	0.0	44.9	44.9	0.00	0.00	0.00
7,205.2	0.50	90.00	7,205.0	0.0	44.9	44.9	0.00	0.00	0.00
Pine Salt									
7,229.2	0.50	90.00	7,229.0	0.0	45.2	45.1	0.00	0.00	0.00
Pine Salt Base									
7,291.2	0.50	90.00	7,291.0	0.0	45.7	45.7	0.00	0.00	0.00
Opeche Salt									
7,300.0	0.50	90.00	7,299.8	0.0	45.8	45.7	0.00	0.00	0.00
7,371.2	0.50	90.00	7,371.0	0.0	46.4	46.4	0.00	0.00	0.00
Opeche Salt Base									
7,400.0	0.50	90.00	7,399.8	0.0	46.6	46.6	0.00	0.00	0.00
7,500.0	0.50	90.00	7,499.8	0.0	47.5	47.5	0.00	0.00	0.00
7,600.0	0.50	90.00	7,599.8	0.0	48.4	48.4	0.00	0.00	0.00
7,615.2	0.50	90.00	7,615.0	0.0	48.5	48.5	0.00	0.00	0.00
Amsden									
7,700.0	0.50	90.00	7,699.8	0.0	49.3	49.2	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
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Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
7,771.2	0.50	90.00	7,771.0	0.0	49.9	49.9	0.00	0.00	0.00
Tyler									
7,800.0	0.50	90.00	7,799.8	0.0	50.1	50.1	0.00	0.00	0.00
7,900.0	0.50	90.00	7,899.8	0.0	51.0	51.0	0.00	0.00	0.00
7,994.2	0.50	90.00	7,994.0	0.0	51.8	51.8	0.00	0.00	0.00
Otter/Base Minnelusa									
8,000.0	0.50	90.00	7,999.8	0.0	51.9	51.9	0.00	0.00	0.00
8,100.0	0.50	90.00	8,099.8	0.0	52.8	52.7	0.00	0.00	0.00
8,200.0	0.50	90.00	8,199.8	0.0	53.6	53.6	0.00	0.00	0.00
8,300.0	0.50	90.00	8,299.8	0.0	54.5	54.5	0.00	0.00	0.00
8,336.3	0.50	90.00	8,336.0	0.0	54.8	54.8	0.00	0.00	0.00
Kibbey Lime									
8,400.0	0.50	90.00	8,399.8	0.0	55.4	55.3	0.00	0.00	0.00
8,484.3	0.50	90.00	8,484.0	0.0	56.1	56.1	0.00	0.00	0.00
Charles Salt									
8,500.0	0.50	90.00	8,499.8	0.0	56.2	56.2	0.00	0.00	0.00
8,600.0	0.50	90.00	8,599.8	0.0	57.1	57.1	0.00	0.00	0.00
8,700.0	0.50	90.00	8,699.7	0.0	58.0	58.0	0.00	0.00	0.00
8,800.0	0.50	90.00	8,799.7	0.0	58.9	58.8	0.00	0.00	0.00
8,895.6	0.50	90.00	8,895.3	0.0	59.7	59.7	0.00	0.00	0.00
Start Drop -5.00									
8,900.0	0.50	90.00	8,899.7	0.0	59.7	59.7	0.00	0.00	0.00
8,905.6	0.50	90.00	8,905.3	0.0	59.8	59.8	0.00	0.00	0.00
Start 1094.7 hold at 8905.6 MD									
8,925.6	0.50	90.00	8,925.3	0.0	60.0	59.9	0.00	0.00	0.00
8,935.6	0.00	0.00	8,935.3	0.0	60.0	60.0	5.00	-5.00	0.00
9,000.0	0.00	0.00	8,999.7	0.0	60.0	60.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.7	0.0	60.0	60.0	0.00	0.00	0.00
9,163.3	0.00	0.00	9,163.0	0.0	60.0	60.0	0.00	0.00	0.00
Base Last Salt									
9,200.0	0.00	0.00	9,199.7	0.0	60.0	60.0	0.00	0.00	0.00
9,300.0	0.00	0.00	9,299.7	0.0	60.0	60.0	0.00	0.00	0.00
9,377.3	0.00	0.00	9,377.0	0.0	60.0	60.0	0.00	0.00	0.00
Mission Canyon									
9,400.0	0.00	0.00	9,399.7	0.0	60.0	60.0	0.00	0.00	0.00
9,500.0	0.00	0.00	9,499.7	0.0	60.0	60.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.7	0.0	60.0	60.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.7	0.0	60.0	60.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.7	0.0	60.0	60.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.7	0.0	60.0	60.0	0.00	0.00	0.00
9,926.3	0.00	0.00	9,926.0	0.0	60.0	60.0	0.00	0.00	0.00
Lodgepole									
10,000.3	0.00	0.00	10,000.0	0.0	60.0	60.0	0.00	0.00	0.00
Start 292.5 hold at 10000.3 MD									
10,100.0	0.00	0.00	10,099.7	0.0	60.0	60.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.7	0.0	60.0	60.0	0.00	0.00	0.00
10,292.8	0.00	0.00	10,292.5	0.0	60.0	60.0	0.00	0.00	0.00
Start Build 12.00 KOP									
10,300.0	0.86	112.38	10,299.7	0.0	60.1	60.0	12.00	12.00	0.00
10,325.0	3.86	112.38	10,324.7	-0.4	61.0	61.0	12.00	12.00	0.00
10,350.0	6.86	112.38	10,349.6	-1.3	63.2	63.2	12.00	12.00	0.00
10,375.0	9.86	112.38	10,374.3	-2.7	66.5	66.6	12.00	12.00	0.00
10,400.0	12.86	112.38	10,398.8	-4.6	71.1	71.2	12.00	12.00	0.00
10,425.0	15.86	112.38	10,423.1	-6.9	76.8	77.0	12.00	12.00	0.00

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Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,450.0	18.86	112.38	10,446.9	-9.8	83.7	84.0	12.00	12.00	0.00
10,475.0	21.86	112.38	10,470.3	-13.1	91.8	92.1	12.00	12.00	0.00
10,500.0	24.86	112.38	10,493.3	-16.9	100.9	101.4	12.00	12.00	0.00
10,525.0	27.86	112.38	10,515.7	-21.1	111.2	111.8	12.00	12.00	0.00
10,550.0	30.86	112.38	10,537.5	-25.7	122.5	123.3	12.00	12.00	0.00
10,575.0	33.86	112.38	10,558.6	-30.8	134.9	135.8	12.00	12.00	0.00
10,600.0	36.86	112.38	10,579.0	-36.4	148.3	149.4	12.00	12.00	0.00
10,625.0	39.86	112.38	10,598.6	-42.3	162.6	163.9	12.00	12.00	0.00
10,650.0	42.86	112.38	10,617.3	-48.6	177.9	179.4	12.00	12.00	0.00
10,675.0	45.86	112.38	10,635.2	-55.2	194.1	195.7	12.00	12.00	0.00
10,700.0	48.86	112.38	10,652.1	-62.2	211.1	213.0	12.00	12.00	0.00
10,705.9	49.58	112.38	10,656.0	-63.9	215.2	217.2	12.00	12.00	0.00
False Bakken									
10,724.9	51.85	112.38	10,668.0	-69.5	228.8	230.9	12.00	12.00	0.00
Upper Bakken Shale									
10,725.0	51.86	112.38	10,668.1	-69.5	228.9	231.0	12.00	12.00	0.00
10,750.0	54.86	112.38	10,683.0	-77.2	247.4	249.8	11.98	11.98	0.00
Middle Bakken									
10,775.0	57.86	112.38	10,696.8	-85.1	266.7	269.3	12.02	12.02	0.00
10,800.0	60.86	112.38	10,709.6	-93.3	286.5	289.4	12.00	12.00	0.00
10,820.1	63.28	112.38	10,719.0	-100.1	303.0	306.1	12.00	12.00	0.00
Lower Bakken Shale									
10,825.0	63.86	112.38	10,721.2	-101.7	307.0	310.1	12.00	12.00	0.00
10,841.1	65.79	112.38	10,728.0	-107.3	320.5	323.8	12.00	12.00	0.00
Pronghorn									
10,850.0	66.86	112.38	10,731.6	-110.4	328.0	331.4	12.00	12.00	0.00
10,875.0	69.86	112.38	10,740.8	-119.2	349.5	353.2	12.00	12.00	0.00
10,884.6	71.01	112.38	10,744.0	-122.7	357.8	361.6	12.00	12.00	0.00
Threeforks									
10,900.0	72.86	112.38	10,748.8	-128.2	371.4	375.4	12.00	12.00	0.00
10,925.0	75.86	112.38	10,755.5	-137.4	393.7	397.9	12.00	12.00	0.00
10,950.0	78.86	112.38	10,761.0	-146.7	416.2	420.7	12.00	12.00	0.00
10,950.1	78.86	112.38	10,761.0	-146.7	416.3	420.8	0.00	0.00	0.00
Threeforks(Top of Target)									
10,975.0	81.86	112.38	10,765.2	-156.1	439.0	443.8	12.03	12.03	0.00
11,000.0	84.86	112.38	10,768.1	-165.5	462.0	467.1	12.00	12.00	0.00
11,025.0	87.86	112.38	10,769.7	-175.0	485.0	490.4	12.00	12.00	0.00
11,039.9	89.65	112.38	10,770.0	-180.7	498.8	504.4	12.00	12.00	0.00
Start 1.3 hold at 11039.9 MD									
11,041.0	89.65	112.38	10,770.0	-181.1	499.8	505.4	0.00	0.00	0.00
7"									
11,041.2	89.65	112.38	10,770.0	-181.2	500.0	505.6	0.00	0.00	0.00
Start DLS 3.02 TFO 269.93 7" csg pt									
11,044.7	89.65	112.38	10,770.0	-182.5	503.3	508.9	0.00	0.00	0.00
Threeforks(Base of Target)									
11,100.0	89.65	110.61	10,770.4	-202.8	554.7	561.0	3.21	0.00	-3.21
11,200.0	89.65	107.59	10,771.0	-235.5	649.2	656.5	3.02	0.00	-3.02
11,300.0	89.64	104.57	10,771.6	-263.1	745.3	753.4	3.02	0.00	-3.02
11,400.0	89.64	101.55	10,772.2	-285.7	842.7	851.5	3.02	0.00	-3.02
11,500.0	89.64	98.53	10,772.8	-303.2	941.1	950.4	3.02	0.00	-3.02
11,600.0	89.64	95.51	10,773.5	-315.4	1,040.4	1,050.0	3.02	0.00	-3.02
11,700.0	89.65	92.49	10,774.1	-322.3	1,140.1	1,149.9	3.02	0.00	-3.02
11,782.3	89.65	90.00	10,774.6	-324.1	1,222.4	1,232.2	3.02	0.00	-3.02
Start 8807.7 hold at 11782.3 MD									

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Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
11,800.0	89.65	90.00	10,774.7	-324.1	1,240.1	1,249.9	0.00	0.00	0.00
11,900.0	89.65	90.00	10,775.3	-324.1	1,340.1	1,349.9	0.00	0.00	0.00
12,000.0	89.65	90.00	10,775.9	-324.1	1,440.1	1,449.8	0.00	0.00	0.00
12,100.0	89.65	90.00	10,776.5	-324.1	1,540.1	1,549.8	0.00	0.00	0.00
12,200.0	89.65	90.00	10,777.1	-324.1	1,640.1	1,649.7	0.00	0.00	0.00
12,300.0	89.65	90.00	10,777.8	-324.1	1,740.1	1,749.6	0.00	0.00	0.00
12,400.0	89.65	90.00	10,778.4	-324.1	1,840.1	1,849.6	0.00	0.00	0.00
12,500.0	89.65	90.00	10,779.0	-324.1	1,940.1	1,949.5	0.00	0.00	0.00
12,600.0	89.65	90.00	10,779.6	-324.1	2,040.1	2,049.5	0.00	0.00	0.00
12,700.0	89.65	90.00	10,780.2	-324.1	2,140.1	2,149.4	0.00	0.00	0.00
12,800.0	89.65	90.00	10,780.8	-324.1	2,240.1	2,249.4	0.00	0.00	0.00
12,900.0	89.65	90.00	10,781.4	-324.1	2,340.1	2,349.3	0.00	0.00	0.00
13,000.0	89.65	90.00	10,782.0	-324.1	2,440.1	2,449.3	0.00	0.00	0.00
13,100.0	89.65	90.00	10,782.6	-324.1	2,540.1	2,549.2	0.00	0.00	0.00
13,200.0	89.65	90.00	10,783.3	-324.1	2,640.1	2,649.2	0.00	0.00	0.00
13,300.0	89.65	90.00	10,783.9	-324.1	2,740.1	2,749.1	0.00	0.00	0.00
13,400.0	89.65	90.00	10,784.5	-324.1	2,840.1	2,849.1	0.00	0.00	0.00
13,500.0	89.65	90.00	10,785.1	-324.1	2,940.1	2,949.0	0.00	0.00	0.00
13,600.0	89.65	90.00	10,785.7	-324.1	3,040.1	3,048.9	0.00	0.00	0.00
13,700.0	89.65	90.00	10,786.3	-324.1	3,140.1	3,148.9	0.00	0.00	0.00
13,800.0	89.65	90.00	10,786.9	-324.1	3,240.1	3,248.8	0.00	0.00	0.00
13,900.0	89.65	90.00	10,787.5	-324.1	3,340.1	3,348.8	0.00	0.00	0.00
14,000.0	89.65	90.00	10,788.1	-324.1	3,440.1	3,448.7	0.00	0.00	0.00
14,100.0	89.65	90.00	10,788.8	-324.1	3,540.1	3,548.7	0.00	0.00	0.00
14,200.0	89.65	90.00	10,789.4	-324.1	3,640.1	3,648.6	0.00	0.00	0.00
14,300.0	89.65	90.00	10,790.0	-324.1	3,740.0	3,748.6	0.00	0.00	0.00
14,400.0	89.65	90.00	10,790.6	-324.1	3,840.0	3,848.5	0.00	0.00	0.00
14,500.0	89.65	90.00	10,791.2	-324.1	3,940.0	3,948.5	0.00	0.00	0.00
14,600.0	89.65	90.00	10,791.8	-324.1	4,040.0	4,048.4	0.00	0.00	0.00
14,700.0	89.65	90.00	10,792.4	-324.1	4,140.0	4,148.3	0.00	0.00	0.00
14,800.0	89.65	90.00	10,793.0	-324.1	4,240.0	4,248.3	0.00	0.00	0.00
14,900.0	89.65	90.00	10,793.6	-324.1	4,340.0	4,348.2	0.00	0.00	0.00
15,000.0	89.65	90.00	10,794.3	-324.1	4,440.0	4,448.2	0.00	0.00	0.00
15,100.0	89.65	90.00	10,794.9	-324.1	4,540.0	4,548.1	0.00	0.00	0.00
15,200.0	89.65	90.00	10,795.5	-324.1	4,640.0	4,648.1	0.00	0.00	0.00
15,300.0	89.65	90.00	10,796.1	-324.1	4,740.0	4,748.0	0.00	0.00	0.00
15,400.0	89.65	90.00	10,796.7	-324.1	4,840.0	4,848.0	0.00	0.00	0.00
15,500.0	89.65	90.00	10,797.3	-324.1	4,940.0	4,947.9	0.00	0.00	0.00
15,600.0	89.65	90.00	10,797.9	-324.1	5,040.0	5,047.9	0.00	0.00	0.00
15,700.0	89.65	90.00	10,798.5	-324.1	5,140.0	5,147.8	0.00	0.00	0.00
15,800.0	89.65	90.00	10,799.2	-324.1	5,240.0	5,247.8	0.00	0.00	0.00
15,900.0	89.65	90.00	10,799.8	-324.1	5,340.0	5,347.7	0.00	0.00	0.00
16,000.0	89.65	90.00	10,800.4	-324.1	5,440.0	5,447.6	0.00	0.00	0.00
16,100.0	89.65	90.00	10,801.0	-324.1	5,540.0	5,547.6	0.00	0.00	0.00
16,200.0	89.65	90.00	10,801.6	-324.1	5,640.0	5,647.5	0.00	0.00	0.00
16,300.0	89.65	90.00	10,802.2	-324.1	5,740.0	5,747.5	0.00	0.00	0.00
16,400.0	89.65	90.00	10,802.8	-324.1	5,840.0	5,847.4	0.00	0.00	0.00
16,500.0	89.65	90.00	10,803.4	-324.1	5,940.0	5,947.4	0.00	0.00	0.00
16,600.0	89.65	90.00	10,804.0	-324.1	6,040.0	6,047.3	0.00	0.00	0.00
16,700.0	89.65	90.00	10,804.7	-324.1	6,140.0	6,147.3	0.00	0.00	0.00
16,800.0	89.65	90.00	10,805.3	-324.1	6,240.0	6,247.2	0.00	0.00	0.00
16,900.0	89.65	90.00	10,805.9	-324.1	6,340.0	6,347.2	0.00	0.00	0.00
17,000.0	89.65	90.00	10,806.5	-324.1	6,440.0	6,447.1	0.00	0.00	0.00
17,100.0	89.65	90.00	10,807.1	-324.1	6,540.0	6,547.1	0.00	0.00	0.00
17,200.0	89.65	90.00	10,807.7	-324.1	6,640.0	6,647.0	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
17,300.0	89.65	90.00	10,808.3	-324.1	6,740.0	6,746.9	0.00	0.00	0.00
17,400.0	89.65	90.00	10,808.9	-324.1	6,840.0	6,846.9	0.00	0.00	0.00
17,500.0	89.65	90.00	10,809.5	-324.1	6,940.0	6,946.8	0.00	0.00	0.00
17,600.0	89.65	90.00	10,810.2	-324.1	7,040.0	7,046.8	0.00	0.00	0.00
17,700.0	89.65	90.00	10,810.8	-324.1	7,140.0	7,146.7	0.00	0.00	0.00
17,800.0	89.65	90.00	10,811.4	-324.1	7,240.0	7,246.7	0.00	0.00	0.00
17,900.0	89.65	90.00	10,812.0	-324.1	7,340.0	7,346.6	0.00	0.00	0.00
18,000.0	89.65	90.00	10,812.6	-324.1	7,440.0	7,446.6	0.00	0.00	0.00
18,100.0	89.65	90.00	10,813.2	-324.1	7,540.0	7,546.5	0.00	0.00	0.00
18,200.0	89.65	90.00	10,813.8	-324.1	7,640.0	7,646.5	0.00	0.00	0.00
18,300.0	89.65	90.00	10,814.4	-324.1	7,740.0	7,746.4	0.00	0.00	0.00
18,400.0	89.65	90.00	10,815.0	-324.1	7,840.0	7,846.4	0.00	0.00	0.00
18,500.0	89.65	90.00	10,815.7	-324.1	7,940.0	7,946.3	0.00	0.00	0.00
18,600.0	89.65	90.00	10,816.3	-324.1	8,040.0	8,046.2	0.00	0.00	0.00
18,700.0	89.65	90.00	10,816.9	-324.1	8,140.0	8,146.2	0.00	0.00	0.00
18,800.0	89.65	90.00	10,817.5	-324.1	8,240.0	8,246.1	0.00	0.00	0.00
18,900.0	89.65	90.00	10,818.1	-324.1	8,340.0	8,346.1	0.00	0.00	0.00
19,000.0	89.65	90.00	10,818.7	-324.1	8,440.0	8,446.0	0.00	0.00	0.00
19,100.0	89.65	90.00	10,819.3	-324.1	8,540.0	8,546.0	0.00	0.00	0.00
19,200.0	89.65	90.00	10,819.9	-324.1	8,640.0	8,645.9	0.00	0.00	0.00
19,300.0	89.65	90.00	10,820.5	-324.1	8,740.0	8,745.9	0.00	0.00	0.00
19,400.0	89.65	90.00	10,821.2	-324.1	8,840.0	8,845.8	0.00	0.00	0.00
19,500.0	89.65	90.00	10,821.8	-324.1	8,940.0	8,945.8	0.00	0.00	0.00
19,600.0	89.65	90.00	10,822.4	-324.1	9,039.9	9,045.7	0.00	0.00	0.00
19,700.0	89.65	90.00	10,823.0	-324.1	9,139.9	9,145.6	0.00	0.00	0.00
19,800.0	89.65	90.00	10,823.6	-324.1	9,239.9	9,245.6	0.00	0.00	0.00
19,900.0	89.65	90.00	10,824.2	-324.1	9,339.9	9,345.5	0.00	0.00	0.00
20,000.0	89.65	90.00	10,824.8	-324.1	9,439.9	9,445.5	0.00	0.00	0.00
20,100.0	89.65	90.00	10,825.4	-324.1	9,539.9	9,545.4	0.00	0.00	0.00
20,200.0	89.65	90.00	10,826.1	-324.1	9,639.9	9,645.4	0.00	0.00	0.00
20,300.0	89.65	90.00	10,826.7	-324.1	9,739.9	9,745.3	0.00	0.00	0.00
20,400.0	89.65	90.00	10,827.3	-324.1	9,839.9	9,845.3	0.00	0.00	0.00
20,500.0	89.65	90.00	10,827.9	-324.1	9,939.9	9,945.2	0.00	0.00	0.00
20,590.1	89.65	90.00	10,828.4	-324.1	10,030.0	10,035.2	0.00	0.00	0.00
TD at 20590.1									

Casing Points

Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)
2,050.0	2,050.0	13 3/8"	13.375	17.500
11,041.0	10,770.0	7"	7.000	8.750

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Formations

Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,920.0	1,920.0	Pierre			
4,566.1	4,566.0	Greenhorn			
4,969.1	4,969.0	Mowry			
5,391.1	5,391.0	Dakota			
6,407.2	6,407.0	Rierdon			
6,896.2	6,896.0	Dunham Salt			
6,942.2	6,942.0	Dunham Salt Base			
7,205.2	7,205.0	Pine Salt			
7,229.2	7,229.0	Pine Salt Base			
7,291.2	7,291.0	Opeche Salt			
7,371.2	7,371.0	Opeche Salt Base			
7,615.2	7,615.0	Amsden			
7,771.2	7,771.0	Tyler			
7,994.2	7,994.0	Otter/Base Minnelusa			
8,336.3	8,336.0	Kibbey Lime			
8,484.3	8,484.0	Charles Salt			
9,163.3	9,163.0	Base Last Salt			
9,377.3	9,377.0	Mission Canyon			
9,926.3	9,926.0	Lodgepole			
10,705.9	10,656.0	False Bakken			
10,724.9	10,668.0	Upper Bakken Shale			
10,750.0	10,683.0	Middle Bakken			
10,820.1	10,719.0	Lower Bakken Shale			
10,841.1	10,728.0	Pronghorn			
10,884.6	10,744.0	Threeforks			
10,950.1	10,761.0	Threeforks(Top of Target)			
11,044.7	10,770.0	Threeforks(Base of Target)			

Plan Annotations

Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates +N/-S (ft)	+E/-W (ft)	Comment
2,020.0	2,020.0	0.0	0.0	Start Build 5.00
2,030.0	2,030.0	0.0	0.0	Start 6865.6 hold at 2030.0 MD
8,895.6	8,895.3	0.0	60.0	Start Drop -5.00
8,905.6	8,905.3	0.0	60.0	Start 1094.7 hold at 8905.6 MD
10,000.3	10,000.0	0.0	60.0	Start 292.5 hold at 10000.3 MD
10,292.8	10,292.5	0.0	60.0	Start Build 12.00 KOP
11,039.9	10,770.0	-180.7	498.8	Start 1.3 hold at 11039.9 MD
11,041.2	10,770.0	-181.2	500.0	Start DLS 3.02 TFO 269.93 7" csg pt
11,782.3	10,774.6	-324.1	1,222.4	Start 8807.7 hold at 11782.3 MD
20,590.1	10,828.4	-324.1	10,030.0	TD at 20590.1

Azimuths to True North
Magnetic North: 8.18°

Magnetic Field
Strength: 56488.5snT
Dip Angle: 72.95°
Date: 1/28/2014
Model: IGRF200510

OASIS PETROLEUM

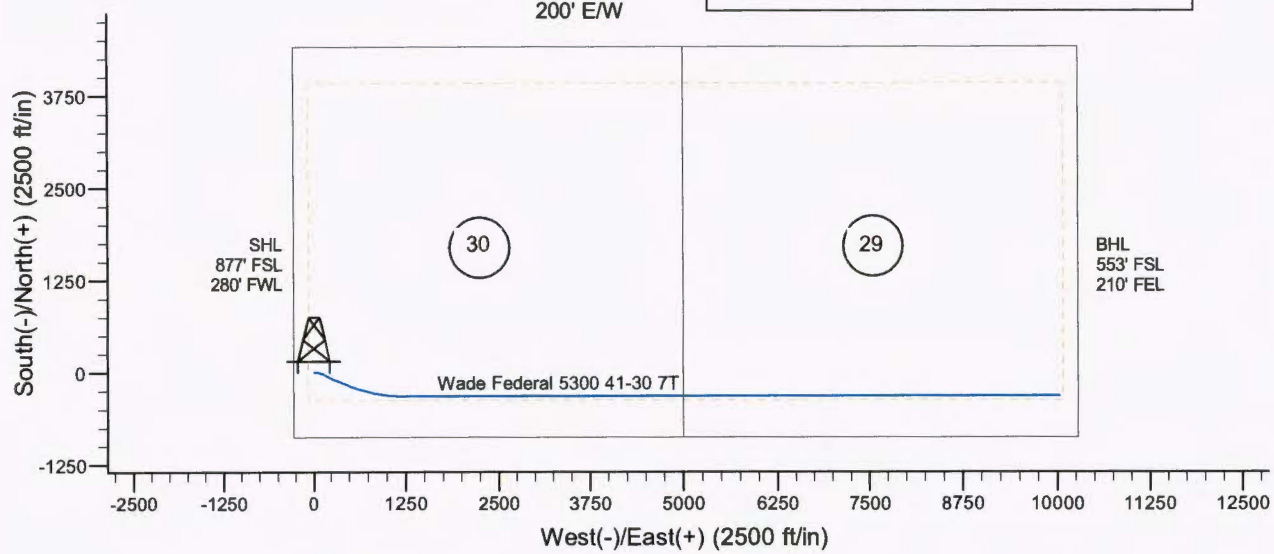
Project: Indian Hills
Site: 153N-100W-29/30
Well: Wade Federal 5300 41-30 7T
Wellbore: Wade Federal 5300 41-30 7T
Design: Design #1

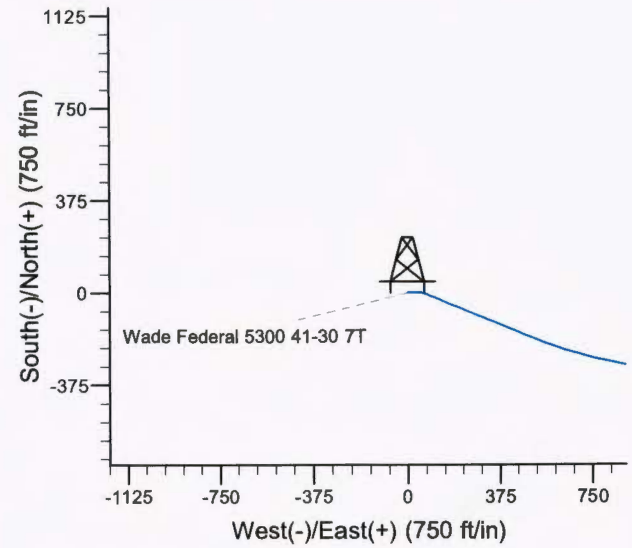
SITE DETAILS: 153N-100W-29/30

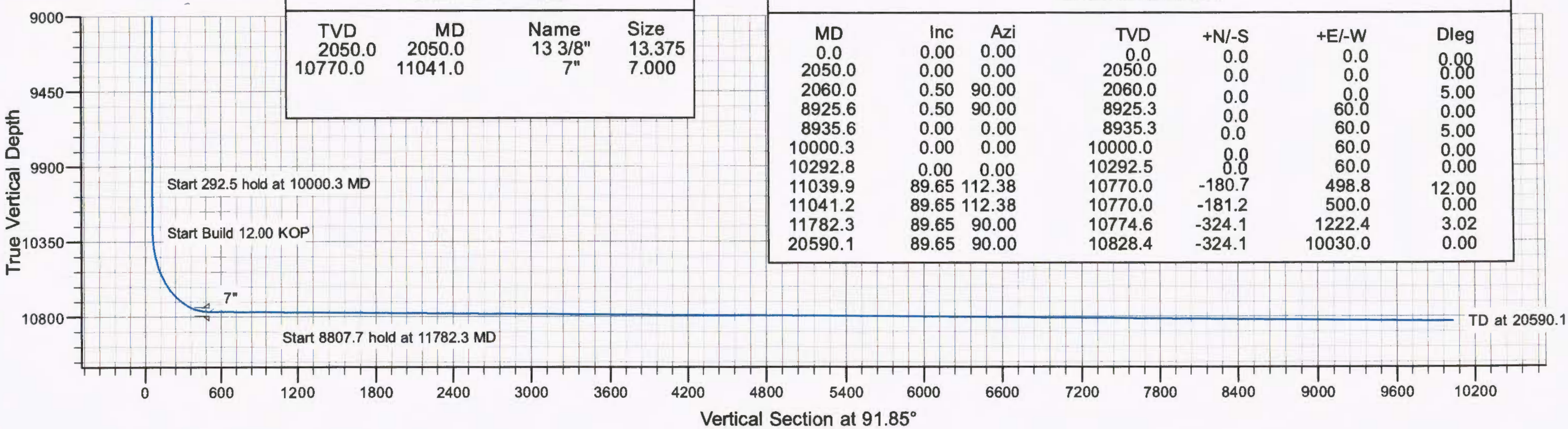
Site Centre Latitude: 48° 2' 28.320 N
Longitude: 103° 36' 10.820 W

Positional Uncertainity: 0.0
Convergence: -2.31
Local North: True

Setbacks
500' N/S
200' E/W







CASING DETAILS

TVD	MD	Name	Size
2050.0	2050.0	13 3/8"	13.375
10770.0	11041.0	7"	7.000

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg
0.0	0.00	0.00	0.0	0.0	0.0	0.00
2050.0	0.00	0.00	2050.0	0.0	0.0	0.00
2060.0	0.50	90.00	2060.0	0.0	0.0	5.00
8925.6	0.50	90.00	8925.3	0.0	60.0	0.00
8935.6	0.00	0.00	8935.3	0.0	60.0	5.00
10000.3	0.00	0.00	10000.0	0.0	60.0	0.00
10292.8	0.00	0.00	10292.5	0.0	60.0	0.00
11039.9	89.65	112.38	10770.0	-180.7	498.8	12.00
11041.2	89.65	112.38	10770.0	-181.2	500.0	0.00
11782.3	89.65	90.00	10774.6	-324.1	1222.4	3.02
20590.1	89.65	90.00	10828.4	-324.1	10030.0	0.00

Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 7T
Section 30 T153N R100W
McKenzie County, ND

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
13-3/8"	0' to 2,050'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 2,050'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.18	2730 / 2.84	514 / 2.62

API Rating & Safety Factor

a) Based on full casing evacuation with 9 ppg fluid on backside (2,050' setting depth).
b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,050' setting depth).
c) Based on string weight in 9 ppg fluid at 2,050' TVD plus 100k# overpull. (Buoyed weight equals 96k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **596 sks** (308 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% $CaCl_2$, 4% D079 Extender and 2% D053 Expanding Agent.

Tail Slurry: **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25% $CaCl_2$ and 0.25 lb/sk Lost Circulation Control Agent

Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 7T
Section 30 T153N R100W
McKenzie County, ND

CONTINGENCY SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
9-5/8"	0' to 6,400'	40	L-80	LTC	8.835"	8.75"	5,450	7,270	9,090

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 6,400'	9-5/8", 40#, L-80, LTC, 8rd	3090 / 3.71	5750 / 1.24	837 / 3.86

API Rating & Safety Factor

a) Collapse pressure based on 11.5 ppg fluid on the backside and 9 ppg fluid inside of casing.

b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000 psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft facture gradient. Backup of 9 ppg fluid.

c) Yield based on string weight in 10 ppg fluid, (217k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are based on 9-5/8" casing set in 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **592 sks** (210 bbls) Conventional system with 75 lb/sk cement, 0.5 lb/sk lost circulation, 10% expanding agent, 2% extender, 2% $CaCl_2$, 0.2% anti-foam and 0.4% fluid loss agent.

Tail Slurry: **521 sks** (108 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti-foam and 0.1% retarder.

Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 7T
Section 30 T153N R100W
McKenzie County, ND

INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
7"	0' – 11,041'	32	HCP-110	LTC	6.094"	6.000"	6730	8970	9870

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 11,041'	11,041'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.11*	12460 / 1.28	897 / 2.24
6,696' – 9,163'	2,467'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07*	12460 / 1.30	

API Rating & Safety Factor

a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.

b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,770' TVD.

c) Based on string weight in 10 ppg fluid, (290k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls Saltwater**
40 bbls Weighted MudPush Express

Lead Slurry: **185 sks** (85 bbls) 2.55 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 extender, 3.0% KCl, 3.0% D154 extender, 0.3% D208 viscosifier, 0.07% retarder, 0.2% anti-foam, 0.5 lb/sk, D130 LCM.

Tail Slurry: **575 sks** (168 bbls) 1.55 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% retarder, 0.2% fluid loss, 0.2% anti-foam and 0.5 lb/sk LCM.

Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 7T
Section 30 T153N R100W
McKenzie County, ND

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Estimated Torque
4-1/2"	10,243' – 20,591'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	Description	Collapse	Burst	Tension
		(psi) **a**	(psi) **b**	(1000 lbs) **c**
10,243' – 20,591'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10670 / 1.99	12410 / 1.28	443 / 2.01

<u>API Rating & Safety Factor</u>

a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,828' TVD.

b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10,828' TVD.

c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 120k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)

NORTH DAKOTA
OIL AND GAS DIVISION

Oil and Gas Division

Lynn D. Helms - Director   Bruce E. Hicks - Assistant Director
Department of Mineral Resources
Lynn D. Helms - Director
North Dakota Industrial Commission
www.dmr.nd.gov/oilgas

28557

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: **6/9/2014**

RE: CORES AND SAMPLES

Well Name:	**WADE FEDERAL 5300 41-30 7T** Well File No.: **28557**
Location:	**LOT4 30-153-100** County: MCKENZIE
Permit Type:	**Development - HORIZONTAL**
Field:	**BAKER** Target Horizon: THREE FORKS B1

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.

2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
   - Sample cuttings shall be collected at:
     - o 30' maximum intervals through all vertical and build sections.
     - o 100' maximum intervals through any horizontal sections.
   - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
   - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.

3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed $12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Stephen Fried
Geologist

June 5, 2014

Brandi Terry
REGULATORY SPECIALIST
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Street, Suite 1500
Houston, TX 77002

RE: HORIZONTAL WELL
WADE FEDERAL 5300 41-30 7T
LOT4 Section 30-153N-100W
McKenzie County
Well File # 28557

Dear Brandi:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500'** setback from the north & south boundaries and **200'** setback from the east & west boundaries within the 1280 acre spacing unit consisting of SECTIONS 29 & 30 T153N R100W. **Tool error is not required pursuant to order.**

PERMIT STIPULATIONS: Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. Due to drainage adjacent to the well site, a dike is required surrounding the entire location. OASIS PETROLEUM NORTH AMERICA LLC must take into consideration NDAC 43-02-03-28 (Safety Regulation) when contemplating simultaneous operations on the above captioned location. Pursuant to NDAC 43-02-03-28 (Safety Regulation) "No boiler, portable electric lighting generator, or treater shall be placed nearer than 150 feet to any producing well or oil tank." OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. Based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 377'S and 10042'E.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of $100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Alice Webber
Engineering Tech

APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work	Type of Well	Approximate Date Work Will Start	Confidential Status
New Location	**Oil & Gas**	**5 / 30 / 2014**	**No**

Operator	Telephone Number
OASIS PETROLEUM NORTH AMERICA LLC	**281-404-9491**

Address	City	State	Zip Code
1001 Fannin Street, Suite 1500	**Houston**	**TX**	**77002**

☑ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☑ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name	Well Number
WADE FEDERAL	**5300 41-30 7T**

Surface Footages		Qtr-Qtr	Section	Township	Range	County
877 F **S** L	**280** F **W** L	**LOT4**	**30**	**153 N**	**100 W**	**McKenzie**

Longstring Casing Point Footages		Qtr-Qtr	Section	Township	Range	County
696 F **S** L	**780** F **W** L	**LOT4**	**30**	**153 N**	**100 W**	**McKenzie**

Longstring Casing Point Coordinates From Well Head		Azimuth	Longstring Total Depth	
181 S From WH	**500 E** From WH	**112.3 °**	**11041** Feet MD	**10770** Feet TVD

Bottom Hole Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
553 F **S** L	**212** F **E** L	**LOT6**	**29**	**153 N**	**100 W**	**Williams**

Bottom Hole Coordinates From Well Head		KOP Lateral 1	Azimuth Lateral 1	Estimated Total Depth Lateral 1	
324 S From WH	**10030 E** From WH	**10292** Feet MD	**90.0 °**	**20590** Feet MD	**10828** Feet TVD

Latitude of Well Head	Longitude of Well Head	NAD Reference	Description of Spacing Unit: (Subject to NDIC Approval)
48 ° 02 ' 28.32 "	**-103 ° 36 ' 10.82 "**	**NAD83**	**SECTIONS 29 & 30 T153N R100W**

Ground Elevation	Acres in Spacing/Drilling Unit	Spacing/Drilling Unit Setback Requirement		Industrial Commission Order
2069 Feet Above S.L.	**1280**	**500** Feet N/S	**200** Feet E/W	**23752**

North Line of Spacing/Drilling Unit	South Line of Spacing/Drilling Unit	East Line of Spacing/Drilling Unit	West Line of Spacing/Drilling Unit
10513 Feet	**10522** Feet	**5082** Feet	**5236** Feet

Objective Horizons					Pierre Shale Top
Three Forks B1					**1920**

Proposed Surface Casing	Size	Weight	Depth	Cement Volume	
	9 - 5/8 "	**36** Lb./Ft.	**2050** Feet	**598** Sacks	**NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.**

Proposed Longstring Casing	Size	Weight(s)	Longstring Total Depth		Cement Volume	Cement Top	Top Dakota Sand
	7 - "	**32** Lb./Ft.	**11041** Feet MD	**10770** Feet TVD	**760** Sacks	**3922** Feet	**5391** Feet

Base Last Charles Salt (If Applicable)	
9163 Feet	**NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.**

Proposed Logs
TRIPLE COMBO: KOP TO KIBBEY GR/RES TO BSC GR TO SURF CND THROUGH THE DAKOTA

Drilling Mud Type (Vertical Hole - Below Surface Casing)	Drilling Mud Type (Lateral)
Invert	**Salt Water Gel**

Survey Type in Vertical Portion of Well	Survey Frequency: Build Section	Survey Frequency: Lateral	Survey Contractor
MWD Every 100 Feet	**30** Feet	**90** Feet	**RYAN**

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, $100 fee.

See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Lateral 2

KOP Lateral 2	Azimuth Lateral 2	Estimated Total Depth Lateral 2		KOP Coordinates From Well Head	
Feet MD	°	Feet MD	Feet TVD	From WH	From WH

Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head	
From WH	From WH	From WH	From WH

KOP Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	

Bottom Hole Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	

Lateral 3

KOP Lateral 3	Azimuth Lateral 3	Estimated Total Depth Lateral 3		KOP Coordinates From Well Head	
Feet MD	°	Feet MD	Feet TVD	From WH	From WH

Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head	
From WH	From WH	From WH	From WH

KOP Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	

Bottom Hole Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	

Lateral 4

KOP Lateral 4	Azimuth Lateral 4	Estimated Total Depth Lateral 4		KOP Coordinates From Well Head	
Feet MD	°	Feet MD	Feet TVD	From WH	From WH

Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head	
From WH	From WH	From WH	From WH

KOP Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	

Bottom Hole Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	

Lateral 5

KOP Lateral 5	Azimuth Lateral 5	Estimated Total Depth Lateral 5		KOP Coordinates From Well Head	
Feet MD	°	Feet MD	Feet TVD	From WH	From WH

Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head	
From WH	From WH	From WH	From WH

KOP Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	

Bottom Hole Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 03 / 31 / 2014
ePermit	Printed Name **Brandi Terry**	Title **REGULATORY SPECIALIST**

FOR STATE USE ONLY

Permit and File Number	API Number
28557	**33 - 053 - 05998**
Field **BAKER**	
Pool **BAKKEN**	Permit Type **DEVELOPMENT**

FOR STATE USE ONLY

Date Approved
6 / 5 / 2014
By **Alice Webber**
Title **Engineering Tech**

NORTH DAKOTA
OIL AND GAS DIVISION

Oil and Gas Division

Lynn D. Helms - Director   Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

RE: **Filter Socks and Other Filter Media**
Leakproof Container Required
Oil and Gas Wells

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

Thank you for your cooperation.

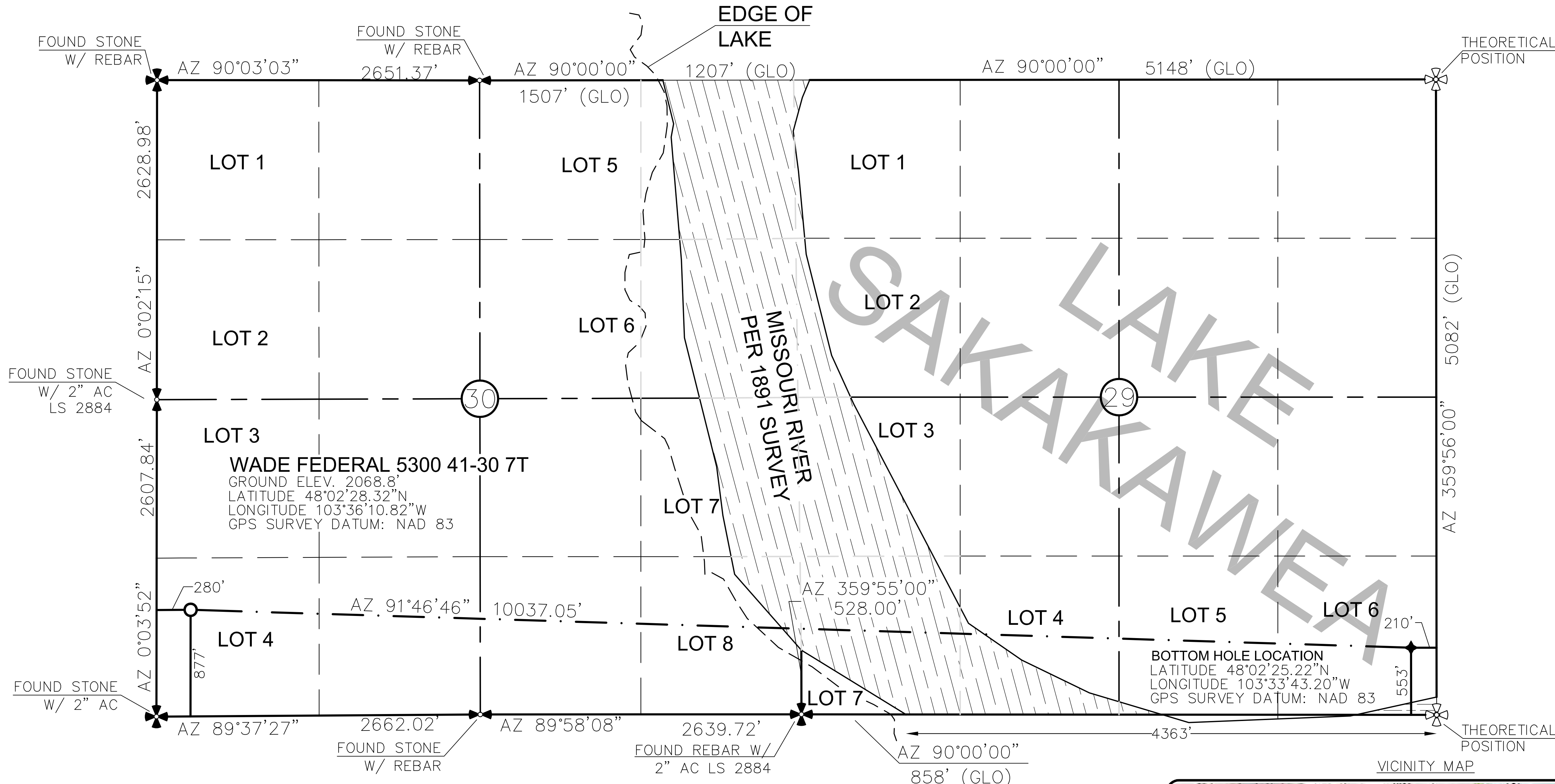
Sincerely,

Bruce E. Hicks
Assistant Director

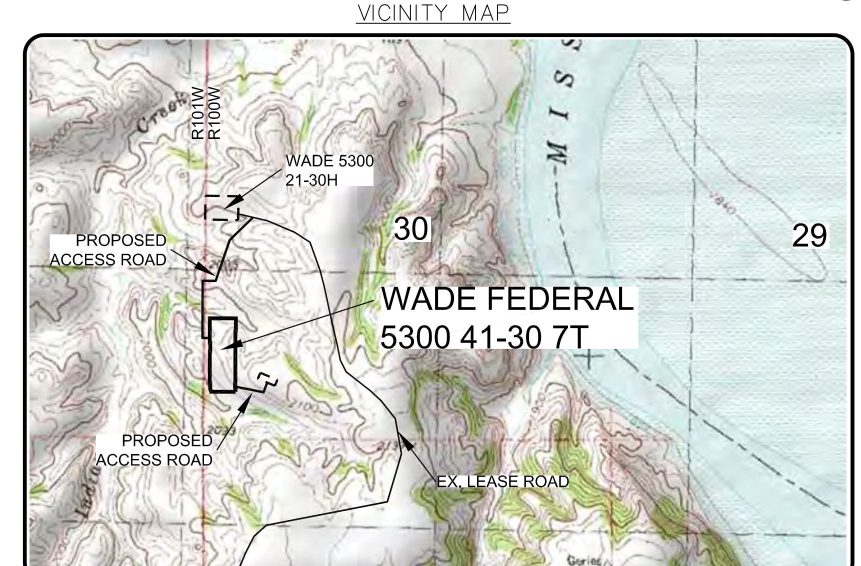
600 E Boulevard Ave - Dept 405, Bismarck, North Dakota 58505-0840 Phone (701)328-8020 Fax (701)328-8022

WELL LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 7T"
877 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



VICINITY MAP



0 1000
1" = 1000'

MONUMENT – RECOVERED
MONUMENT – NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 3/18/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

STAKED ON 1/10/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/18/14
NORTH DAKOTA

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

Q:\2013\S13-09-381.06 Oasis Petroleum – Wade 5300 41-30 7T Well Location 8-Well Pad\CAD\REVISED WADE 7T.dwg – 3/19/2014 6:59 AM josh schmierer

OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.06
Checked By: D.D.K. Date: JAN. 2014

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

INTERSTATE ENGINEERING
Professionals you need, people you trust

SHEET NO. 1/8

DRILLING PLAN

OPERATOR	Oasis Petroleum	**COUNTY/STATE**	McKenzie Co. , ND
WELL NAME	Wade Federal 5300 41-30 7T	**RIG**	Nabors 149
WELL TYPE	Horizontal Three Forks		
LOCATION	SW SW 30-153N-100W	Surface Location (survey plat): 877' FSL	280' FWL
EST. T.D.	20,590'	**GROUND ELEV:** 2,045'	**Sub Height:** 25'
TOTAL LATERAL:	9,549'	**KB ELEV:** 2,070'	

MARKER		**TVD**	**Subsea TVD**
Pierre	NDIC MAP	1,920	150
Greenhorn		4,566	-2,496
Mowry		4,969	-2,899
Dakota		5,391	-3,321
Rierdon		6,407	-4,337
Dunham Salt		6,896	-4,826
Dunham Salt Base		6,942	-4,872
Pine Salt		7,205	-5,135
Pine Salt Base		7,229	-5,159
Opeche Salt		7,291	-5,221
Opeche Salt Base		7,371	-5,301
Amsden		7,615	-5,545
Tyler		7,771	-5,701
Otter/Base Minnelusa		7,994	-5,924
Kibbey Lime		8,336	-6,266
Charles Salt		8,484	-6,414
Base Last Salt		9,163	-7,093
Mission Canyon		9,377	-7,307
Lodgepole		9,926	-7,856
False Bakken		10,656	-8,586
Upper Bakken Shale		10,668	-8,598
Middle Bakken		10,683	-8,613
Lower Bakken Shale		10,719	-8,649
Pronghorn		10,728	-8,658
Threeforks		10,744	-8,674
Threeforks(Top of Target)		10,761	-8,691
Threeforks(Base of Target)		10,770	-8,700
Claystone		10,770	-8,700
Est. Dip Rate:	-0.35		
Max. Anticipated BHP:	4667		

LOGS:	**Type**	**Interval**
OH Logs:	Request Log waiver based on the Wade Federal 5300 21-30H 2,150' N of surface location	
CBL/GR:	Above top of cement/GR to base of casing	
MWD GR:	KOP to lateral TD	
DEVIATION:	Surf:	3 deg. max., 1 deg / 100'; srvy every 500'
	Prod:	5 deg. max., 1 deg / 100'; srvy every 100'
DST'S:	None planned	
CORES:	None planned	
MUDLOGGING:	Two-Man:	Begin 200' above Kibbey 30' samples in curve and lateral
BOP:	11" 5000 psi blind, pipe & annular	

Surface Formation: *Glacial till*

MUD:	**Interval**		**Type**	**WT**	**Vis**	**WL**	**Remarks**
Surface:	0' -	2,050'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks
Intermediate:	2,050' -	11,041'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks
Lateral:	11,041' -	20,590'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks

CASING:	**Size**	**Wt ppf**	**Hole**	**Depth**		**Cement**	**WOC**	**Remarks**
Surface:	9-5/8"	36#	13-1/2"	2,050'		To Surface	12	100' into Pierre
Intermediate:	7"	29/32#	8-3/4"	11,041'		3891	24	1500' above Dakota
Production Liner:	4.5"	11.6#	6"	20,590'	**TOL @ 10,243'**			50' above KOP

PROBABLE PLUGS, IF REQ'D:

OTHER:	**MD**	**TVD**	**FNL/FSL**	**FEL/FWL**	**S-T-R**	**AZI**	
Surface:	2,050	2,050	877 FSL	280 FWL	**30-T153-R100**		Survey Company:
KOP:	10,293'	10,293'	877 FSL	340 FWL	**30-T153-R100**		Build Rate: 12 deg /100'
EOC:	11,039'	10,770'	696 FSL	779 FWL	**30-T153-R100**	112.4	
Casing Point:	11,041'	10,770'	696 FSL	780 FWL	**30-T153-R100**	112.4	
Middle Bakken Lateral TD:	20,590'	10,828'	553 FSL	210 FEL	**29-T153-R100**	90.0	

Comments:

Request Log waiver based on the Wade Federal 5300 21-30H 2,150' N of surface location

No frac string planned

35 packers and 25 sleeves planned 3.6MM lbs 30% ceramic

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)

OASIS PETROLEUM

Geology: N. Gabelman 1/20/2014 **Engineering:** SMG3.26

Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 7T
Secion 30 T153N R100W
McKenzie Co, ND

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
9-5/8"	0' - 2050	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 2050	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.13	3520 / 3.72	453 / 2.78

API Rating & Safety Factor

a) Based on full casing evacuation with 9 ppg fluid on backside 2050 setting depth).
b) Burst pressure based on 9 ppg fluid with no fluid on backside 2050 setting depth).
c) Based on string weight in 9 ppg fluid at 2050 TVD plus 100k# overpull. (Buoyed weight equals 63k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2 " hole with 60% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **425 sks** (225 bbls), 11.5 lb/gal, 2.97 cu. Ft./sk Varicem Cement with 0.125 il/sk Lost Circulation Additive

Tail Slurry: **173 sks** (62 bbls), 13.0 lb/gal, 2.01 cu.ft./sk Varicem with .125 lb/sk Lost Circulation Agent

Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 7T
Secion 30 T153N R100W
McKenzie Co, ND

INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift****	**Minimum**	**Optimum**	**Max**
7"	0' - 11041'	32	HCP-110	LTC	6.094"	6.000"**	6730	8970	9870

****Special Drift 7" 32# to 6.0"**

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' - 11041'	11041'	7", 32#, P-110, LTC, 8rd	11820 / 2.11*	12460 / 1.28	897 / 2.24
6696' - 9163'	2467'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07**	12460 / 1.30	

API Rating & Safety Factor

a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.

b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10770' TVD.

c) Based on string weight in 10 ppg fluid, (300k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.
Mix and pump the following slurry

Pre-flush (Spacer):	**100 bbls**	Saltwater
	20bbls	CW8
	20bbls	Fresh Water
Lead Slurry:	**185 sks**	(85 bbls), 11.8 ppg, 2.55 cu. ft./sk Econocem Cement with .3% Fe-2 and .25 lb/sk Lost Circulation Additive
Tail Slurry:	**575 sks**	(168 bbls), 14.0 ppg, 1.55 cu. ft./sk Extendcem System with .2% HR-5 Retarder and .25 lb/sk Lost Circulation Additive

Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 7T
Secion 30 T153N R100W
McKenzie Co, ND

PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
4-1/2"	10243' - 20591'	13.5	P-110	BTC	3.920"	3.795"	2270	3020	3780

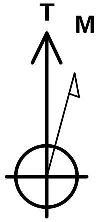
Interval	Length	Desctiption	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
10243' - 20591'	10348	4-1/2", 13.5 lb, P-110, BTC	10670 / 1.99	12410 / 1.28	443 / 2.01

API Rating & Safety Factor

a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10828' TVD.

b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10828' TVD.

c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 120k lbs.) plus 100k lbs overpull.



Azimuths to True North
Magnetic North: 8.18°

Magnetic Field
Strength: 56488.5snT
Dip Angle: 72.95°
Date: 1/28/2014
Model: IGRF200510

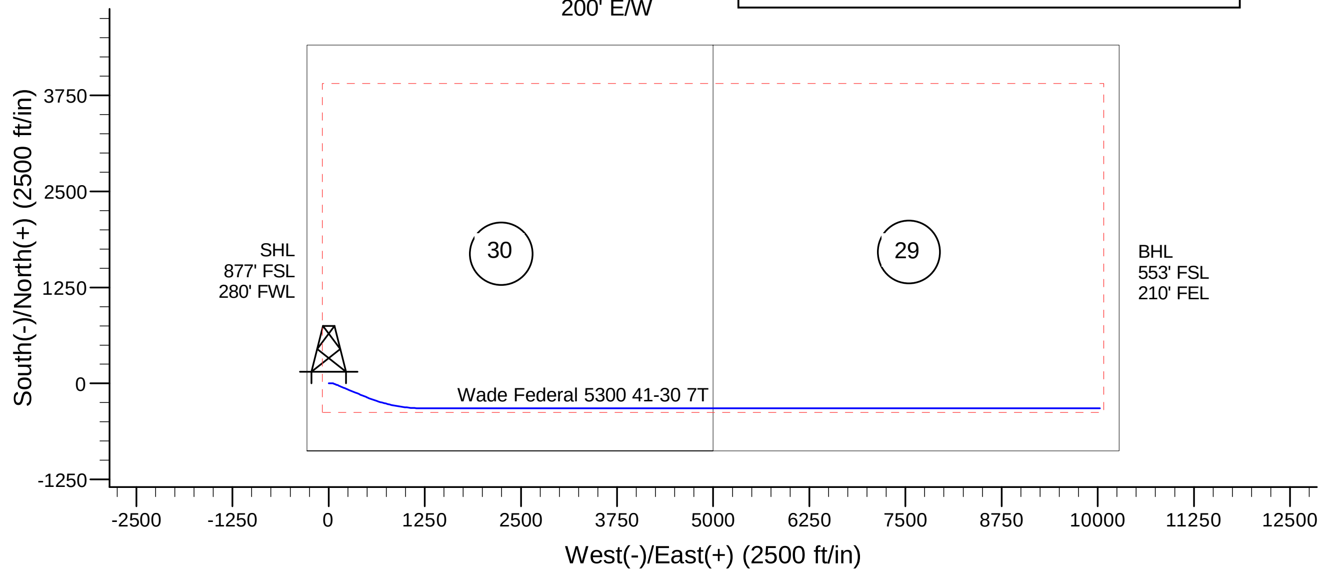
PETROLEUM

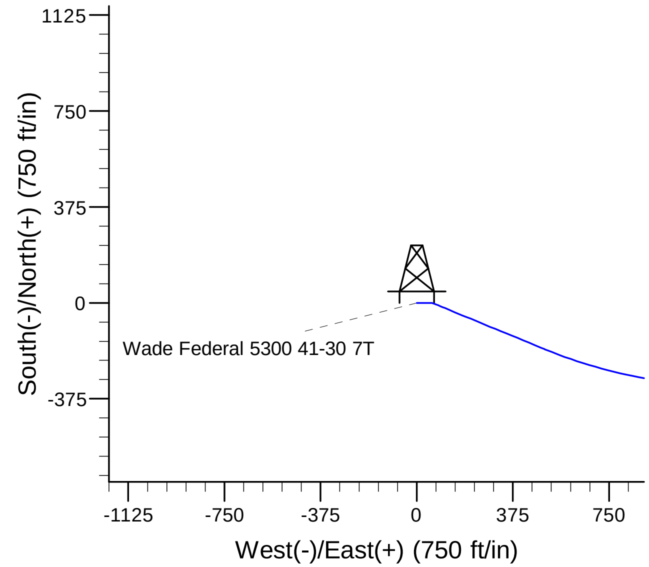
Project: Indian Hills
Site: 153N-100W-29/30
Well: Wade Federal 5300 41-30 7T
Wellbore: Wade Federal 5300 41-30 7T
Design: Design #1

SITE DETAILS: 153N-100W-29/30

Site Centre Latitude: 48° 2' 28.320 N
Longitude: 103° 36' 10.820 W

Positional Uncertainity: 0.0
Convergence: -2.31
Local North: True



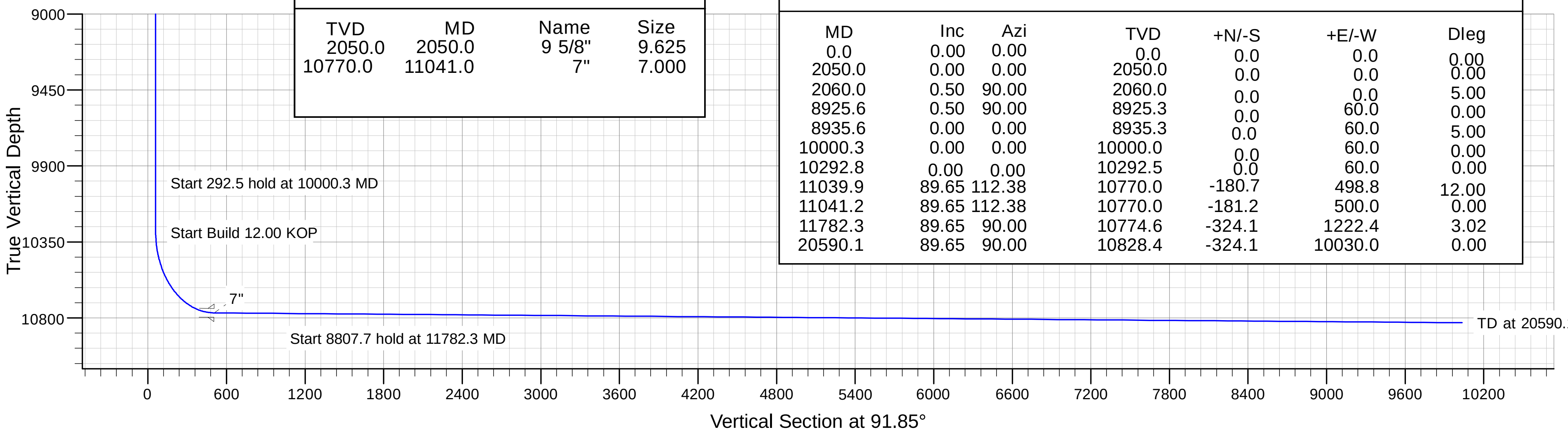


CASING DETAILS

TVD	MD	Name	Size
2050.0	2050.0	9 5/8"	9.625
10770.0	11041.0	7"	7.000

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg
0.0	0.00	0.00	0.0	0.0	0.0	0.00
2050.0	0.00	0.00	2050.0	0.0	0.0	0.00
2060.0	0.50	90.00	2060.0	0.0	0.0	5.00
8925.6	0.50	90.00	8925.3	0.0	60.0	0.00
8935.6	0.00	0.00	8935.3	0.0	60.0	5.00
10000.3	0.00	0.00	10000.0	0.0	60.0	0.00
10292.8	0.00	0.00	10292.5	0.0	60.0	0.00
11039.9	89.65	112.38	10770.0	-180.7	498.8	12.00
11041.2	89.65	112.38	10770.0	-181.2	500.0	0.00
11782.3	89.65	90.00	10774.6	-324.1	1222.4	3.02
20590.1	89.65	90.00	10828.4	-324.1	10030.0	0.00



Oasis

Indian Hills
153N-100W-29/30
Wade Federal 5300 41-30 7T

Wade Federal 5300 41-30 7T

Plan: Design #1

Standard Planning Report

19 May, 2014

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Project Indian Hills

Map System:	US State Plane 1983	**System Datum:**	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site 153N-100W-29/30

Site Position:		**Northing:**	395,521.43 ft	**Latitude:**	48° 2' 32.580 N
From:	Lat/Long	**Easting:**	1,209,621.64 ft	**Longitude:**	103° 36' 11.410 W
Position Uncertainty:	0.0 ft	**Slot Radius:**	13.200 in	**Grid Convergence:**	-2.31 °

Well Wade Federal 5300 41-30 7T

Well Position	**+N/-S**	-431.7 ft	**Northing:**	395,088.50 ft	**Latitude:**	48° 2' 28.320 N
	+E/-W	40.1 ft	**Easting:**	1,209,644.31 ft	**Longitude:**	103° 36' 10.820 W
Position Uncertainty		0.0 ft	**Wellhead Elevation:**		**Ground Level:**	2,045.0 ft

Wellbore Wade Federal 5300 41-30 7T

Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	1/28/2014	8.18	72.95	56,489

Design Design #1

Audit Notes:

Version:		**Phase:**	PROTOTYPE	**Tie On Depth:**	0.0

Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)
	0.0	0.0	0.0	91.85

Plan Sections

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,050.0	0.00	0.00	2,050.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,060.0	0.50	90.00	2,060.0	0.0	0.0	5.00	5.00	0.00	90.00	
8,925.6	0.50	90.00	8,925.3	0.0	60.0	0.00	0.00	0.00	0.00	
8,935.6	0.00	0.00	8,935.3	0.0	60.0	5.00	-5.00	0.00	180.00	
10,000.3	0.00	0.00	10,000.0	0.0	60.0	0.00	0.00	0.00	0.00	
10,292.8	0.00	0.00	10,292.5	0.0	60.0	0.00	0.00	0.00	0.00	
11,039.9	89.65	112.38	10,770.0	-180.7	498.8	12.00	12.00	0.00	112.38	
11,041.2	89.65	112.38	10,770.0	-181.2	500.0	0.00	0.00	0.00	0.00	
11,782.3	89.65	90.00	10,774.6	-324.1	1,222.4	3.02	0.00	-3.02	269.93	
20,590.1	89.65	90.00	10,828.4	-324.1	10,030.0	0.00	0.00	0.00	0.00	

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,920.0	0.00	0.00	1,920.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,020.0	0.00	0.00	2,020.0	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 5.00									
2,030.0	0.00	0.00	2,030.0	0.0	0.0	0.0	0.00	0.00	0.00
Start 6865.6 hold at 2030.0 MD									
2,050.0	0.00	0.00	2,050.0	0.0	0.0	0.0	0.00	0.00	0.00
9 5/8"									
2,060.0	0.50	90.00	2,060.0	0.0	0.0	0.0	5.00	5.00	0.00
2,100.0	0.50	90.00	2,100.0	0.0	0.4	0.4	0.00	0.00	0.00
2,200.0	0.50	90.00	2,200.0	0.0	1.3	1.3	0.00	0.00	0.00
2,300.0	0.50	90.00	2,300.0	0.0	2.1	2.1	0.00	0.00	0.00
2,400.0	0.50	90.00	2,400.0	0.0	3.0	3.0	0.00	0.00	0.00
2,500.0	0.50	90.00	2,500.0	0.0	3.9	3.9	0.00	0.00	0.00
2,600.0	0.50	90.00	2,600.0	0.0	4.8	4.8	0.00	0.00	0.00
2,700.0	0.50	90.00	2,700.0	0.0	5.6	5.6	0.00	0.00	0.00
2,800.0	0.50	90.00	2,800.0	0.0	6.5	6.5	0.00	0.00	0.00
2,900.0	0.50	90.00	2,900.0	0.0	7.4	7.4	0.00	0.00	0.00
3,000.0	0.50	90.00	3,000.0	0.0	8.2	8.2	0.00	0.00	0.00
3,100.0	0.50	90.00	3,100.0	0.0	9.1	9.1	0.00	0.00	0.00
3,200.0	0.50	90.00	3,200.0	0.0	10.0	10.0	0.00	0.00	0.00
3,300.0	0.50	90.00	3,300.0	0.0	10.9	10.9	0.00	0.00	0.00
3,400.0	0.50	90.00	3,399.9	0.0	11.7	11.7	0.00	0.00	0.00
3,500.0	0.50	90.00	3,499.9	0.0	12.6	12.6	0.00	0.00	0.00
3,600.0	0.50	90.00	3,599.9	0.0	13.5	13.5	0.00	0.00	0.00
3,700.0	0.50	90.00	3,699.9	0.0	14.4	14.3	0.00	0.00	0.00
3,800.0	0.50	90.00	3,799.9	0.0	15.2	15.2	0.00	0.00	0.00
3,900.0	0.50	90.00	3,899.9	0.0	16.1	16.1	0.00	0.00	0.00
4,000.0	0.50	90.00	3,999.9	0.0	17.0	17.0	0.00	0.00	0.00
4,100.0	0.50	90.00	4,099.9	0.0	17.8	17.8	0.00	0.00	0.00
4,200.0	0.50	90.00	4,199.9	0.0	18.7	18.7	0.00	0.00	0.00
4,300.0	0.50	90.00	4,299.9	0.0	19.6	19.6	0.00	0.00	0.00
4,400.0	0.50	90.00	4,399.9	0.0	20.5	20.5	0.00	0.00	0.00
4,500.0	0.50	90.00	4,499.9	0.0	21.3	21.3	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
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Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,566.1	0.50	90.00	4,566.0	0.0	21.9	21.9	0.00	0.00	0.00
Greenhorn									
4,600.0	0.50	90.00	4,599.9	0.0	22.2	22.2	0.00	0.00	0.00
4,700.0	0.50	90.00	4,699.9	0.0	23.1	23.1	0.00	0.00	0.00
4,800.0	0.50	90.00	4,799.9	0.0	24.0	23.9	0.00	0.00	0.00
4,900.0	0.50	90.00	4,899.9	0.0	24.8	24.8	0.00	0.00	0.00
4,969.1	0.50	90.00	4,969.0	0.0	25.4	25.4	0.00	0.00	0.00
Mowry									
5,000.0	0.50	90.00	4,999.9	0.0	25.7	25.7	0.00	0.00	0.00
5,100.0	0.50	90.00	5,099.9	0.0	26.6	26.6	0.00	0.00	0.00
5,200.0	0.50	90.00	5,199.9	0.0	27.4	27.4	0.00	0.00	0.00
5,300.0	0.50	90.00	5,299.9	0.0	28.3	28.3	0.00	0.00	0.00
5,391.1	0.50	90.00	5,391.0	0.0	29.1	29.1	0.00	0.00	0.00
Dakota									
5,400.0	0.50	90.00	5,399.9	0.0	29.2	29.2	0.00	0.00	0.00
5,500.0	0.50	90.00	5,499.9	0.0	30.1	30.0	0.00	0.00	0.00
5,600.0	0.50	90.00	5,599.9	0.0	30.9	30.9	0.00	0.00	0.00
5,700.0	0.50	90.00	5,699.9	0.0	31.8	31.8	0.00	0.00	0.00
5,800.0	0.50	90.00	5,799.9	0.0	32.7	32.7	0.00	0.00	0.00
5,900.0	0.50	90.00	5,899.9	0.0	33.6	33.5	0.00	0.00	0.00
6,000.0	0.50	90.00	5,999.8	0.0	34.4	34.4	0.00	0.00	0.00
6,100.0	0.50	90.00	6,099.8	0.0	35.3	35.3	0.00	0.00	0.00
6,200.0	0.50	90.00	6,199.8	0.0	36.2	36.2	0.00	0.00	0.00
6,300.0	0.50	90.00	6,299.8	0.0	37.0	37.0	0.00	0.00	0.00
6,400.0	0.50	90.00	6,399.8	0.0	37.9	37.9	0.00	0.00	0.00
6,407.2	0.50	90.00	6,407.0	0.0	38.0	38.0	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	90.00	6,499.8	0.0	38.8	38.8	0.00	0.00	0.00
6,600.0	0.50	90.00	6,599.8	0.0	39.7	39.6	0.00	0.00	0.00
6,700.0	0.50	90.00	6,699.8	0.0	40.5	40.5	0.00	0.00	0.00
6,800.0	0.50	90.00	6,799.8	0.0	41.4	41.4	0.00	0.00	0.00
6,896.2	0.50	90.00	6,896.0	0.0	42.2	42.2	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.50	90.00	6,899.8	0.0	42.3	42.3	0.00	0.00	0.00
6,942.2	0.50	90.00	6,942.0	0.0	42.6	42.6	0.00	0.00	0.00
Dunham Salt Base									
7,000.0	0.50	90.00	6,999.8	0.0	43.2	43.1	0.00	0.00	0.00
7,100.0	0.50	90.00	7,099.8	0.0	44.0	44.0	0.00	0.00	0.00
7,200.0	0.50	90.00	7,199.8	0.0	44.9	44.9	0.00	0.00	0.00
7,205.2	0.50	90.00	7,205.0	0.0	44.9	44.9	0.00	0.00	0.00
Pine Salt									
7,229.2	0.50	90.00	7,229.0	0.0	45.2	45.1	0.00	0.00	0.00
Pine Salt Base									
7,291.2	0.50	90.00	7,291.0	0.0	45.7	45.7	0.00	0.00	0.00
Opeche Salt									
7,300.0	0.50	90.00	7,299.8	0.0	45.8	45.7	0.00	0.00	0.00
7,371.2	0.50	90.00	7,371.0	0.0	46.4	46.4	0.00	0.00	0.00
Opeche Salt Base									
7,400.0	0.50	90.00	7,399.8	0.0	46.6	46.6	0.00	0.00	0.00
7,500.0	0.50	90.00	7,499.8	0.0	47.5	47.5	0.00	0.00	0.00
7,600.0	0.50	90.00	7,599.8	0.0	48.4	48.4	0.00	0.00	0.00
7,615.2	0.50	90.00	7,615.0	0.0	48.5	48.5	0.00	0.00	0.00
Amsden									
7,700.0	0.50	90.00	7,699.8	0.0	49.3	49.2	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
7,771.2	0.50	90.00	7,771.0	0.0	49.9	49.9	0.00	0.00	0.00
Tyler									
7,800.0	0.50	90.00	7,799.8	0.0	50.1	50.1	0.00	0.00	0.00
7,900.0	0.50	90.00	7,899.8	0.0	51.0	51.0	0.00	0.00	0.00
7,994.2	0.50	90.00	7,994.0	0.0	51.8	51.8	0.00	0.00	0.00
Otter/Base Minnelusa									
8,000.0	0.50	90.00	7,999.8	0.0	51.9	51.9	0.00	0.00	0.00
8,100.0	0.50	90.00	8,099.8	0.0	52.8	52.7	0.00	0.00	0.00
8,200.0	0.50	90.00	8,199.8	0.0	53.6	53.6	0.00	0.00	0.00
8,300.0	0.50	90.00	8,299.8	0.0	54.5	54.5	0.00	0.00	0.00
8,336.3	0.50	90.00	8,336.0	0.0	54.8	54.8	0.00	0.00	0.00
Kibbey Lime									
8,400.0	0.50	90.00	8,399.8	0.0	55.4	55.3	0.00	0.00	0.00
8,484.3	0.50	90.00	8,484.0	0.0	56.1	56.1	0.00	0.00	0.00
Charles Salt									
8,500.0	0.50	90.00	8,499.8	0.0	56.2	56.2	0.00	0.00	0.00
8,600.0	0.50	90.00	8,599.8	0.0	57.1	57.1	0.00	0.00	0.00
8,700.0	0.50	90.00	8,699.7	0.0	58.0	58.0	0.00	0.00	0.00
8,800.0	0.50	90.00	8,799.7	0.0	58.9	58.8	0.00	0.00	0.00
8,895.6	0.50	90.00	8,895.3	0.0	59.7	59.7	0.00	0.00	0.00
Start Drop -5.00									
8,900.0	0.50	90.00	8,899.7	0.0	59.7	59.7	0.00	0.00	0.00
8,905.6	0.50	90.00	8,905.3	0.0	59.8	59.8	0.00	0.00	0.00
Start 1094.7 hold at 8905.6 MD									
8,925.6	0.50	90.00	8,925.3	0.0	60.0	59.9	0.00	0.00	0.00
8,935.6	0.00	0.00	8,935.3	0.0	60.0	60.0	5.00	-5.00	0.00
9,000.0	0.00	0.00	8,999.7	0.0	60.0	60.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.7	0.0	60.0	60.0	0.00	0.00	0.00
9,163.3	0.00	0.00	9,163.0	0.0	60.0	60.0	0.00	0.00	0.00
Base Last Salt									
9,200.0	0.00	0.00	9,199.7	0.0	60.0	60.0	0.00	0.00	0.00
9,300.0	0.00	0.00	9,299.7	0.0	60.0	60.0	0.00	0.00	0.00
9,377.3	0.00	0.00	9,377.0	0.0	60.0	60.0	0.00	0.00	0.00
Mission Canyon									
9,400.0	0.00	0.00	9,399.7	0.0	60.0	60.0	0.00	0.00	0.00
9,500.0	0.00	0.00	9,499.7	0.0	60.0	60.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.7	0.0	60.0	60.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.7	0.0	60.0	60.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.7	0.0	60.0	60.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.7	0.0	60.0	60.0	0.00	0.00	0.00
9,926.3	0.00	0.00	9,926.0	0.0	60.0	60.0	0.00	0.00	0.00
Lodgepole									
10,000.3	0.00	0.00	10,000.0	0.0	60.0	60.0	0.00	0.00	0.00
Start 292.5 hold at 10000.3 MD									
10,100.0	0.00	0.00	10,099.7	0.0	60.0	60.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.7	0.0	60.0	60.0	0.00	0.00	0.00
10,292.8	0.00	0.00	10,292.5	0.0	60.0	60.0	0.00	0.00	0.00
Start Build 12.00 KOP									
10,300.0	0.86	112.38	10,299.7	0.0	60.1	60.0	12.00	12.00	0.00
10,325.0	3.86	112.38	10,324.7	-0.4	61.0	61.0	12.00	12.00	0.00
10,350.0	6.86	112.38	10,349.6	-1.3	63.2	63.2	12.00	12.00	0.00
10,375.0	9.86	112.38	10,374.3	-2.7	66.5	66.6	12.00	12.00	0.00
10,400.0	12.86	112.38	10,398.8	-4.6	71.1	71.2	12.00	12.00	0.00
10,425.0	15.86	112.38	10,423.1	-6.9	76.8	77.0	12.00	12.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,450.0	18.86	112.38	10,446.9	-9.8	83.7	84.0	12.00	12.00	0.00
10,475.0	21.86	112.38	10,470.3	-13.1	91.8	92.1	12.00	12.00	0.00
10,500.0	24.86	112.38	10,493.3	-16.9	100.9	101.4	12.00	12.00	0.00
10,525.0	27.86	112.38	10,515.7	-21.1	111.2	111.8	12.00	12.00	0.00
10,550.0	30.86	112.38	10,537.5	-25.7	122.5	123.3	12.00	12.00	0.00
10,575.0	33.86	112.38	10,558.6	-30.8	134.9	135.8	12.00	12.00	0.00
10,600.0	36.86	112.38	10,579.0	-36.4	148.3	149.4	12.00	12.00	0.00
10,625.0	39.86	112.38	10,598.6	-42.3	162.6	163.9	12.00	12.00	0.00
10,650.0	42.86	112.38	10,617.3	-48.6	177.9	179.4	12.00	12.00	0.00
10,675.0	45.86	112.38	10,635.2	-55.2	194.1	195.7	12.00	12.00	0.00
10,700.0	48.86	112.38	10,652.1	-62.2	211.1	213.0	12.00	12.00	0.00
10,705.9	49.58	112.38	10,656.0	-63.9	215.2	217.2	12.00	12.00	0.00
False Bakken									
10,724.9	51.85	112.38	10,668.0	-69.5	228.8	230.9	12.00	12.00	0.00
Upper Bakken Shale									
10,725.0	51.86	112.38	10,668.1	-69.5	228.9	231.0	12.00	12.00	0.00
10,750.0	54.86	112.38	10,683.0	-77.2	247.4	249.8	11.98	11.98	0.00
Middle Bakken									
10,775.0	57.86	112.38	10,696.8	-85.1	266.7	269.3	12.02	12.02	0.00
10,800.0	60.86	112.38	10,709.6	-93.3	286.5	289.4	12.00	12.00	0.00
10,820.1	63.28	112.38	10,719.0	-100.1	303.0	306.1	12.00	12.00	0.00
Lower Bakken Shale									
10,825.0	63.86	112.38	10,721.2	-101.7	307.0	310.1	12.00	12.00	0.00
10,841.1	65.79	112.38	10,728.0	-107.3	320.5	323.8	12.00	12.00	0.00
Pronghorn									
10,850.0	66.86	112.38	10,731.6	-110.4	328.0	331.4	12.00	12.00	0.00
10,875.0	69.86	112.38	10,740.8	-119.2	349.5	353.2	12.00	12.00	0.00
10,884.6	71.01	112.38	10,744.0	-122.7	357.8	361.6	12.00	12.00	0.00
Threeforks									
10,900.0	72.86	112.38	10,748.8	-128.2	371.4	375.4	12.00	12.00	0.00
10,925.0	75.86	112.38	10,755.5	-137.4	393.7	397.9	12.00	12.00	0.00
10,950.0	78.86	112.38	10,761.0	-146.7	416.2	420.7	12.00	12.00	0.00
10,950.1	78.86	112.38	10,761.0	-146.7	416.3	420.8	0.00	0.00	0.00
Threeforks(Top of Target)									
10,975.0	81.86	112.38	10,765.2	-156.1	439.0	443.8	12.03	12.03	0.00
11,000.0	84.86	112.38	10,768.1	-165.5	462.0	467.1	12.00	12.00	0.00
11,025.0	87.86	112.38	10,769.7	-175.0	485.0	490.4	12.00	12.00	0.00
11,039.9	89.65	112.38	10,770.0	-180.7	498.8	504.4	12.00	12.00	0.00
Start 1.3 hold at 11039.9 MD									
11,041.0	89.65	112.38	10,770.0	-181.1	499.8	505.4	0.00	0.00	0.00
7"									
11,041.2	89.65	112.38	10,770.0	-181.2	500.0	505.6	0.00	0.00	0.00
Start DLS 3.02 TFO 269.93 7" csg pt									
11,044.7	89.65	112.38	10,770.0	-182.5	503.3	508.9	0.00	0.00	0.00
Threeforks(Base of Target)									
11,100.0	89.65	110.61	10,770.4	-202.8	554.7	561.0	3.21	0.00	-3.21
11,200.0	89.65	107.59	10,771.0	-235.5	649.2	656.5	3.02	0.00	-3.02
11,300.0	89.64	104.57	10,771.6	-263.1	745.3	753.4	3.02	0.00	-3.02
11,400.0	89.64	101.55	10,772.2	-285.7	842.7	851.5	3.02	0.00	-3.02
11,500.0	89.64	98.53	10,772.8	-303.2	941.1	950.4	3.02	0.00	-3.02
11,600.0	89.64	95.51	10,773.5	-315.4	1,040.4	1,050.0	3.02	0.00	-3.02
11,700.0	89.65	92.49	10,774.1	-322.3	1,140.1	1,149.9	3.02	0.00	-3.02
11,782.3	89.65	90.00	10,774.6	-324.1	1,222.4	1,232.2	3.02	0.00	-3.02
Start 8807.7 hold at 11782.3 MD									

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
11,800.0	89.65	90.00	10,774.7	-324.1	1,240.1	1,249.9	0.00	0.00	0.00
11,900.0	89.65	90.00	10,775.3	-324.1	1,340.1	1,349.9	0.00	0.00	0.00
12,000.0	89.65	90.00	10,775.9	-324.1	1,440.1	1,449.8	0.00	0.00	0.00
12,100.0	89.65	90.00	10,776.5	-324.1	1,540.1	1,549.8	0.00	0.00	0.00
12,200.0	89.65	90.00	10,777.1	-324.1	1,640.1	1,649.7	0.00	0.00	0.00
12,300.0	89.65	90.00	10,777.8	-324.1	1,740.1	1,749.6	0.00	0.00	0.00
12,400.0	89.65	90.00	10,778.4	-324.1	1,840.1	1,849.6	0.00	0.00	0.00
12,500.0	89.65	90.00	10,779.0	-324.1	1,940.1	1,949.5	0.00	0.00	0.00
12,600.0	89.65	90.00	10,779.6	-324.1	2,040.1	2,049.5	0.00	0.00	0.00
12,700.0	89.65	90.00	10,780.2	-324.1	2,140.1	2,149.4	0.00	0.00	0.00
12,800.0	89.65	90.00	10,780.8	-324.1	2,240.1	2,249.4	0.00	0.00	0.00
12,900.0	89.65	90.00	10,781.4	-324.1	2,340.1	2,349.3	0.00	0.00	0.00
13,000.0	89.65	90.00	10,782.0	-324.1	2,440.1	2,449.3	0.00	0.00	0.00
13,100.0	89.65	90.00	10,782.6	-324.1	2,540.1	2,549.2	0.00	0.00	0.00
13,200.0	89.65	90.00	10,783.3	-324.1	2,640.1	2,649.2	0.00	0.00	0.00
13,300.0	89.65	90.00	10,783.9	-324.1	2,740.1	2,749.1	0.00	0.00	0.00
13,400.0	89.65	90.00	10,784.5	-324.1	2,840.1	2,849.1	0.00	0.00	0.00
13,500.0	89.65	90.00	10,785.1	-324.1	2,940.1	2,949.0	0.00	0.00	0.00
13,600.0	89.65	90.00	10,785.7	-324.1	3,040.1	3,048.9	0.00	0.00	0.00
13,700.0	89.65	90.00	10,786.3	-324.1	3,140.1	3,148.9	0.00	0.00	0.00
13,800.0	89.65	90.00	10,786.9	-324.1	3,240.1	3,248.8	0.00	0.00	0.00
13,900.0	89.65	90.00	10,787.5	-324.1	3,340.1	3,348.8	0.00	0.00	0.00
14,000.0	89.65	90.00	10,788.1	-324.1	3,440.1	3,448.7	0.00	0.00	0.00
14,100.0	89.65	90.00	10,788.8	-324.1	3,540.1	3,548.7	0.00	0.00	0.00
14,200.0	89.65	90.00	10,789.4	-324.1	3,640.1	3,648.6	0.00	0.00	0.00
14,300.0	89.65	90.00	10,790.0	-324.1	3,740.0	3,748.6	0.00	0.00	0.00
14,400.0	89.65	90.00	10,790.6	-324.1	3,840.0	3,848.5	0.00	0.00	0.00
14,500.0	89.65	90.00	10,791.2	-324.1	3,940.0	3,948.5	0.00	0.00	0.00
14,600.0	89.65	90.00	10,791.8	-324.1	4,040.0	4,048.4	0.00	0.00	0.00
14,700.0	89.65	90.00	10,792.4	-324.1	4,140.0	4,148.3	0.00	0.00	0.00
14,800.0	89.65	90.00	10,793.0	-324.1	4,240.0	4,248.3	0.00	0.00	0.00
14,900.0	89.65	90.00	10,793.6	-324.1	4,340.0	4,348.2	0.00	0.00	0.00
15,000.0	89.65	90.00	10,794.3	-324.1	4,440.0	4,448.2	0.00	0.00	0.00
15,100.0	89.65	90.00	10,794.9	-324.1	4,540.0	4,548.1	0.00	0.00	0.00
15,200.0	89.65	90.00	10,795.5	-324.1	4,640.0	4,648.1	0.00	0.00	0.00
15,300.0	89.65	90.00	10,796.1	-324.1	4,740.0	4,748.0	0.00	0.00	0.00
15,400.0	89.65	90.00	10,796.7	-324.1	4,840.0	4,848.0	0.00	0.00	0.00
15,500.0	89.65	90.00	10,797.3	-324.1	4,940.0	4,947.9	0.00	0.00	0.00
15,600.0	89.65	90.00	10,797.9	-324.1	5,040.0	5,047.9	0.00	0.00	0.00
15,700.0	89.65	90.00	10,798.5	-324.1	5,140.0	5,147.8	0.00	0.00	0.00
15,800.0	89.65	90.00	10,799.2	-324.1	5,240.0	5,247.8	0.00	0.00	0.00
15,900.0	89.65	90.00	10,799.8	-324.1	5,340.0	5,347.7	0.00	0.00	0.00
16,000.0	89.65	90.00	10,800.4	-324.1	5,440.0	5,447.6	0.00	0.00	0.00
16,100.0	89.65	90.00	10,801.0	-324.1	5,540.0	5,547.6	0.00	0.00	0.00
16,200.0	89.65	90.00	10,801.6	-324.1	5,640.0	5,647.5	0.00	0.00	0.00
16,300.0	89.65	90.00	10,802.2	-324.1	5,740.0	5,747.5	0.00	0.00	0.00
16,400.0	89.65	90.00	10,802.8	-324.1	5,840.0	5,847.4	0.00	0.00	0.00
16,500.0	89.65	90.00	10,803.4	-324.1	5,940.0	5,947.4	0.00	0.00	0.00
16,600.0	89.65	90.00	10,804.0	-324.1	6,040.0	6,047.3	0.00	0.00	0.00
16,700.0	89.65	90.00	10,804.7	-324.1	6,140.0	6,147.3	0.00	0.00	0.00
16,800.0	89.65	90.00	10,805.3	-324.1	6,240.0	6,247.2	0.00	0.00	0.00
16,900.0	89.65	90.00	10,805.9	-324.1	6,340.0	6,347.2	0.00	0.00	0.00
17,000.0	89.65	90.00	10,806.5	-324.1	6,440.0	6,447.1	0.00	0.00	0.00
17,100.0	89.65	90.00	10,807.1	-324.1	6,540.0	6,547.1	0.00	0.00	0.00
17,200.0	89.65	90.00	10,807.7	-324.1	6,640.0	6,647.0	0.00	0.00	0.00

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Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
17,300.0	89.65	90.00	10,808.3	-324.1	6,740.0	6,746.9	0.00	0.00	0.00
17,400.0	89.65	90.00	10,808.9	-324.1	6,840.0	6,846.9	0.00	0.00	0.00
17,500.0	89.65	90.00	10,809.5	-324.1	6,940.0	6,946.8	0.00	0.00	0.00
17,600.0	89.65	90.00	10,810.2	-324.1	7,040.0	7,046.8	0.00	0.00	0.00
17,700.0	89.65	90.00	10,810.8	-324.1	7,140.0	7,146.7	0.00	0.00	0.00
17,800.0	89.65	90.00	10,811.4	-324.1	7,240.0	7,246.7	0.00	0.00	0.00
17,900.0	89.65	90.00	10,812.0	-324.1	7,340.0	7,346.6	0.00	0.00	0.00
18,000.0	89.65	90.00	10,812.6	-324.1	7,440.0	7,446.6	0.00	0.00	0.00
18,100.0	89.65	90.00	10,813.2	-324.1	7,540.0	7,546.5	0.00	0.00	0.00
18,200.0	89.65	90.00	10,813.8	-324.1	7,640.0	7,646.5	0.00	0.00	0.00
18,300.0	89.65	90.00	10,814.4	-324.1	7,740.0	7,746.4	0.00	0.00	0.00
18,400.0	89.65	90.00	10,815.0	-324.1	7,840.0	7,846.4	0.00	0.00	0.00
18,500.0	89.65	90.00	10,815.7	-324.1	7,940.0	7,946.3	0.00	0.00	0.00
18,600.0	89.65	90.00	10,816.3	-324.1	8,040.0	8,046.2	0.00	0.00	0.00
18,700.0	89.65	90.00	10,816.9	-324.1	8,140.0	8,146.2	0.00	0.00	0.00
18,800.0	89.65	90.00	10,817.5	-324.1	8,240.0	8,246.1	0.00	0.00	0.00
18,900.0	89.65	90.00	10,818.1	-324.1	8,340.0	8,346.1	0.00	0.00	0.00
19,000.0	89.65	90.00	10,818.7	-324.1	8,440.0	8,446.0	0.00	0.00	0.00
19,100.0	89.65	90.00	10,819.3	-324.1	8,540.0	8,546.0	0.00	0.00	0.00
19,200.0	89.65	90.00	10,819.9	-324.1	8,640.0	8,645.9	0.00	0.00	0.00
19,300.0	89.65	90.00	10,820.5	-324.1	8,740.0	8,745.9	0.00	0.00	0.00
19,400.0	89.65	90.00	10,821.2	-324.1	8,840.0	8,845.8	0.00	0.00	0.00
19,500.0	89.65	90.00	10,821.8	-324.1	8,940.0	8,945.8	0.00	0.00	0.00
19,600.0	89.65	90.00	10,822.4	-324.1	9,039.9	9,045.7	0.00	0.00	0.00
19,700.0	89.65	90.00	10,823.0	-324.1	9,139.9	9,145.6	0.00	0.00	0.00
19,800.0	89.65	90.00	10,823.6	-324.1	9,239.9	9,245.6	0.00	0.00	0.00
19,900.0	89.65	90.00	10,824.2	-324.1	9,339.9	9,345.5	0.00	0.00	0.00
20,000.0	89.65	90.00	10,824.8	-324.1	9,439.9	9,445.5	0.00	0.00	0.00
20,100.0	89.65	90.00	10,825.4	-324.1	9,539.9	9,545.4	0.00	0.00	0.00
20,200.0	89.65	90.00	10,826.1	-324.1	9,639.9	9,645.4	0.00	0.00	0.00
20,300.0	89.65	90.00	10,826.7	-324.1	9,739.9	9,745.3	0.00	0.00	0.00
20,400.0	89.65	90.00	10,827.3	-324.1	9,839.9	9,845.3	0.00	0.00	0.00
20,500.0	89.65	90.00	10,827.9	-324.1	9,939.9	9,945.2	0.00	0.00	0.00
20,590.1	89.65	90.00	10,828.4	-324.1	10,030.0	10,035.2	0.00	0.00	0.00
TD at 20590.1									

Casing Points

Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)
2,050.0	2,050.0	9 5/8"	9.625	13.500
11,041.0	10,770.0	7"	7.000	8.750

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Wade Federal 5300 41-30 7T
Company:	Oasis	**TVD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2070.0ft (Original Well Elev)
Site:	153N-100W-29/30	**North Reference:**	True
Well:	Wade Federal 5300 41-30 7T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 7T		
Design:	Design #1		

Formations

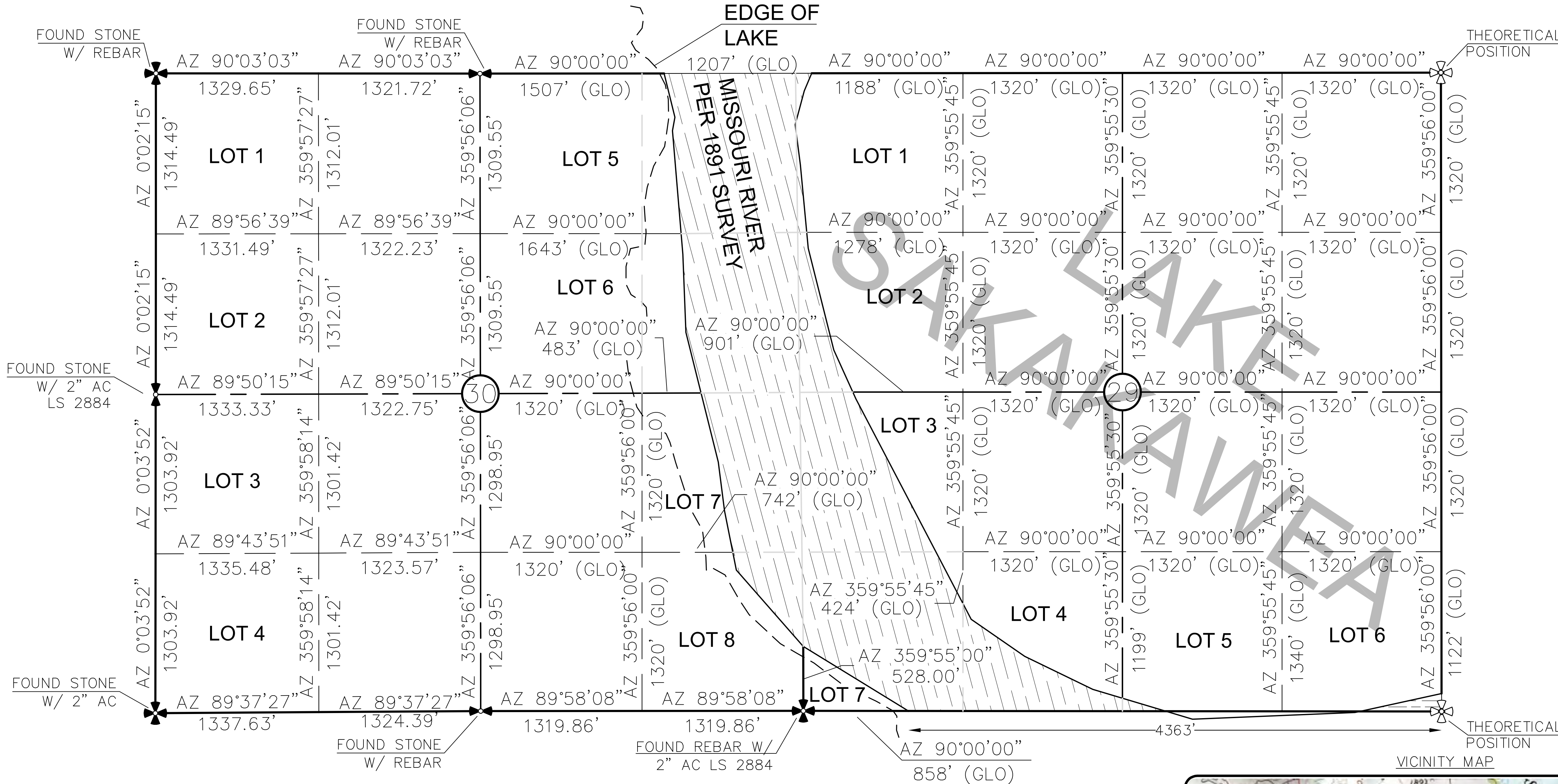
Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,920.0	1,920.0	Pierre			
4,566.1	4,566.0	Greenhorn			
4,969.1	4,969.0	Mowry			
5,391.1	5,391.0	Dakota			
6,407.2	6,407.0	Rierdon			
6,896.2	6,896.0	Dunham Salt			
6,942.2	6,942.0	Dunham Salt Base			
7,205.2	7,205.0	Pine Salt			
7,229.2	7,229.0	Pine Salt Base			
7,291.2	7,291.0	Opeche Salt			
7,371.2	7,371.0	Opeche Salt Base			
7,615.2	7,615.0	Amsden			
7,771.2	7,771.0	Tyler			
7,994.2	7,994.0	Otter/Base Minnelusa			
8,336.3	8,336.0	Kibbey Lime			
8,484.3	8,484.0	Charles Salt			
9,163.3	9,163.0	Base Last Salt			
9,377.3	9,377.0	Mission Canyon			
9,926.3	9,926.0	Lodgepole			
10,705.9	10,656.0	False Bakken			
10,724.9	10,668.0	Upper Bakken Shale			
10,750.0	10,683.0	Middle Bakken			
10,820.1	10,719.0	Lower Bakken Shale			
10,841.1	10,728.0	Pronghorn			
10,884.6	10,744.0	Threeforks			
10,950.1	10,761.0	Threeforks(Top of Target)			
11,044.7	10,770.0	Threeforks(Base of Target)			

Plan Annotations

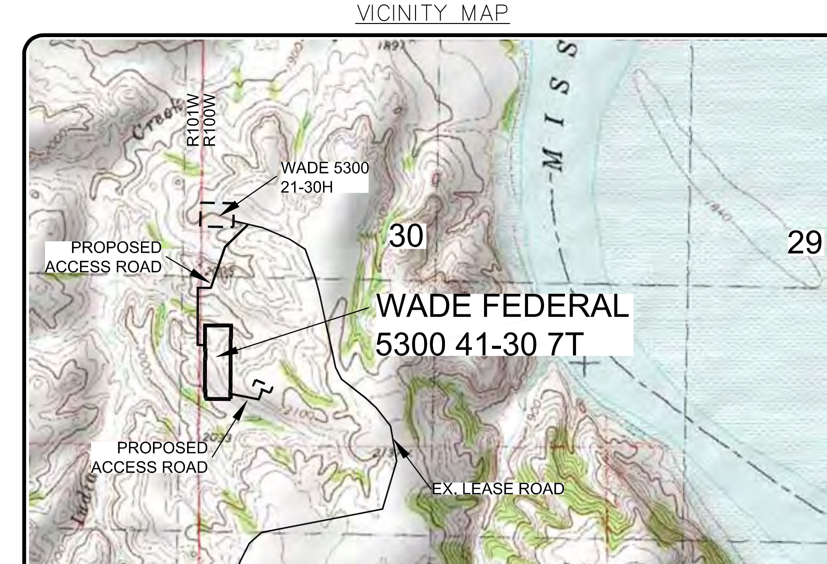
Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates +N/-S (ft)	+E/-W (ft)	Comment
2,020.0	2,020.0	0.0	0.0	Start Build 5.00
2,030.0	2,030.0	0.0	0.0	Start 6865.6 hold at 2030.0 MD
8,895.6	8,895.3	0.0	60.0	Start Drop -5.00
8,905.6	8,905.3	0.0	60.0	Start 1094.7 hold at 8905.6 MD
10,000.3	10,000.0	0.0	60.0	Start 292.5 hold at 10000.3 MD
10,292.8	10,292.5	0.0	60.0	Start Build 12.00 KOP
11,039.9	10,770.0	-180.7	498.8	Start 1.3 hold at 11039.9 MD
11,041.2	10,770.0	-181.2	500.0	Start DLS 3.02 TFO 269.93 7" csg pt
11,782.3	10,774.6	-324.1	1,222.4	Start 8807.7 hold at 11782.3 MD
20,590.1	10,828.4	-324.1	10,030.0	TD at 20590.1

SECTION BREAKDOWN

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 7T"
877 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTIONS 29 & 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



VICINITY MAP



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ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1897. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY 0°03'.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/18/14
NORTH DAKOTA

— MONUMENT – RECOVERED
— MONUMENT – NOT RECOVERED

0 1000
1" = 1000'

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTIONS 29 & 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.06
Checked By: D.D.K. Date: JAN 2014

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SHEET NO. 2/8

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
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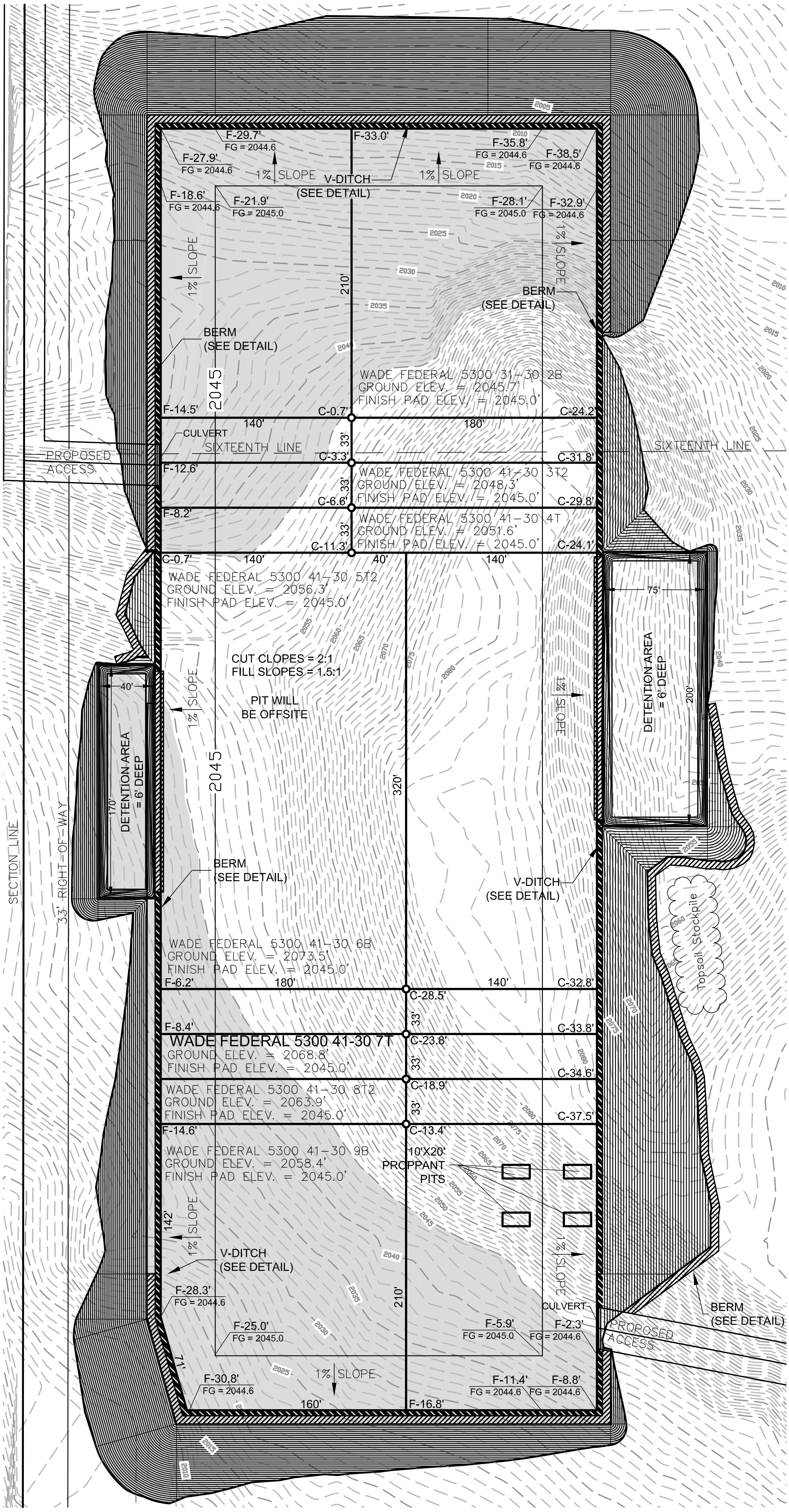
"WADE FEDERAL 5300 41-30 7T"
877 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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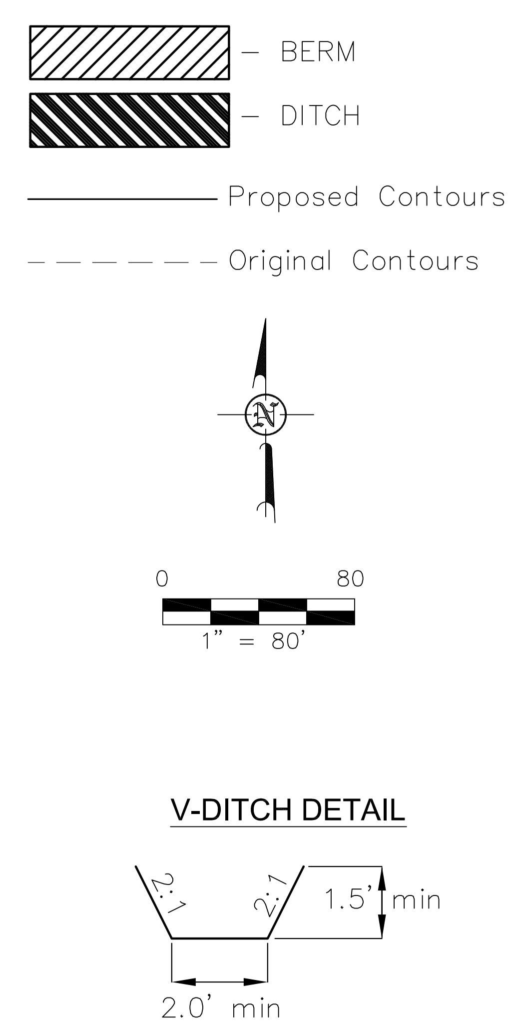
REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 3/18/14
NORTH DAKOTA

NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

3/8
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OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S13-09-381.06
Checked By:	D.D.K.	Date:	JAN. 2014

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

Q:\2013\S13-09-\381.06 Oasis Petroleum - Wade 5300 41-30 7T Well Location 8-Well Pad\CAD\REVISED WADE 7T.dwg - 3/19/2014 7:00 AM josh schmierer

WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 7T"
877 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

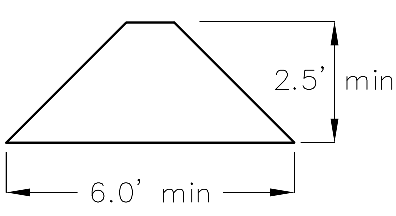
WELL SITE ELEVATION	2068.8
WELL PAD ELEVATION	2045.0

EXCAVATION	149,966
PLUS PIT	0
	149,966
EMBANKMENT	113,402
PLUS SHRINKAGE (25%)	28,351
	141,753
STOCKPILE PIT	0
STOCKPILE TOP SOIL (6")	7,955
BERMS	2,533 LF = 821 CY
DITCHES	1,655 LF = 253 CY
DETENTION AREA	4,219 CY
STOCKPILE MATERIAL	3,909
DISTURBED AREA FROM PAD	10.18 ACRES

NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 2:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION
877' FSL
280' FWL

BERM DETAIL



DITCH DETAIL

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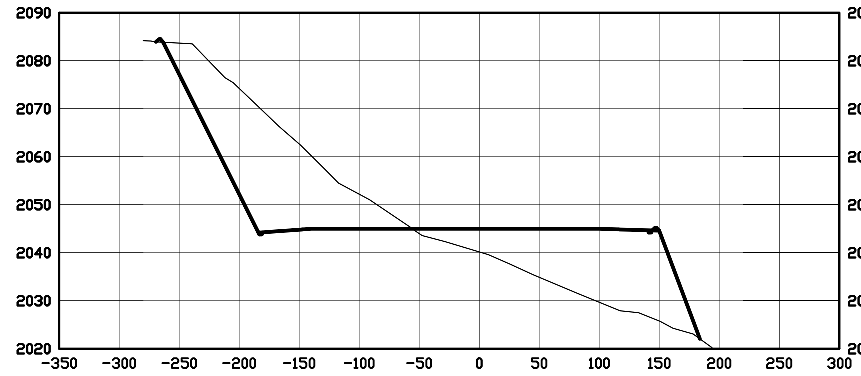
OASIS PETROLEUM NORTH AMERICA, LLC
QUANTITIES
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.06
Checked By: D.D.K. Date: JAN. 2014

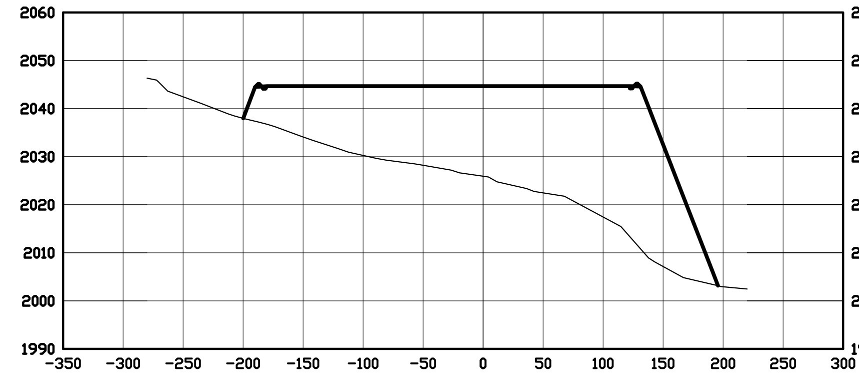
Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

CROSS SECTIONS

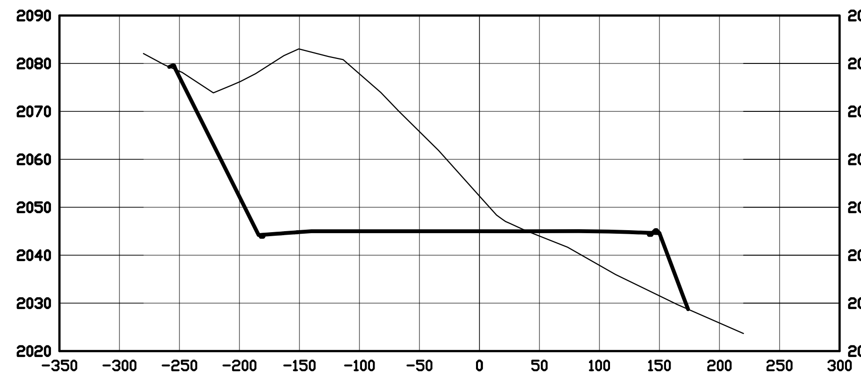
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 7T"
877 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



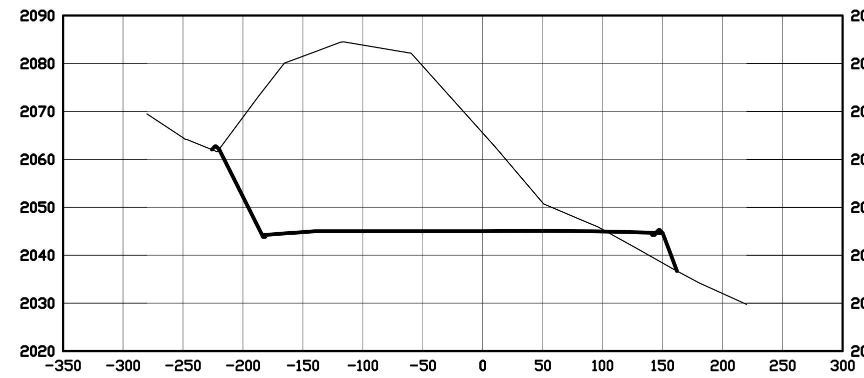
STA. 8+00



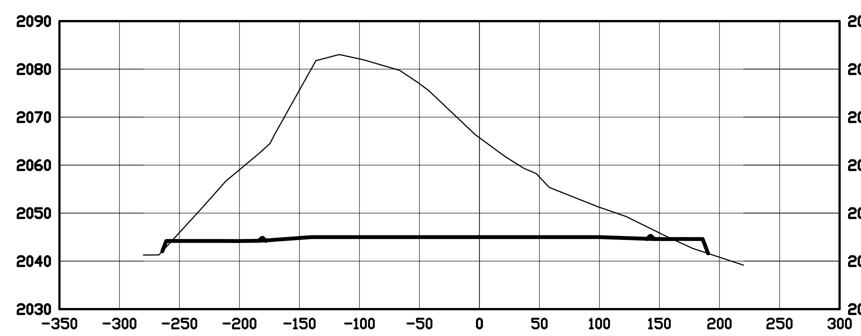
STA. 9+40



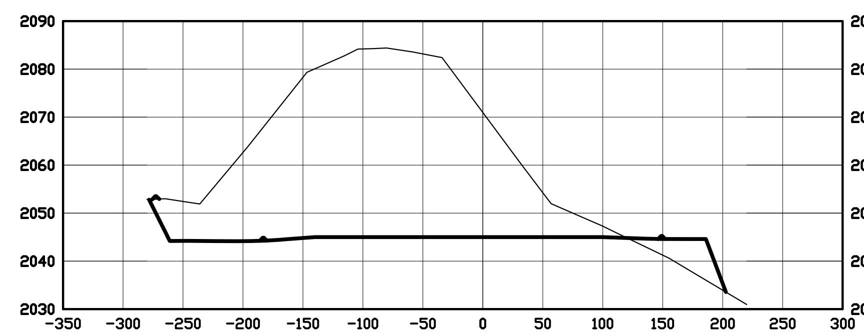
STA. 7+00



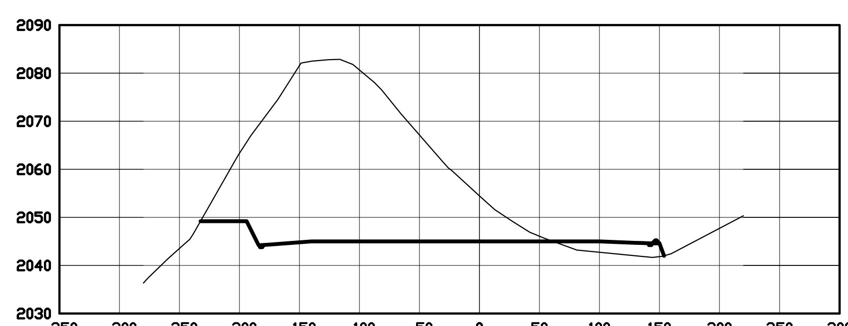
STA. 6+00



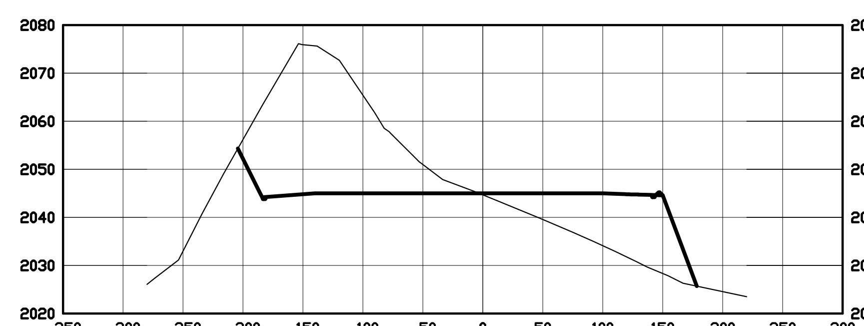
STA. 4+00



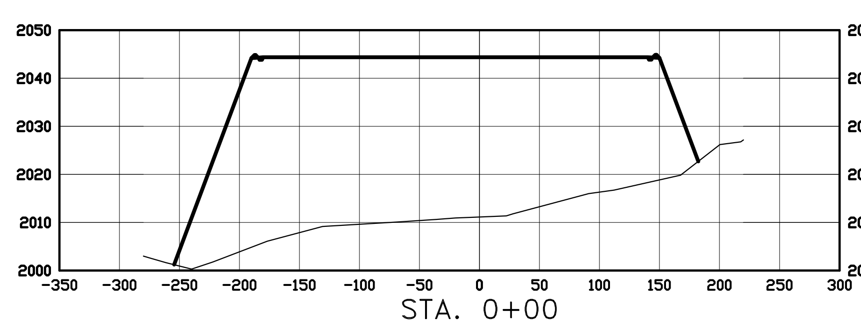
STA. 5+00



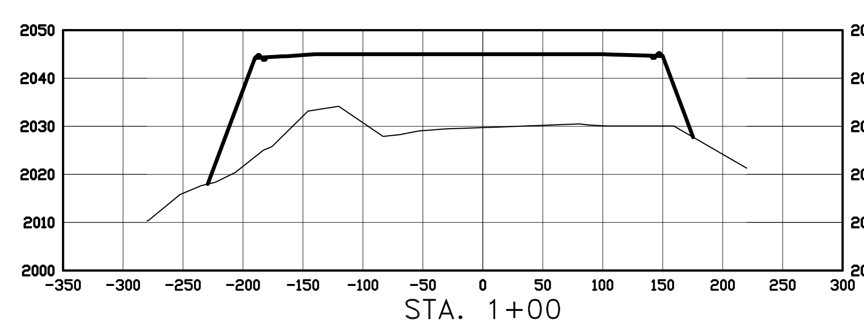
STA. 3+00



STA. 2+00



STA. 0+00



STA. 1+00

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS–3880
DATE: 3/18/14
NORTH DAKOTA

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SCALE
HORIZ 1"=160'
VERT 1"=40'

7/8
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OASIS PETROLEUM NORTH AMERICA, LLC
CROSS SECTIONS
SECTION 30, T153N, R100W

MCKENZIE COUNTY, NORTH DAKOTA

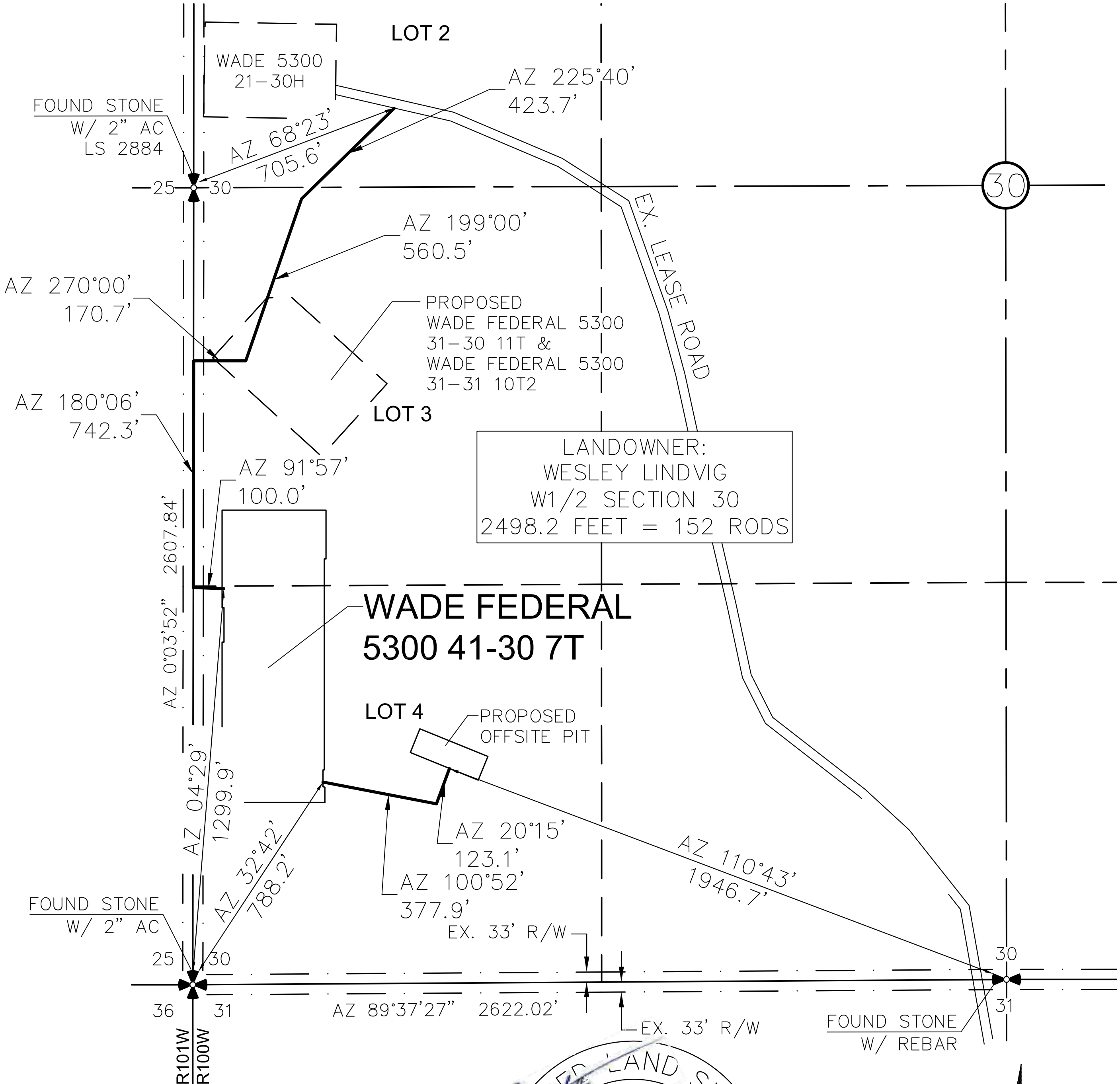
Drawn By:	B.H.H.	Project No.:	S13-09-381.06
Checked By:	D.D.K.	Date:	JAN. 2014

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

Q:\2013\S13–09–\381.06 Oasis Petroleum – Wade 5300 41–30 7T Well Location 8–Well Pad\CAD\REVISED WADE 7T.dwg – 3/19/2014 7:01 AM josh schmierer

ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 7T"
877 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



REGISTERED LAND SURVEYOR
DARYL D.
KASEMAN
LS-3880
DATE: 3/18/14
NORTH DAKOTA

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N

0 500

1" = 500'

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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ACCESS APPROACH
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S13-09-381.06
Checked By:	D.D.K.	Date:	JAN. 2014

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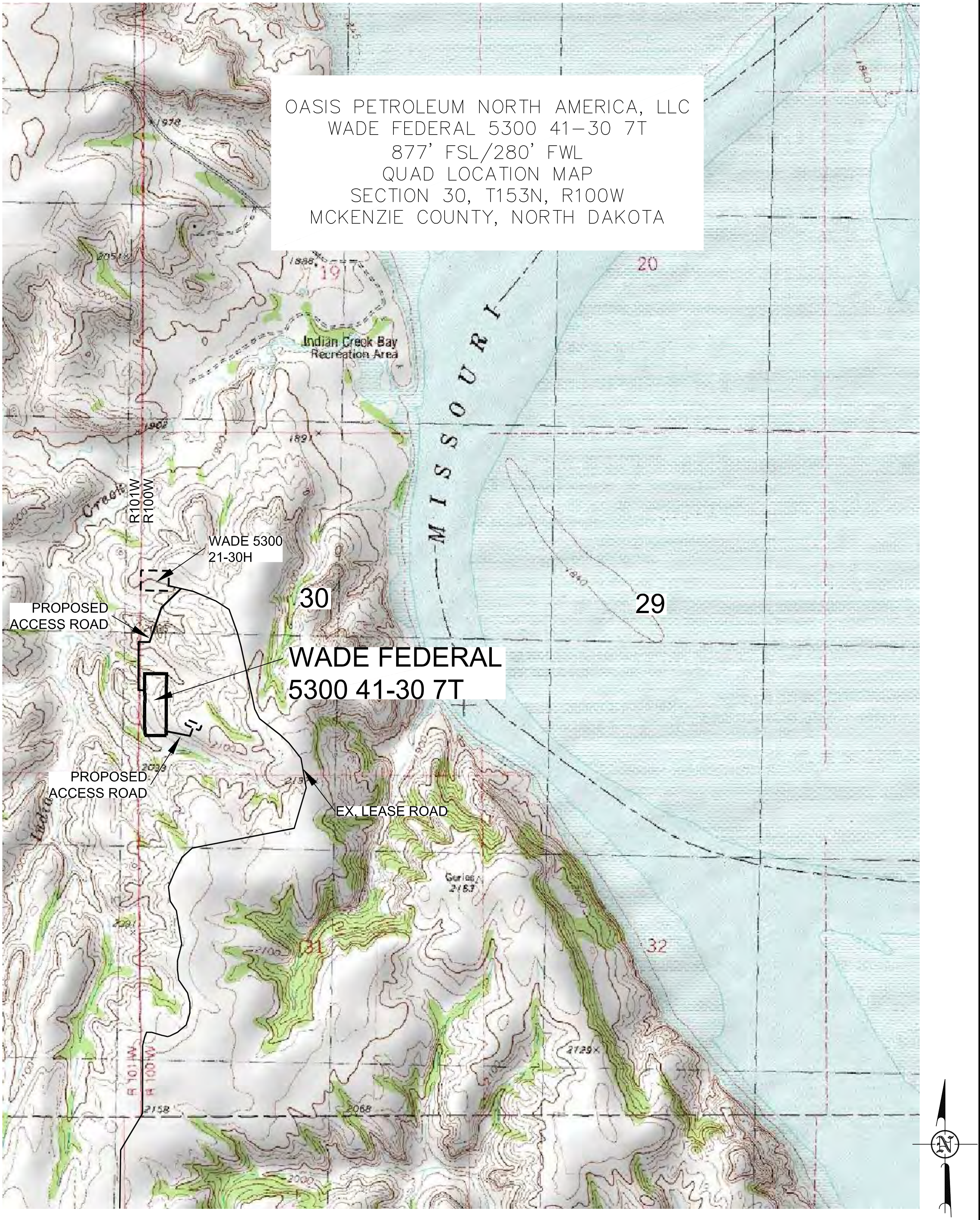
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD



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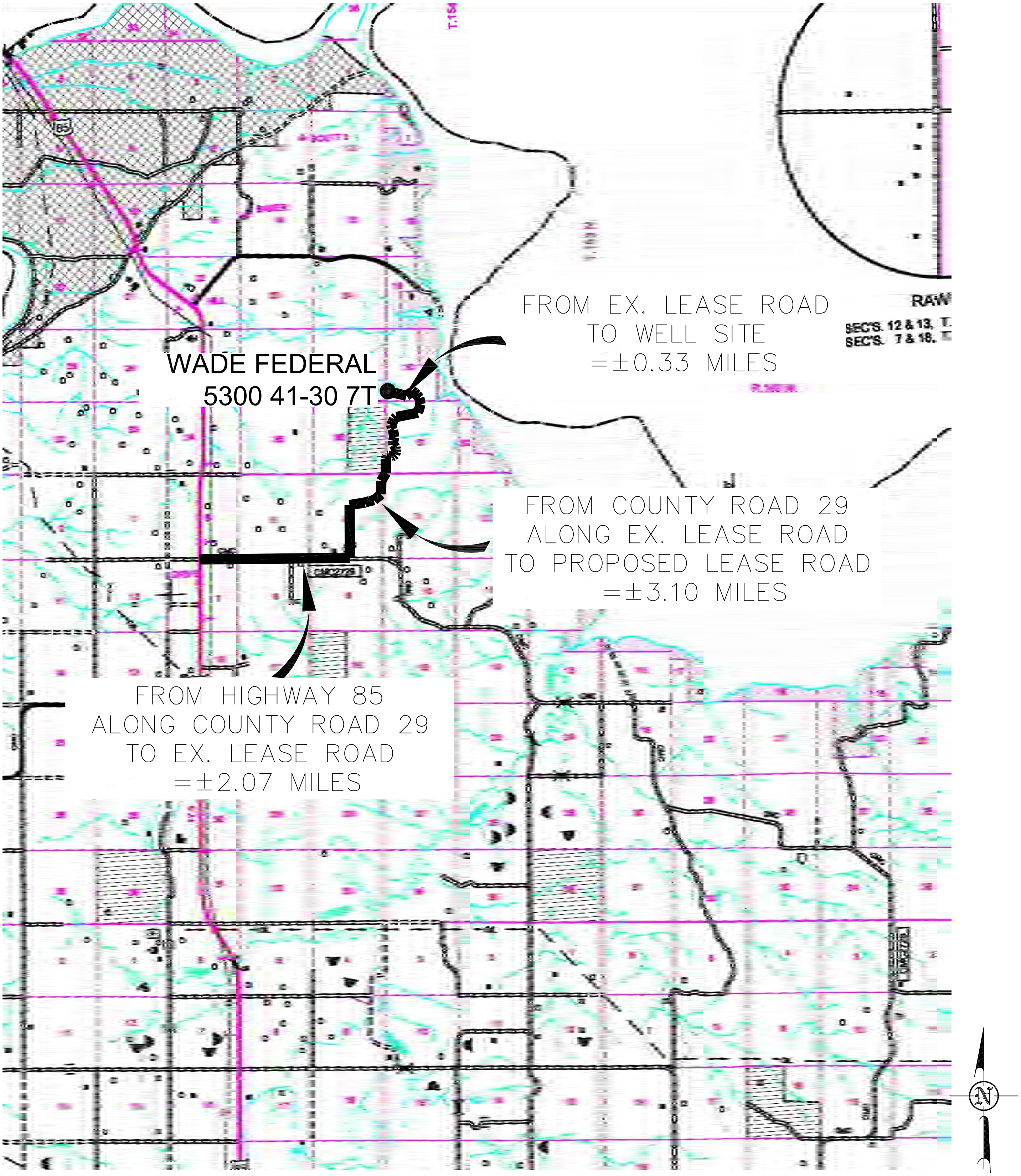
OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S13-09-381.06
Checked By:	D.D.K.	Date:	JAN. 2014

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 7T"
877 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



SCALE: 1" = 2 MILE

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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S13-09-381.06
Checked By:	D.D.K.	Date:	JAN. 2014

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

STATEMENT

This statement is being sent in order to comply with NDAC 43-02-03-16 (Application for permit to drill and recomplete) which states (in part that) "confirmation that a legal street address has been requested for the well site, and well facility if separate from the well site, and the proposed road access to the nearest existing public road". On the date noted below a legal street address was requested from the appropriate county office.

McKenzie County
Aaron Chisolm – McKenzie County Dept.

Wade Federal 5300 31-30 2B
Wade Federal 5300 31-30 3T2
Wade Federal 5300 41-30 4T
Wade Federal 5300 41-30 5T2
Wade Federal 5300 41-30 6B
Wade Federal 5300 41-30 7T
Wade Federal 5300 41-30 8T2
Wade Federal 5300 41-30 9B

Brandi Terry

Brandi Terry
Regulatory Specialist
Oasis Petroleum North America, LLC

From:	Heather McCowan
To:	Webber, Alice D.; Chelsea Covington; Brandi Terry
Subject:	RE: Wade Federal 8 Well Pad
Date:	Thursday, June 05, 2014 9:25:20 AM
Attachments:	image002.png

Hell Alice,

The cuttings will be buried offsite for this well. I believe a sundry was sent in.

Thanks,

Heather McCowan

Regulatory Assistant | 1001 Fannin, Suite 1500, Houston, Texas 77002 | 281-404-9563 Direct | hmccowan@oasispetroleum.com

From: Webber, Alice D. [mailto:adwebber@nd.gov]
Sent: Thursday, June 05, 2014 9:12 AM
To: Chelsea Covington; Brandi Terry; Heather McCowan
Subject: RE: Wade Federal 8 Well Pad

Good morning Ladies,

Just checking in to see if we know where the cuttings are being hauled to for this pad. That is the only thing we are waiting for and I would like to get these permitted for Oasis today.

Thanks,
Alice